

AGENTS

Helix Constitution — Universal AGENTS.md

Field	Value
Revision	43
Created	2026-05-14
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Status	active

Status summary	Added §11.4.192 — Continuous multi-track auto-backfill: a FREE track MUST be IMMEDIATELY re-assigned its next-highest-priority domain work-set, never left idle while its domain has actionable items (User mandate 2026-07-11, verbatim "once a Track is free, we shall assign next set of workable items to it ... full continuous multi-track development ... no stopping ... this is MANDATORY"): the moment any track's worker COMPLETES its current work-set, that FREE track MUST be IMMEDIATELY assigned the NEXT-highest-priority actionable work-set from its OWN domain (its §11.4.191 logic_group / canonical branch) — a track is NEVER idle while its domain still holds actionable items, and the conductor/ruler CONTINUOUSLY re-assigns with NO stopping; PROMOTES the §11.4.103(B) ≥3-parallel-background-streams-with-auto-backfill invariant into the multi-track (§11.4.176/.178/.179/.187) layer (§11.4.103(B) backfills a STREAM the moment it is done, §11.4.192 backfills a TRACK the moment its worker is done, keyed to that track's canonical (track, branch) domain §11.4.191); (1) on completion the ruler §11.4.187 looks up + CLAIMs exactly-once §11.4.176 + DISPATCHes the next §11.4.182-labelled stream; (2) priority per §11.4.42/.72 (audio-first); (3) SOURCE-SIDE advances even when on-device VALIDATION is build/device-gated (§11.4.108/.185, honest §11.4.3/.52 SKIP-with-reason ONLY the device-gated portion — NOT a stop); (4) honest boundary §11.4.6 — a track idles ONLY when its domain genuinely has zero actionable items OR all remaining is externally blocked with no source-side advancement possible (§11.4.94(A)/.97/.101), "the merges are done"/"nothing visible right now" is NEVER a valid idle reason. Classification: universal (§11.4.17). Composes §11.4.42/.58/.72/.94/.97/.101/.103/.167/.176/.178/.179/.182/.187. Propagation gate CM-COVENANT-114-192-PROPAGATION + recommended gate CM-CONTINUOUS-MULTITRACK-BACKFILL + paired §1.1 mutation. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.191 — Work-to-track/branch binding enforcement: a logic-group's work can NEVER be committed OR dispatched onto the wrong track/branch (research-derived 2026-07-10): every logical group of workable items binds to EXACTLY ONE canonical (track, branch) recorded ONCE in the §11.4.93 workable-items DB (logic_groups.destination authoritative branch + a canonical_track column + a group_paths file-scope manifest), and NO change belonging to a group may LAND on nor be DISPATCHED to a track/branch other than that group's canonical one — the file-scope generalisation of §11.4.181/.182 from branch-NAME to work-PLACEMENT consistency (forensic FACT 2026-07-10: scrcpy/Remote-app work — group mistiq-vader , canonical feature/mistiq-vader /T2 — was committed onto main /T1; §11.4.181's guard-branch-consistency.sh never fired because committing onto an existing branch is not a feature-branch CREATE, no gate checked which FILES land on which BRANCH); (1) single source of truth looked-up-never-reinvented (§11.4.6); (2) preventive PreToolUse hook guard-work-track-binding.sh (§11.4.109/.177,
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BLOCKS a git-commit-class Bash call OR an Agent/Task dispatch whose file-scope/label track+branch ≠ canonical; fail-closed on unreadable registry); (3) detective gate CM-WORK-TRACK-BINDING-ENFORCED (§11.4.110/.108, asserts on merge-base(HEAD,main)..HEAD that every group_paths -owned file's group destination == current branch); (4) risk-free reconciliation by MERGE-onto-canonical-branch, ff-only, NEVER force-push (§9.2/.113/.84). Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-191-PROPAGATION + recommended gate CM-WORK-TRACK-BINDING-ENFORCED + paired §1.1 mutation. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.190 — Website engineering-quality mandate (User mandate 2026-07-10): every website / web-UI a project ships — INCLUDING this project's OWN website — MUST satisfy AND PROVE with captured physical evidence (§11.4.5/.69/.107/.170, never a bare claim §11.4.6) ALL of — (A) FULL RESPONSIVENESS across all major browser engines (Chromium/Firefox/WebKit) + all OSes + all device classes (desktop/laptop/tablet/phone/large-display) + the full range of screen sizes/orientations/pixel-densities, NO broken layout/overflow/clipping/overlap/off-screen/unusable control at ANY breakpoint; (B) COMPLETE SEO OPTIMIZATION — semantic HTML + document outline, per-page unique title + meta description, Open Graph + Twitter cards, canonical URLs, schema.org/JSON-LD structured data, robots + XML sitemap, crawlable/indexable markup, WCAG AA + Core Web Vitals; (C) UNIQUE OpenDesign-AUTHORED templates/layouts (§11.4.162), NOT generic/off-the-shelf/templated, visually unique/innovative/creative/brand-distinct; (D) BLEEDING-EDGE ENTERPRISE VISUAL QUALITY, stunning polished look-and-feel in BOTH light+dark themes; (E) ANTI-BLUFF PROOF — responsiveness by device-independent host-rendered screenshots across the real breakpoint×engine matrix + an OCR/vision layout oracle, SEO by automated audit (Lighthouse SEO + structured-data validation + meta/OG/sitemap/robots presence) meeting a defined score floor, visual quality by §11.4.170 host-rendered pixel proof per screen×state×{light,dark}, uniqueness by OpenDesign-token provenance; a website reported responsive/SEO-optimized/unique/enterprise-quality WITHOUT these captured proofs is a §11.4 PASS-bluff at the web-UI layer; strict WEB-quality expansion of §11.4.162 (OpenDesign) binding §11.4.170 host-rendered pixel proof as the correctness oracle. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-190-PROPAGATION + recommended gates CM-WEBSITE-RESPONSIVE-PROVEN / CM-WEBSITE-SEO-OPTIMIZED / CM-WEBSITE-OPENDSIGN-UNIQUE + paired §1.1 mutation. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.189 — Most-reopened cases get extra-depth live-testing scrutiny FIRST (User mandate 2026-07-10): all LIVE TESTING MUST give EXTRA-DEPTH retest + in-depth investigation + FULL validation/verification with REAL PHYSICAL captured evidence (§11.4.5/.69/.107) and NO bluff ANY-

WHERE to the cases REOPENED THE MOST TIMES (highest §11.4.55 reopens-count) — the empirically-most-fragile set gets the DEEPEST live scrutiny FIRST, ahead of the rest of the suite (§11.4.132 risk-descending, most-reopened the load-bearing key); each most-reopened case's live PASS cites captured-evidence on a CLEAN deployment (§11.4.108 runtime-signature) with the §11.4.115 RED → GREEN flip where a §11.4.135 guard exists; STRENGTHENS/REFINES §11.4.132(d) (promotes most-reopened into its OWN extra-depth mandate at the LIVE-TEST layer, not merely one ranking factor) + §11.4.55 + §11.4.129/.130. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-189-PROPAGATION + recommended gate CM-MOST-REOPENED-EXTRA-DEPTH-LIVE-TEST + paired §1.1 mutation. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.188 — Regular main → feature merge cadence (User mandate 2026-07-09): every long-lived feature branch AND every parallel-development track (§11.4.58/.103/.167/.176/.178/.179) MUST regularly `git merge origin/main` INTO its own branch THROUGHOUT the work, NEVER only at the end, so no branch drifts far from main + the back-merge stays small/low-conflict/low-risk; GENERALISES §11.4.167(D) (trunk-into-every-BIG-feature-stream) to EVERY feature branch on EVERY track (§11.4.167(D) remains the big-item specialisation); cadence = after every trunk tag / ≥ daily / before any significant new work chunk (many small merges); HOW = MERGE never rebase a shared/tagged branch (§9/§11.4.113), fetch-first (§11.4.37/.71/.6), §9.2 pre-op backup before large/risky merges, resolve conflicts with ZERO markers + ZERO dropped files + union preserved (§11.4.41 step 3), quiescent-only (§11.4.84), background merge+verify (§11.4.88/.89/.103), NEVER force-push (§11.4.113); ANTI-BLUFF = post-merge smoke GREEN (§11.4.42) + conflict-marker scan empty + no-lost-commit, captured evidence (§11.4.5/.69), a merge claimed "clean" without those checks is a §11.4/.1 integration-layer bluff; honest boundary §11.4.6 — keeps the branch mergeable + low-conflict, does NOT prove its own work correct (still §11.4.108/.40) nor replace the approval-gated back-merge (§11.4.167(I)/.40/.41). Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-188-PROPAGATION + recommended gate CM-REGULAR-MAIN-MERGE-INTO-FEATURE + paired §1.1 mutation. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.187 — Automatic multi-track ruler orchestration (User mandate 2026-07-04; landed 2026-07-09): every project incorporating this constitution MUST ship its multi-track parallel-development orchestration as a UNIVERSAL, AUTOMATIC, OUT-OF-THE-BOX, INHERITED capability, not a per-project reimplementa-tion, via a single conductor session that programmatically spawns/drives/resumes/monitors/controls per-track headless workers (native headless `claude -p --output-format stream-json`, session_id captured from the first `init` event, success read from `result.is_error` never the process exit code, `--resume` invoked from the

same cwd+env); per-subscription auth (unset ANTHROPIC_API_KEY + per-alias CLAUDE_CODE_OAUTH_TOKEN / CLAUDE_CONFIG_DIR) with rate-limit → rebind-to-healthy-alias → provider-fallback → bounded-park (§11.4.101); crash-resilient ruler self-supervisor (durable write-temp-then-rename state + watchdog rehydrate, never tmpfs); idempotent bootstrap auto-installed via the §11.4.164 post_update_hook.sh seam; conductor-stays-home (never bound into a track worktree); the entire mechanism lives at constitution/scripts/multitrack/ inherited BY REFERENCE (§11.4.28(B)/§11.4.177), carrying NO project literal — every consumer supplies its own config/multitrack/<hostname>.yaml as DATA. Honest boundary §11.4.6: this anchor mandates the MECHANISM; real per-subscription quota-isolation proof needs live tokens (an operator-scheduled acceptance window), never conflated with a mock-driven spawn/resume/fallback-contract GREEN. Classification: universal (§11.4.17). Composes §11.4.20/.28/.58/.70/.101/.103/.111/.116/.119/.126/.128/.131/.147/.164/.167/.176/§12.6/§12.8. Propagation gate CM-COVENANT-114-187-PROPAGATION + recommended gates CM-MULTITRACK-ENGINE-IN-CONSTITUTION / CM-MULTITRACK-AUTOINSTALL-WIRED / CM-MULTITRACK-BOOTSTRAP-IDEMPOTENT / CM-MULTITRACK-BIND-ON-START / CM-MULTITRACK-FALLBACK-MONITOR / CM-MULTITRACK-HEADLESS-SPAWN / CM-MULTITRACK-RULER-SUPERVISOR + paired §1.1 mutation. Also landed the previously-missing CM-COVENANT-114-167-PROPAGATION + CM-COVENANT-114-176-PROPAGATION standalone gate scripts (closing the §11.4.167/§11.4.176 gate-code gap flagged since their original landing) + CM-MULTITRACK-ENGINE-IN-CONSTITUTION mechanism gate, each with a paired §1.1 mutation test under constitution/scripts/gates/. CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.184 — Mandatory SonarQube static-analysis CLI + local tooling installed and PATH-discoverable (User mandate 2026-07-06): the SonarQube scanner CLI (sonar-scanner) MUST be installed AND durably PATH-discoverable via .bashrc / .zshrc (exactly like the Firebase/GitHub/GitLab CLIs), PLUS the shared constitution/scripts/sonarqube/ tooling consumed BY REFERENCE (§11.4.28/§11.4.177/§11.4.80-style, NEVER copied); proven GREEN by sonar-qube_install_check.sh exit 0 + command -v sonar-scanner (anti-bluff §11.4.6, never assumed); rootless podman §11.4.161, never rootful docker/sudo; DISTINCT from the repealed §11.4.166 Semgrep mandate — mandates TOOLING availability, not a scan on every build. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-184-PROPAGATION + recommended gate CM-SONARQUBE-CLI-INSTALLED + paired §1.1 mutation. Landed in numerical order between §11.4.183 and the already-present §11.4.185 (union resolution — nothing dropped). CLAUDE.md + AGENTS.md + QWEN.md + GEMINI.md mirrors updated in lockstep (§11.4.157). Prior round: Added §11.4.170 — Device-independent host-side rendered-UI visual-proof mandate (User mandate 2026-06-25): every change to ANY user-facing UI surface MUST be proven by

DEVICE-INDEPENDENT host-side RENDERED PIXELS before claimed correct — the real component rendered to a PNG ON THE HOST (no device/emulator/running app; Compose → Roborazzi/Paparazzi, web → Playwright/Storybook, SwiftUI → snapshot-testing, equiv per stack) for EVERY screen×state×{light,dark} theme, dual-validated by (i) golden image-diff AND (ii) an OCR/vision oracle reading rendered text+labels+control bounds (NO overlap / label-over-label / clipping / off-screen / collapsed-or-giant-unbounded widget). VALUE/token-equality / property-assertion UI tests are FORBIDDEN as the PROOF a UI is correct (forensic FACT 2026-06-25: hex/sp/dp value-equality unit tests stayed GREEN while the operator opened a broken giant-button screen) — may supplement, NEVER substitute the rendered-pixel proof; self-validated golden-good/golden-bad analyzer §11.4.107(10); device-offline is NEVER a valid skip (host-render IS the device-independent path); COMPLEMENTS not replaces §11.4.153/.158/.159/.160 live-device recording + §11.4.162 OpenDesign tokens + §11.4.168 exported-doc visual validation. Propagation gate CM-COVENANT-114-170-PROPAGATION + recommended gate CM-HOST-RENDERED-UI-VISUAL-PROOF + paired §1.1 mutation. Classification: universal (§11.4.17). Prior round: §11.4.166 REPEALED (operator decision 2026-06-22): Universal Semgrep static-analysis mandate repealed — Semgrep no longer mandatory; scaffolding, submodules/semgrep, MCP/pre-commit/PATH wiring, docs_chain semgrep_status context, and CM-COVENANT-114-166-PROPAGATION/CM-SEMGREP-WIRED gates removed. Added §11.4.163 mirror — Universal Media Validation & Verification Mandate: every recorded artifact MUST pass MEDIA VALIDATION pipeline before acceptance (OCR/transcription/text parsing vs SPECIFY-phase patterns, self-validated analyzer, structured verdict with pinpoint on FAIL, post-recording+real-time triggers, paired §1.1 mutation). Added §11.4.164 mirror — Universal Constitution Auto-Propagation & Hook System: post_update_hook.sh detects/registers/installs changed skills/MCP/hooks/scripts; consumer invokes after every constitutional pull. Added §11.4.165 mirror — Universal Independent Verification Agent Mandate: every output passes INDEPENDENT verifier (§11.4.70/.20), iterates to zero-finding GO (§11.4.134). All Classification: universal (§11.4.17). Prior round: Added §11.4.162 mirror — OpenDesign UI design system mandate (User mandate 2026-06-21): every project producing user-facing interfaces MUST use OpenDesign as the mandatory UI design-and-refinement system — NOT ad-hoc CSS or one-off design tools; install as a project dependency and use its design tokens/themes for color palette (light+dark), typography, spacing, and component-level tokens; extend upstream per §11.4.74 for missing patterns; every UI component ships light+dark variants with project brand colors from canonical assets; elements MUST NOT overlap or overlay labels; all UI changes covered by standard test types including visual regression. Classification: universal (§11.4.17). Prior round: Added §11.4.161 mirror — Rootless container runtime mandate (User mandate 2026-06-21): every project MUST use Podman in rootless mode (or equivalent rootless container

runtime) for ALL containerized workloads — Docker in rootful mode, sudo, or any escalation to root is FORBIDDEN unless the target platform has no rootless option AND that constraint is documented per §11.4.112; the `vasic-digital/containers` submodule (§11.4.76) MUST be used as the sole container orchestration layer — no ad-hoc docker/podman commands outside `pkg/boot / pkg/compose / pkg/health` ; missing capabilities extend the container Submodule upstream per §11.4.74; integration tests boot infra on-demand via the Submodule. Classification: universal (§11.4.17). Prior round: Added §11.4.160 mirror — Vision-verified recording + HelixQA bridge mandate (User mandate 2026-06-21): every video recording for feature/QA evidence MUST be processed through a vision/OCR pipeline that reads on-screen content and confirms expected results BEFORE acceptance; the recording system MUST provide a bridge feeding captured frames to HelixQA's test infrastructure (or equivalent) for content verification — automated read-the-screen against specified expected patterns; the bridge MUST capture frames at ≤5s intervals, run OCR/vision analysis with self-validated analyzer (§11.4.107(10)), compare against SPECIFY-phase patterns, produce per-frame PASS/FAIL with evidence path, surface failures immediately for re-record per §11.4.159(L). Honest boundary: vision verification confirms on-screen content — does NOT replace §11.4.5 quality analysis nor §11.4.108 runtime-signature; FLAG_SECURE surfaces use the §11.4.117 proxy oracle. Classification: universal (§11.4.17). Prior round: Added §11.4.153 mirror — Crashlytics-recorded-data continuous monitoring + systematic-debug + regression-test-coverage mandate (User mandate 2026-06-13): every project with Firebase Crashlytics enabled/wired MUST continuously monitor ALL four recorded surfaces (fatal crashes, ANRs, performance traces, AND non-fatals), systematic-debug each (reproduce-before-fix §11.4.102/.115), fix/improve, and cover every closure with a permanent §11.4.135 regression guard (RED-on-broken → GREEN-on-fixed) + a closure log citing the console issue id/URL + the validation/verification test paths; regular cadence (the §11.4.47 five-trigger set), no false results, no bluff — a console-resolved Crashlytics issue with no falsifiable regression test is FORBIDDEN (the silent-recurrence vector). STRENGTHENS+COMPLETES §11.4.47 (review/dedup finds it; §11.4.152 fixes it + proves it stays fixed). Project-level reference instantiation = a consuming project's §6.O + §6.AC. Composes §11.4/.1/.5/.6/.9/.34/.40/.43/.47/.69/.90/.102/.107/.108/.115/.118/.123/.130/.135/.138/.150/ §1.1. Propagation gate `CM-COVENANT-114-152-PROPAGATION` + recommended gate `CM-CRASHLYTICS-ISSUE-FULLY-COVERED` . Classification: universal (§11.4.17). Prior round: Added §11.4.151 mirror — Project-prefixed release-tag/version-naming mandate (User mandate 2026-06-12): every release tag AND version name on the main repo AND every owned submodule MUST be prefixed `<PREFIX>-<version>` (e.g. `myproject-1.0.0-dev-0.0.1`); prefix resolution order (1) `HELIX_RELEASE_PREFIX` from `.env` (authoritative, git-ignored §11.4.30, documented in tracked `.env.example`

Field	Value
	§11.4.77) else (2) lowercased snake_case project-root dir name §11.4.29; SAME prefix across main + all owned submodules in one release = greppable across every repo; version codes increment monotonically. Composes §2.1/§11.4.28/.29/.30/.40/.77/.113/.126. Propagation gate CM-COVENANT-114-151-PROPAGATION + recommended gate CM-RELEASE-PREFIX-NAMING . Classification: universal (§11.4.17). Prior round: Added §11.4.148 + §11.4.149 mirrors — workable-item integrity + testing-diary family (User mandate 2026-06-10). §11.4.148 BINDS+STRENGTHENS §11.4.15/.16/.21/.54/.91/.93/.95/.106 into one DB↔docs↔external-tracker integrity contract: (D1) no item without a valid status+type+stable id on ALL three surfaces; (D2) comprehensive structured description (what/manifest/repro/acceptance); (D3) BLOCKED items carry WHY + enumerated unblock CHOICES (tightens §11.4.21; Blocked / BLOCKED = documented alias of canonical operator-blocked); (D4) regular never-missed bidirectional DB↔docs↔tracker sync, §11.4.86-fingerprinted + §11.4.106 docs_chain-bound; (D5) generic idempotent external-tracker push (statuses/types/assignee-from-env/sub-tasks, sink-side proof §11.4.69, machinery project-agnostic §11.4.28). §11.4.149 per-workable-item TESTING DIARY (date/time + tested-by + result + observations + action+why), distinct from item_history/Reopens: append-only test_diary table with PASS-requires-evidence schema constraint + in-depth diary + derived summary VIEW + four-format per-item exports §11.4.86-fingerprinted §11.4.106-bound + external-tracker SUB-TASK `TODO
Is-sues	none
Is-sues summary	—
Fixed	§11.4.108 mirror
Fixed summary	§11.4.107 lands in lockstep with the Constitution.md §11.4.107 addition.
Continuation	—

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- Documentation always-sync composite covenant (§11.4.60, User mandate 2026-05-19)
- Credentials-handling (§11.4.10)
- Host-session safety (§12)
- Continuation document (§12.10)
- Data safety (§9)
- CLI workflow expectations
- Hierarchy of project authority
- When stuck

This is the **base AGENTS.md** for any CLI agent (Codex, Cursor, Aider, Continue, Gemini CLI, future LLMs) working on a project that includes the Helix Constitution submodule. Project-level `AGENTS.md` may extend or tighten rules but **MUST NOT** weaken them.

Last revision: 2026-05-14

How inheritance works

Because not every CLI agent honours the Markdown `@import` syntax, each consuming project's root `AGENTS.md` **MUST** do one of:

1. **Reference + restate critical rules** (safest — works for every agent):

```
# <Project> — AGENTS.md
```

```
> Base agent rules: `constitution/AGENTS.md` — READ IT FIRST.
```

```
> The base file is authoritative for any topic not covered here.
```

```
## Critical base rules restated (for agents that don't follow links)
```

- Never commit secrets.
- All commits go through the project's commit wrapper.
- Anti-bluff covenant binds every test (Constitution §11.4).

```
## Project-specific
```

```
- ...
```

2. **Generate at build time** — keep one source-of-truth in this submodule, concatenate with a project-specific delta file. A `compose-agents.sh` script in this submodule's `Upstreams/` (or `scripts/` if added) would output `constitution/AGENTS.md` followed by `---` followed by `AGENTS.project.md`.

Identity & posture

You are working inside a project that has opted into the Helix Constitution. The Constitution is the source of truth for engineering discipline (test coverage, anti-bluff covenant, data safety, host safety, credentials handling, documentation discipline). Whenever this AGENTS.md mentions a `§X.Y` reference, it points to `constitution/Constitution.md`.

The Constitution applies recursively to every submodule of every project that includes it — you **MUST** treat the Constitution's rules as in-force regardless of how deep into the submodule tree you are working.

Top-level invariants for every agent

1. **Never assume; verify by reading files.** Especially before any destructive action.
2. **Never bluff.** Do not invent file paths, command outputs, test results, or library APIs you have not verified. If you are unsure, say so explicitly and verify.

3. **No guessing language** (§11.4.6). `likely` , `probably` , `maybe` , `might` , `appears` , `seems` are forbidden when reporting causes. Use captured evidence or mark explicitly `UNCONFIRMED:` .
4. **Test coverage for every change** (§1) — pre-build gate, post-build gate, runtime test, AND a meta-test mutation that proves the gate catches regressions.
5. **Commit through the project's official wrapper.** Direct `git add` / `git commit` / `git push` on the main repo is forbidden in normal workflow.
6. **Never skip hooks** (`--no-verify` , `--no-gpg-sign`).
7. **Never force-push.** Force-push requires explicit per-session human authorization AND a green §9.1.5 post-op gate.
8. **Hardlinked backup before any destructive op** (§9). Zero excuse.
9. **Host safety** (§12) — abort if pre-flight check fails; wrap heavy work in bounded execution scopes; never exceed 60% of host RAM.
10. **Credentials NEVER tracked** (§11.4.10) — `.env` patterns git-ignored; runtime-load only; per-service file separation. **Pre-store leak audit** (§11.4.10.A, 2026-05-17) — before storing operator-provided credentials in any gitignored config, grep the entire tracked tree AND git history for the literal value(s); surface any findings to operator before storing.
11. **CONTINUATION.md kept in sync** (§12.10) — every non-trivial state change updates the continuation document in the same commit.

Critical base rules restated (for agents that don't honour @imports)

Anti-bluff covenant — END-USER QUALITY GUARANTEE (§11.4)

Forensic anchor — verbatim user mandate (2026-04-28):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

The bar for shipping is **not** "tests pass" but **"users can use the feature."** Every PASS **MUST** carry positive evidence captured during execution that the feature works for the end user. Metadata-only PASS, configuration-only PASS, absence-of-error PASS, and grep-based PASS without runtime evidence are critical defects.

Mutation-paired gates (§1.1)

Every new gate **MUST** be paired with a mutation entry in the project's meta-test harness that mutates the asserted condition and asserts the gate now FAILs. A gate without a paired mutation is a bluff gate and is forbidden.

Recorded-evidence requirement (§11.4.2)

Every PASS for a user-visible feature **MUST** be cross-checked against a captured recording + action timeline. A PASS that lacks at least one matched timeline event in the analyzer findings is a §11.4 PASS-bluff.

Test-interrupt-on-discovery (§11.4.4)

The moment any defect is re-discovered, re-produced, or newly identified during a test cycle, the cycle **MUST** stop. Then: systematic debugging → fix at root cause → four-layer test coverage (pre-build / post-build / runtime / meta-test paired mutation) → full rebuild → re-deploy on every target → full retest from beginning.

No-guessing mandate (§11.4.6)

Forensic anchor — verbatim user mandate (2026-05-08):

"'LIKELY' is guessing, we MUST NOT have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We MUST always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!"

Forbidden vocabulary when reporting causes: likely , probably , maybe , might , possibly , presumably , seems , appears to , guess , seemingly , apparently , perhaps , supposedly , conjectured , and synonyms.

Item-type tracking (§11.4.16)

Every active item in the project Issues file MUST carry a `**Type:**` line within eight non-blank lines of its heading. Three-value CLOSED vocabulary: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream — refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The Issues_Summary file carries the Type column for every active item. All three Issues / Issues_Summary / Fixed file types kept in sync (Markdown + HTML + PDF). Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION enforce the mandate. No escape hatch.

Universal-vs-project classification (§11.4.17, User mandate 2026-05-14)

Before adding ANY new rule / mandatory constraint / covenant clause / gate / "MUST"-bearing statement, classify it: **universal** (any project → constitution submodule) or **project-specific** (specific hardware / vendor / package / region → project / submodule layer). Commit message MUST carry Classification: line + one-sentence rationale. When uncertain, default to project-specific. Gate CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION audits new rule commits. Paired mutation enforces the gate is not a bluff.

Subagent-driven-by-default (§11.4.20, User mandate 2026-05-14)

When the runtime supports subagent delegation (Agent tool / task runner / sub-session), the primary agent MUST default to subagent delegation for multi-step scope (≥ 3 phases), parallelisable independent subtasks, long-running diagnostic loops, OR specialised domain workflows. Foreground-only is reserved for single-file edits, mid-execution operator clarification, critical-state sequencing (commits / pushes / tags), or tasks under ~ 30 s. Discipline: tight scope (4-6 tasks), checkpoint commits per task, anti-stall protection in prompts, anti-bluff verification of subagent claims via repo state. Parallel subagents MUST partition non-overlapping files; parent `commit_all.sh --auto-cascade` bundles via `git add -A`. Gate `CM-SUBAGENT-DELEGATION-AUDIT` (per consuming project) flags foreground multi-step work as a violation. Paired mutation enforces the gate is not a bluff.

Script documentation mandate (§11.4.18, User mandate 2026-05-14)

Every Bash / shell / POSIX-sh script under `scripts/` / `bin/` / `tests/` / `library` dirs / CI hooks (depth-N recursive) MUST carry (1) an in-source documentation block at the top (Purpose / Usage / Inputs / Outputs / Side-effects / Dependencies / Cross-references) AND (2) an external user guide under `docs/scripts/<script-name>.md` (Overview / Prerequisites / Usage examples / Edge cases / Internal behaviour / Related scripts / Last verified date). When a script is modified, BOTH the in-source block AND the external user guide MUST be updated in the SAME commit. **No documentation ever out of sync with its codebase.** Gate `CM-SCRIPT-DOCS-SYNC` enforces. Paired mutation strips the invariant.

Fixed-document column-alignment (§11.4.19, User mandate 2026-05-14)

The closed-archive tracker (`Fixed.md`) MUST mirror the open-work tracker (`Issues.md` + `Issues_Summary.md`) along the same lifecycle axes — at minimum **Status** and **Type**. Concretely: (1) every `###` / `####` heading in `Fixed.md` carries `**Status:**` (from closed-set `{Fixed ( $\rightarrow$  Fixed.md) | Fixed – pending device verification | Fixed – RECLASSIFIED}`) and `**Type:**` (`{Bug | Feature | Task}`) within 8 non-blank lines of the heading; (2) `Fixed_Summary.md` companion exists with column structure `# | Level |`

Status | Type | One-line description matching Issues_Summary.md exactly; (3) all three formats (.md , .html , .pdf) for BOTH Fixed and Fixed_Summary stay in sync via the same single-shot wrapper that handles Issues + Issues_Summary; (4) closure is atomic — resolved items migrate from Issues.md to Fixed.md in the same commit. Pre-build gate CM-FIXED-COLUMN-ALIGNMENT (5+ invariants: file-exists, header-has-Status+Type, mtime ≥ Fixed.md, generator script present, sync wrapper invokes it, HTML+PDF exports present). Paired mutation strips Status column from Fixed_Summary header → gate FAILs. Classification: universal (per §11.4.17).

Operator-blocked status + self-resolution exhaustion (§11.4.21, User mandate 2026-05-14)

Operator-blocked is the §11.4.15 Status closed-set's 7th value alongside {Queued | In progress | Ready for testing | In testing | Reopened | Fixed (→ Fixed.md)} . **Last-resort classification** — agent MUST first exhaust applicable self-resolution paths: (a) CLI / ADB / SSH / API access already available, (b) subagent delegation per §11.4.20, (c) existing repo tooling, (d) captured fallback (synthetic event / asset substitution / §11.4.3 SKIP), (e) external research per §11.4.8. Every operator-blocked item MUST carry ****Operator-Block-Details:**** within 8 non-blank lines of its heading stating WHAT (action) / WHY (each exhausted alternative) / UNBLOCK CONDITION (observable signal) / WHO (contact or doc pointer). Issues_Summary.md lists it as a sortable Status value. Items re-evaluated every Nth tag cycle (≥3rd recommended). Fake operator-blocked without exhaustion audit = §11.4 covenant violation at planning layer (PASS-bluff equivalent). Gates CM-ITEM-OPERATOR-BLOCKED-DETAILS + CM-OPERATOR-BLOCKED-SELF-RESOLUTION-AUDIT (every NEW Operator-blocked commit carries an "Attempted: a — ...; b — ...; c — ..." trail). Classification: universal (per §11.4.17). No escape hatch.

Document-sync commit discipline (§11.4.22, User mandate 2026-05-14)

Every project tracking work items through an Issues / Fixed lifecycle MUST provide a **lightweight commit path** distinct from the full-repo commit wrapper. Stages, commits, pushes ONLY the status-tracking doc set: Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + their HTML + PDF exports + auto-

generated audit artifact. Wrapper MUST: (a) auto-invoke the project's export-regeneration pipeline first so Markdown + HTML + PDF stay in sync; (b) stage ONLY the explicit doc-set list — NEVER `git add -A`; (c) use a separate flock disjoint from the full-tree wrapper; (d) push to every parent-repo remote; (e) exit 3 on nothing-to-commit (informational). MUST be invocable standalone OR via a delegation flag on the full-tree wrapper (e.g. `commit_all.sh --docs-only`). Inherits §9 preflight. Gate `CM-COMMIT-DOCS-EXISTS` + paired mutation. Composes with §11.4.12 / §11.4.15 / §11.4.18 / §12.10. Classification: universal (per §11.4.17). No escape hatch — doc-status drift is a §11.4 PASS-bluff at the documentation layer.

Build-resource stats tracking (§11.4.24, User mandate 2026-05-14)

Every project under this Constitution with a build exceeding 1 minute wall-clock MUST run a host-side resource sampler for every build — samples `/proc/meminfo` + `/proc/loadavg` + `/proc/stat` + `/proc/diskstats` at a fixed interval (recommended 5 s); writes TSV. On stop, computes **min / max / mean / p95** per metric; appends one TSV row per build to a registry; regenerates a Markdown report (`Stats.md`) whose top surfaces **ever-values** (min / max / mean across all tracked builds) and whose body lists per-build entries most-recent-first with `SUCCESS` / `FAIL` / `UNKNOWN` + reason. `Stats.md` exported to `Stats.html` + `Stats.pdf` through the project's normal export pipeline per §11.4.12. Triple committed via the §11.4.22 lightweight doc-sync wrapper. Sampler MUST stay under 50 MB RSS / 5% CPU. Stop hook called from both success AND failure paths of the build wrapper. Gate `CM-BUILD-RESOURCE-STATS-TRACKER`

- paired mutation. Classification: mixed (per §11.4.17). Composes with §11.4.12 / §11.4.18 / §11.4.22 / §12.6 / §12.7 / §12.9. No escape hatch — build-resource debugging without time-series data is the bluff this anchor forbids.

Full-Automation-Coverage (§11.4.25, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered!"

No feature / functionality / flow / use case / edge case / service / application on any supported platform may be considered deliverable until automation tests prove six invariants: (1) anti-bluff posture with captured runtime evidence (§7.1 + §11.4); (2) proof of working capability end-to-end on target topology (§11.4.3, no mocks); (3) implementation matches documented promise; (4) no open issues/bugs surfaced (cross-checked vs §11.4.15 / §11.4.16); (5) full documentation (user manual + §11.4.18 for scripts) kept in sync via §11.4.12; (6) four-layer test floor per §1 (pre-build + post-build + runtime + paired mutation). Consuming projects publish a coverage ledger (feature × platform × invariant × status) regenerated at release-gate sweep. Classification: universal (§11.4.17). Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. See Constitution §11.4.25 for the full mandate.

Constitution-Submodule Update Workflow (§11.4.26, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every time we add something into our root (constitution Submodule) Constitution, CLAUDE.MD and AGENTS.MD we MUST FIRST fetch and pull all new changes / work from constitution Submodule first! All changes we apply MUST BE committed and pushed to all constitution Submodule upstreams! In case of conflict, IT MUST BE carefully resolved! Nothing can be broken, made faulty, corrupted or unusable! After merging full validation and verification MUST BE done!"

Before any modification to `constitution/{Constitution, CLAUDE, AGENTS}.md` , execute in order: (1) fetch + pull (--ff-only or --rebase; never strategy=ours / unrelated-histories without authorization) so the submodule is at upstream tip

before editing; (2) apply the change with §11.4.17 classification + verbatim mandate quote; (3) validate (`meta_test_inheritance.sh` + no merge-conflict markers + cross-file consistency); (4) commit (governance files only, no `git add -A`)

- push to EVERY configured upstream remote (§2.1 violation if not); (5) careful conflict resolution preserving union of governance content — force-push forbidden (§9.2); (6) post-merge validation: `git submodule update --remote --init` + re-run cascade verifier (CONST-047) confirming the new clause reaches every owned submodule; (7) bump consuming project's `.gitmodules` pointer to new HEAD in the SAME commit as cascade work — out-of-sync pointer = §11.4.26 violation. Classification: universal (§11.4.17). Severity-equivalent to force-push without §9.2 authorization. No escape hatch. See Constitution §11.4.26 for the full pipeline (operational scope, cross-cutting reach).

Submodule-dependency-manifest (§11.4.31, User mandate 2026-05-15)

Every owned-by-us submodule MUST ship `helix-deps.yaml` listing its own-org dependencies: `{name, ssh_url, ref, why, layout: flat|grouped}`. Tooling `incorporate-submodule <ssh-url>` adds the submodule at the parent project's canonical path (CONST-051(C)) + reads its `helix-deps.yaml` + recurses + aborts on conflicting refs + emits `<root>/helix-manifest.yaml` audit record. Anti-bluff: each manifest paired with a Challenge that bootstraps from scratch, asserts layout matches manifest, runs submodule tests, captures wire evidence. Without the proof, the manifest is a §11.4.31 violation. This rule is the operational complement of §11.4.28 / CONST-051(C) — manifests are the bridge that lets consumers reconstruct the dependency graph at the parent root. Classification: universal (§11.4.17). See Constitution §11.4.31 for the full mandate.

Post-constitution-pull validation (§11.4.32, User mandate 2026-05-15)

Whenever a project's constitution submodule is fetched + pulled with any content change, run `scripts/verify-all-constitution-rules.sh` BEFORE the new HEAD is treated as canonical for other work. The sweep re-runs the governance-cascade verifier + every implementable rule gate (CONST-053 `.gitignore` audit,

CONST-051(C) nested-own-org audit, CONST-052 case audit, CONST-050(A) mock-from-production audit, CONST-035 anti-bluff smoke). Failures populate Issues per §11.4.15 (Status: Reopened, Type: Bug); closure requires positive-evidence per §11.4. Pull-time invocation auto-triggered by `git submodule update --remote constitution`. Anti-bluff: sweep's own meta-test (paired mutation §1.1) plants a violation per gate, asserts sweep FAILs. This is the **enforcement engine** for every other rule — without it, new rules are decorative anchors rather than enforced gates. Classification: universal (§11.4.17). See Constitution §11.4.32 for the full mandate.

.gitignore + no-versioned-build-artifacts (§11.4.30, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"every project module, every Submodule, every service and application MUST HAVE proper .gitignore file! We MUST NOT git version build artifacts, cache files, tmp files, main .env file(s) or any files containing sensitive data, API keys or token! ... If any violation is detected it MUST be fixed before commit is executed!"

Every project module / submodule / service / application MUST ship a proper `.gitignore`. Forbidden-from-version-control: build artefacts (`bin/` , `build/` , `dist/` , `target/` , `*.exe` , `*.so` , `*.class` , `*.pyc`), cache files (`__pycache__/` , `node_modules/` , `.gradle/` , `.terraform/`), temp files (`*.tmp` , `*.swp` , `.DS_Store`), sensitive data (`.env` , `.env.*` except `.env.example` placeholder, `*.pem` , `*.key` , `id_rsa*` , `.netrc` , `secrets/` , `api_keys.sh`), generated logs (`*.log` , `coverage.out` , `htmlcov/`), OS/IDE personal state (`.idea/` , `.history/`).

Anti-bluff invariant: ignore-line alone is not enough — no file matching forbidden patterns may be tracked. Pre-commit attention: every author inspects `git diff --staged + git status` BEFORE commit; violations abort the commit (un-stage, add to ignore, scrub if already-tracked). Gate `CM-GITIGNORE-`

`PRECOMMIT-AUDIT` + paired mutation. Recreatable-content test: if a documented mechanism regenerates the file from sources, it's a build derivative and MUST be ignored.

Secret-leak intersection (§11.4.10 / CONST-042 / §12.1): a `.env` leak is BOTH a §11.4.30 and a §11.4.10 violation; rotation + post-mortem required.

Classification: universal (§11.4.17). Severity-equivalent to §11.4 PASS-bluff at the repository-hygiene layer. Composes with §1, §2, §9.1, §11.4.10, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, §11.4.29, CONST-047. See Constitution §11.4.30 for the full mandate.

Lowercase-snake_case-naming (§11.4.29, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules MSUT HAVE lowercase names with space separator between the words of '_' character (snake-case)! All existing Submodules and directories which are not following this rule MUST BE renamed! ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!"

Every directory / submodule / file MUST use lowercase snake_case. Non-compliant names MUST be renamed; every reference (configs, docs, source imports, governance files) updated atomically (reference drift = §11.4.29 violation of equal severity). Exceptions (common-sense, technology-preserving): language-mandated case for Java/Kotlin/Android/Apple/C#/Swift inside the language root; vendor third-party submodules; build artefacts.

`Upstreams/` → `upstreams/` transition: constitution submodule's `install_upstreams.sh` MUST support BOTH directory names during migration; lowercase wins when both present.

Project-Toolkit Upstreamable machinery fetched+pulled before any rename batch + itself complies. Lacking BOTH-dir support is a release blocker.

Test coverage of renames: regression test for reference resolution + full CONST-050(B) test-type matrix + anti-bluff wire-evidence.

Phased execution: brainstorming → phase plan → fine-grained tasks/subtasks → every change covered. §11.4.20 subagent delegation for cross-cutting rename sweeps.

Classification: universal (§11.4.17). No escape hatch beyond common-sense exceptions. Severity-equivalent to §11.4 PASS-bluff at the reference-integrity layer. Composes with §1, §11.4.12, §11.4.17, §11.4.18, §11.4.20, §11.4.25, §11.4.26, §11.4.27, §11.4.28, CONST-047. See Constitution §11.4.29 for the full mandate.

Submodules-as-equal-codebase + decoupling + dependency-layout (§11.4.28, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"All existing Submodules in the project that we are controlling and belong to some our organizations (vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — we can ALWAYS check dynamically using GitHub and GitLab CLIs) are equal parts of the project's codebase! We MUST work on that code as much as we do with main project's codebase! ... We MUST NEVER modify Submodules to bring into them any project specific context ... All Submodule dependencies that are used by Submodule MUST BE accessed from the root of the project! We MUST NOT have nested Submodule dependencies."

Three invariants. **(A) Equal-codebase:** every owned-by-us submodule (orgs: vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory — discoverable via gh/ glab) is an equal part of the consuming project's codebase. Same engineering attention: analysis, extension, tests, gap-fill, bug-fix, documentation. Coverage ledgers list each submodule as in-scope. **(B) Decoupling:** NEVER inject project-specific context INTO submodules; they remain project-not-aware, reusable, modular, testable. When a submodule needs parent info, use configuration injection. **(C) Dependency-layout:** every dependency consumed by an owned

submodule lives at `<root>/<name>/` or `<root>/submodules/<name>/` of the parent project. Nested own-org submodule chains FORBIDDEN — add the dependency at the parent root; the submodule reaches it via documented import/SDK/runtime resolver. Third-party submodules exempt.

Gates: `CM-OWNED-SUBMODULE-EQUAL-ENGINEERING` , `CM-OWNED-SUBMODULE-DECOUPLING` , `CM-OWNED-SUBMODULE-LAYOUT` . Paired mutations for each.

Classification: universal (§11.4.17). No escape hatch. Severity-equivalent to a §11.4 PASS-bluff at the codebase- completeness layer. Composes with §1, §3, §11.4.17, §11.4.20, §11.4.25, §11.4.26, §11.4.27, CONST-047. See Constitution §11.4.28 for the full mandate.

No-fakes-beyond-unit-tests + 100%-test-type-coverage (§11.4.27, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist ONLY in Unit tests! All other test types MUST interact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project MUST BE 100% covered with every supported test type."

Two invariants. **(A)** Mocks/stubs/fakes/placeholders/TODOs/ FIXMEs/"for now" patterns PERMITTED only in unit-test sources; non-unit tests (integration, E2E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges, HelixQA) MUST exercise the real, fully implemented system. Production code MUST NOT import mock paths. Gate

`CM-NO-FAKES-BEYOND-UNIT-TESTS` + paired mutation. **(B)** Codebase MUST be covered by every supported test type the domain warrants: unit, integration, E2E, full-automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges (vasic-digital/Challenges submodule fully incorporated), HelixQA (HelixDevelopment/HelixQA submodule fully incorporated, with full autonomous QA sessions executing every registered test bank).

Required dependency submodules (recursive per CONST-047): Challenges
(`git@github.com:vasic-digital/Challenges.git`) + HelixQA
(`git@github.com:HelixDevelopment/HelixQA.git`) + any other functionality
submodules under vasic-digital/HelixDevelopment orgs the project depends on.
Pointers bumped to upstream HEAD in same commit as cascade work (§11.4.26
step 7).

Classification: universal (§11.4.17). Severity-equivalent to a §11.4 PASS-bluff at the
release-gate layer. No escape hatch. Composes with §1, §7.1, §11.4.1–§11.4.26
(esp. §11.4.25 — strict per-type-of-test expansion). See Constitution §11.4.27 for
full mandate.

Type-aware closure-status vocabulary (§11.4.33, User mandate 2026-05-15)

Every project that tracks work items by Type per §11.4.16 MUST close them with
the Type-appropriate closure-status word: `Bug` → `Fixed` (→ `Fixed.md`) ,
`Feature` → `Implemented` (→ `Fixed.md`) , `Task` →
`Completed` (→ `Fixed.md`) . The (→ `Fixed.md`) suffix is preserved across all
three so existing migration tooling (atomic `Issues.md` → `Fixed.md` move per
§11.4.19) keeps working. Generators treat the three terminal values as semantically
equivalent (all closed, positive evidence captured) but preserve the literal in emitted
docs. Closing a `Feature` with `Fixed` (→ `Fixed.md`) or a `Task` with
`Implemented` (→ `Fixed.md`) is a §11.4.33 violation. Pre-build gate `CM-CLOSURE-`
`VOCAB-TYPE-AWARE` . Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.23.
Classification: universal (per §11.4.17). No escape hatch.

Reopened-source attribution (§11.4.34, User mandate 2026-05-15)

Every `Issues.md` heading whose `**Status:**` is `Reopened` MUST carry a
`**Reopened-Details:**` line within 8 non-blank lines of the heading, capturing
four sub-facts: **By:** `AI` or `User` ; **On:** ISO date; **Reason:** one of { `test-failed`
`manual-testing-detected` | `captured-evidence-contradicts` | `end-user-`
`report` | `cycle-re-discovered` | `design-reconsidered` } or explicit free text;
Evidence: path or short description of the captured artefact. Reopens without
evidence are §11.4.6 / §11.4.7 violations: the reopen IS a demotion-from-Fixed
change. `Issues_Summary.md` Status column MUST distinguish Reopened sub-

states by source (e.g., Reopened (AI: test-failed) vs Reopened (User: manual-testing)). Pre-build gate CM-ITEM-REOPENED-DETAILS mirrors §11.4.21 walk pattern. Composes with §11.4.6 / §11.4.7 / §11.4.15 / §11.4.21. Classification: universal (per §11.4.17). No escape hatch.

Canonical-root inheritance clarity (§11.4.35, User mandate 2026-05-15)

The constitution submodule's three files (`constitution/Constitution.md` , `constitution/CLAUDE.md` , `constitution/AGENTS.md`) ARE the canonical root — also called the parent files. Universal rules per §11.4.17 live here.

The consuming project's repository-root files (`<project-root>/CLAUDE.md` , `<project-root>/AGENTS.md` , optionally `<project-root>/Constitution.md` or equivalent) are consumer extensions. They open with the inheritance pointer (either `@constitution/CLAUDE.md` import or `## INHERITED FROM constitution/CLAUDE.md` heading). Project-specific rules per §11.4.17 live here.

When in doubt: universal rule → constitution submodule; project- specific rule → consumer's file. Default consumer-side when uncertain. "Parent CLAUDE.md" / "root Constitution" → constitution submodule file at `constitution/<filename>` . "Project CLAUDE.md" / "this project's AGENTS.md" → consumer file at `<project-root>/<filename>` . Moving a rule between layers MUST be a visible commit — `git mv` + explicit "Lifted from to constitution per §11.4.35" / "Demoted from constitution to per §11.4.35" line in the message. Pre-build gate CM-CANONICAL-ROOT-CLARITY . Composes with §11.4.17. Classification: universal (per §11.4.17). No escape hatch.

Mandatory install_upstreams on clone/add (§11.4.36, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in .bashrc or .zshrc) using install_upstreams command executed from the root of the cloned (added) repository - only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository's upstreams!"

Every clone / add of a Git repository under any consuming project MUST be followed by `install_upstreams` invocation from the repository's root IF its tree contains `upstreams/` (or legacy `Upstreams/` per §11.4.29 transition) populated with `*.sh` recipe files. The utility (installed on operator's `PATH` via `.bashrc` / `.zshrc`, implementation lives in this constitution submodule) reads recipe files + configures every declared upstream as a named git remote + fans out `origin` push URLs.

Skipping the invocation when `upstreams/` is present silently breaks §2.1 (multi-upstream push is the norm). Gate `CM-INSTALL-UPSTREAMS-ON-CLONE` + paired mutation. Automation: `incorporate-submodule` (§11.4.31) auto-invokes; manual invocation also supported. Pre-commit check: `git remote -v | grep -c push` reports expected upstream count.

Classification: universal (§11.4.17). Composes with §2, §2.1, §3, §9.2, §11.4.17, §11.4.20, §11.4.28, §11.4.29, §11.4.30, §11.4.31. See Constitution §11.4.36 for the full mandate.

Fetch-before-edit (§11.4.37, User mandate 2026-05-15)

Forensic anchor — verbatim user mandate (2026-05-15):

"Make sure that `feedback_fetch_before_edit` memory rule is part of our constitution Submodule - the root Consitution, AGENTS.MD and CLAUDE.MD. Validate and verify that Proejct-Toolkit and all Submodules do inherit all of them!"

FIRST git-touching action of any session: `git fetch --all --prune && git log --oneline HEAD..@{u} && git submodule foreach --recursive 'git fetch --all --prune --quiet' .` If `HEAD..@{u}` is non-empty, integrate (ff-merge / rebase / surface per §11.4.4) BEFORE any local edit, scanner, or test. Skipping on "nothing could have changed" is a §11.4.6 (no-guessing) violation — remote state is not knowable without a fetch.

The originating 2026-05-15 incident: agent ran `git remote remove gitee` as a fresh task, but the work had been committed upstream by a parallel agent 25 minutes earlier — local config edit was a no-op echo of already-merged history. A 30-second fetch would have caught it.

Scope: consuming project root + every owned submodule recursively (§11.4.28) + the constitution submodule itself (§11.4.26 step 1) + any dependency cloned via `incorporate-submodule` (§11.4.31) or `git submodule add` (§11.4.36). Gate `CM-FETCH-BEFORE-EDIT-AUDIT` + paired mutation per §1.1. Classification: universal (per §11.4.17). No escape hatch.

Installable-asset evidence (§11.4.38, User mandate 2026-05-17)

For any user-distributable build artifact (package, bundle, installer, or container image), tests/challenges MUST open the artifact and verify each user-visible asset is **present** and **non-degenerate**. A PASS without artifact-opening verification is a §11.4 PASS-bluff. The failure mode: source file exists → build packages it → source-layer checks pass → artifact produced with asset stripped or misconfigured, and no gate opens the artifact to verify.

Required per consuming project: one challenge script per artifact type that opens the produced artifact and verifies every declared user-visible asset. The challenge MUST run as part of the standard QA gate. Classification: universal (per §11.4.17). No escape hatch. See Constitution §11.4.38 for the full mandate.

Full-suite retest before release tag (§11.4.40, User mandate 2026-05-17)

Release tag MUST NOT be created until a COMPLETE retest with ALL existing tests runs on a clean baseline AFTER every workable item in the batch is done, fixed, polished, individually verified. Spot-check retests FORBIDDEN — miss interaction defects. Comprises 7 steps: (1) pre-build full sweep, (2) post-build full

sweep, (3) on-device 4-phase cycle on EVERY owned device, (4) meta-test full mutation sweep, (5) Challenge bank full sweep, (6) Issues.md/Fixed.md state audit, (7) CONTINUATION.md sync check. Typically 12–48h elapsed, NOT optional, NOT abbreviated. Composes with §11.4.4 (per-fix retest still required at fix granularity) + §11.4.7 (full-suite is authoritative baseline for closures) + §11.4.39 (per-feature on-device validation runs as step 3). Classification: universal (per §11.4.17). No escape hatch. See Constitution §11.4.40 for the full mandate.

Pre-force-push merge-first (§11.4.41, User mandate 2026-05-17)

Forensic anchor — verbatim user mandate (2026-05-17):

"make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides!"

Any force-push authorised under CONST-043 MUST be preceded by a mechanical 4-step merge-first pipeline: (1) fetch every remote (`git fetch --all --prune --tags`); (2) integrate every divergent commit locally (rebase / merge / operator-confirmed cherry-pick per appropriate strategy); (3) audit the integrated tree (no conflict markers anywhere in governance + source + test files; no file silently dropped; previously-passing tests still pass; captured-evidence artefacts still validate); (4) only THEN execute `git push --force-with-lease` .

Two-gate composition with CONST-043. §11.4.41 does NOT relax CONST-043 — it adds a SECOND mechanical gate. CONST-043 alone authorises a push that loses remote work; §11.4.41 alone risks pushing without operator awareness. Both required.

Three failure modes prevented: remote-side content loss (parallel sessions overwritten), stale-state acts (`--force-with-lease` reads stale local refs without prior fetch), conflict-driven corruption (markers committed verbatim — observed 2026-05-17 in helix_qa + containers governance files). Verification artefact: `docs/changeLogs/<tag>.md` "Force-push merge-first audit" section captures fetch

output, divergence log, integration strategy, conflict-marker scan, test delta, push output with lease SHA, + CONST-043 authorisation quote. Gate `CM-FORCE-PUSH-MERGE-FIRST`

- paired mutation. Classification: universal (§11.4.17). No escape hatch. Composes with §9.2, §11.4.4, §11.4.6, §11.4.26, §11.4.32, §11.4.37, §11.4.40, CONST-043, CONST-047. See Constitution §11.4.41 for the full mandate.

Iteration-discipline mandate (§11.4.42, User mandate 2026-05-18)

Work proceeds in priority-ordered cycles. Five mandatory steps per cycle: (1) select TOP + MIDDLE critical only (defer LOW until critical batch closed), (2) batch implementation per §11.4.4 + §11.4.9, (3) smoke gate (<30 min), (4) ONLY if smoke GREEN and no new user/operator report → §11.4.40 full retest (12–48 h), (5) release-ready OR loop back. Composes with §11.4.4 / §11.4.7 / §11.4.9 / §11.4.34 / §11.4.40 — §11.4.42 is the meta-loop conductor binding them. Every PASS in smoke and full retest MUST carry positive captured evidence (§11.4.2 + §11.4.5). Tests AND Challenges bound equally. No escape hatch. Classification: universal (per §11.4.17). See Constitution §11.4.42 for the full mandate.

TDD-Fix-Discipline (§11.4.43, User mandate 2026-05-18)

Every fix follows 5-step workflow: **RED** (failing test FIRST, real product defect, captured evidence) → **LIVE-ADB-PROBE** (try on running device when feasible; INFEASIBLE for kernel / framework / HAL / vendor / init.rc / sepolicy / ro.* / Android.bp — must rebuild; cite `LIVE_PROBE_INFEASIBLE: <reason>` in commit) → **GREEN** (source patch, batched per §11.4.9, four-layer coverage per §11.4.4) → **VERIFY** (re-run RED test, PASSes under SAME conditions per §11.4.7, 10-iteration reliability per §11.4.42) → **DOCUMENT** (Issues.md → Fixed.md with type-aware closure §11.4.33, CLAUDE.md Applied Fixes row, changelog, guides, HelixQA bank, CONTINUATION.md §12.10 — all in SAME commit). Test-after-fix is a §11.4 PASS-bluff: test agrees with fix but does not catch the bug. No escape hatch. Classification: universal (§11.4.17). See Constitution §11.4.43 for the full mandate.

Document revision header (§11.4.44, User mandate 2026-05-18)

Every IN-scope tracked Markdown document (Issues.md / Issues_Summary.md / Fixed.md / Fixed_Summary.md / CONTINUATION.md / docs/guides/** / docs/research/** / docs/scripts/** / docs/changelogs/** / docs/superpowers/plans/** / docs/hardware/** / every other docs/*.md) carries `**Revision:** N` (monotonic positive integer)

- `**Last modified:** YYYY-MM-DDTHH:MM:SSZ` directly below the H1 title. CLAUDE.md / AGENTS.md / README / LICENSE / VERSION / rendered HTML / rendered PDF are OUT of scope. Agents MUST read the `**Revision:** N` literal from the document head when reporting document state, NEVER infer freshness from filename mtime or commit log — both can lie. Agents writing to an IN-scope doc MUST invoke `scripts/doc_revision_bump.sh <file>` as the final step before staging, OR rely on the pre-commit hook. Agents MUST NOT manually edit the revision number — only the bump script is authoritative.

Pre-build gates `CM-DOC-REVISION-HEADER-PRESENT` + `CM-COVENANT-114-44-PROPAGATION` + paired mutations. Composes with §12.10 (CONTINUATION.md header reuse), §11.4.12, §11.4.22, §11.4.23, §11.4.18. No escape hatch. Classification: universal (§11.4.17). See Constitution §11.4.44 for the full mandate.

Integration-status-doc maintenance (§11.4.45, User mandate 2026-05-18)

Every non-trivial domain integration MUST have a `docs/<domain>/<integration>/Status.md` carrying the §11.4.44 revision header + auto-synced HTML+PDF + auto-colored per §11.4.23 + sync wrapper + captured-evidence table (every claim cites the test log / recording / sink-probe path) + closed status vocabulary (PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED) + operator-blocked items at the top + reference from CONTINUATION.md §3 when non-terminal. §11.4.45 is the generic form of §12.10 applied to every integration domain.

Pre-build gates `CM-COVENANT-114-45-PROPAGATION` + `CM-AF-INTEGRATION-STATUS-DOCS` + 4 paired mutations. Composes with §11.4.5 / §11.4.12 / §11.4.13 / §11.4.15 / §11.4.22 / §11.4.23 / §11.4.44 / §12.10. No escape hatch. Classification: universal (§11.4.17). See Constitution §11.4.45 for the full mandate.

Validate-recent-work-before-post-flash-tests (§11.4.46, User mandate 2026-05-18)

Every post-flash run MUST first run a recent-work validation pass (targeted tests for Issues.md In-progress / Ready-for-testing / Reopened + Fixed.md last-7-days + CONTINUATION.md §3 active work). Full `test_all_fixes.sh` runs ONLY after 100% green.

Helper: `scripts/testing/recent_work_validate.sh --device <serial>` writes `/data/local/tmp/.recent_work_validated` ; full suite refuses without it. Each recent fix needs paired §11.4.43 RED-then-GREEN — naked GREEN is a bluff. Composes with §11.4.4 + §11.4.7 + §11.4.40

- §11.4.42 + §11.4.43 + §11.4.44 + §12.10. Pre-build gates `CM-COVENANT-114-46-PROPAGATION` + `CM-AF-RECENT-WORK-VALIDATION-GATE` + `CM-AF-VALIDATION-ARTIFACT-FILE` + paired mutations. Classification: universal (§11.4.17). No escape hatch. See Constitution §11.4.46 for the full mandate.

Firebase data review (§11.4.47, User mandate 2026-05-18)

Before every "bigger working round" (pre-build / pre-flash / pre-tag blocking; daily / post-deployment burn-in non-blocking) the operator/loop MUST execute `scripts/firebase/review_round.sh` . The pass queries Crashlytics (fatals + non-fatals + ANRs) + Analytics + Performance, classifies findings by severity, dedup- maps to existing Issues.md entries via a three-tier algorithm (exact Firebase Issue-ID match → stacktrace-similarity cluster hash → operator merge review), and drafts new Issue entries for unrecognised findings with full Firebase Console URL refs + §11.4.4(a) systematic-debugging output.

Five mandatory elements: 5-trigger cadence, 3-source query (all three sources), Issues.md output with Firebase metadata (Issue IDs + URL + Cluster Hash / KPI / Funnel), 3-tier dedup, and comprehensive root-cause analysis per Issue.

Pre-build gates `CM-COVENANT-114-47-PROPAGATION` + `CM-AF-FIREBASE-REVIEW-CADENCE` + `CM-AF-FIREBASE-ISSUE-XREF` + 3 paired mutations. Composes with §11.4.4 / §11.4.4(a) / §11.4.6 / §11.4.7 / §11.4.10 / §11.4.12 / §11.4.14 / §11.4.15 / §11.4.16 / §11.4.34 / §11.4.42 / §11.4.43 / §11.4.44 / §11.4.45 / §11.4.46. No escape hatch — no `--skip-firebase-review`. Classification: universal (§11.4.17). See Constitution §11.4.47 for the full mandate.

UI-driven video testing (§11.4.48, User mandate 2026-05-18)

Every test that asserts video playback on a secondary display MUST traverse the user-equivalent UI path (launcher icon → app home → content list → tile tap → playback → in-app pause/resume → back button stop). NOT Intent/Broadcast shortcuts (`am start -a VIEW` , `cmd media_session play`). Real `uiautomator dump` element resolution

- `input tap X Y` . Coverage: every video-capable app in `PRODUCT_PACKAGES` + every stream type (progressive HTTP / HLS / DASH / RTMP / file-local / DRM) + every codec (H.264/265/VP9/AV1/MPEG-2/MP4V video; AC-3/E-AC-3/TrueHD/DTS/DTS-HD/MLP/Opus/AAC audio). Secondary display verified via `ffprobe-on-captured-mp4` + VOM activeDecoder state. Arvus codec-state cross-check per §11.4.13 + §CG screenshot.

Per §11.4.4 four-layer: pre-build gate `CM-AF-UI-DRIVEN-VIDEO- COVERAGE` + propagation gate `CM-COVENANT-114-48-PROPAGATION` + on-device test framework at `device/rockchip/rk3588/tests/ui_driven/` (Layer 1 helper + Layer 2 per-app drivers + Layer 3 scenarios) + Layer 4 orchestrator `scripts/testing/run_ui_driven_video_suite.sh` + paired meta-test mutations. No escape hatch — no `--use-intent-shortcut` , `--skip-ui-traverse` , `--legacy-intent-mode` flag. Classification: universal (§11.4.17). See Constitution §11.4.48 for the full mandate.

Dual-approach testing (§11.4.49, User mandate 2026-05-18)

Every feature test exercising a user-visible behaviour MUST ship in TWO variants: a UI-driven variant (uiautomator-based, §11.4.48 surfaces A–E) AND an Intent/Broadcast-driven variant (`am start --es` / `am broadcast` -based). Either alone

is a §11.4 PASS-bluff for the OPPOSITE half of the stack — UI catches app-side bugs; Intent catches framework/system-server bugs (MediaCodec.configure hook, IVideoOutputManager bind timing, broadcast permission gating).

Shared assertion base `tests/lib/dual_approach_test_base.sh` — both variants share `dat_init` / `dat_start_capture` / `dat_assert_codec_state` / `dat_assert_video_frames` / `dat_assert_audio_channels` / `dat_arvus_dashboard_capture` / `dat_cleanup` / `dat_report_finding`. Both variants gather identical evidence into mirror dirs `qa-results/dual_approach/<F>/<run-ts>/{ui,intent}/` so orchestrator diffs results and pinpoints which half of the stack contains a bug. Status mismatch between variants is itself a finding.

Kinopoisk 5.1 EAC3 is the canonical first implementation. Both variants are RED per §11.4.43 until the §CN decoder pipeline fix lands. Pre-build gates: `CM-COVENANT-114-49-PROPAGATION` + `CM-AF-DUAL-APPROACH-COVERAGE` + `CM-AF-KINOPOISK-5-1-DUAL-COVERAGE`. Three paired meta-test mutations. No escape hatch — no `--ui-only` / `--intent-only` / `--skip-dual` flag. Classification: universal (§11.4.17). See Constitution §11.4.49 for the full mandate.

Deterministic consistency (§11.4.50, User mandate 2026-05-18)

Every test that PASSES MUST have been executed N times (default N=3 normal tests, N=10 cycle-validation suites) against the same firmware MD5 + same device + same topology and produced IDENTICAL PASS in every iteration. A divergent N-iter run is auto-FAIL — there is no "first PASSED therefore X was a flake" path. §11.4.7's `intermittent` / `transient` / `flake` / `flaky` vocabulary is enforced MECHANICALLY by this mandate, not merely textually.

Coverage: every public API path (Activity / Service / Broadcast receiver / ContentProvider URI / IPC interface / JNI entry / sysprop write / sysfs node / init.rc trigger) MUST have ≥1 dedicated test that drives it. Untested paths surface in a feature-coverage-matrix audit; the threshold ratchets 70 → 85 → 95 → 99 over phases.

Reliability mechanism: `ab_run_n_times` helper in the project anti-bluff library loops, captures evidence-hash per iter, asserts all N hashes + exit codes identical. NO operator-facing escape converts divergence to PASS.

Pre-build gates: `CM-COVENANT-114-50-PROPAGATION` + `CM-AF-RELIABILITY-CHECK-WIRED` + `CM-AF-FEATURE-COVERAGE-MATRIX` . Three paired meta-test mutations. No escape hatch — no `--allow-flake` , `--first-pass-suffices` , `--skip-n-iter` , `--skip-coverage-audit` flag. Classification: universal (§11.4.17). See Constitution §11.4.50 for the full mandate.

Live-ADB-First maximization (§11.4.51, User mandate 2026-05-18)

Every fix MUST be classified by rebuild-requirement before commit using the project's per-file-class decision matrix. If `LIVE_ADB_TESTABLE` (on-device test scripts, host scripts, *atmosphere-.sh boot scripts, persist.* properties, markdown docs, test fixture assets, HelixQA YAML banks), the operator MUST first apply the fix to the running device via `adb push` / `setprop` / `pm install -r` / `mount -o remount, rw` , run the §11.4.43 RED test live, capture PASS, THEN commit + rebuild + reflash. Commit footer: `LIVE_ADB_VALIDATED: yes` . If

`REQUIRES_REBUILD` (kernel, framework Java/AIDL, native C++ in APEX, sepolicy, `init.rc`, `ro.*` properties, XML overlays, codec XML in APEX, `Android.bp/.mk`), the operator proceeds directly to source-side + rebuild. Commit footer:

`REQUIRES_REBUILD: <reason>` . Mixed batches use partial.

§11.4.51 REFINES §11.4.43 step 2 with mechanical enforcement. Helper:

`classify_fix_rebuild_requirement.sh` walks `git diff --name-only` , looks up each file against the matrix, emits per- file classification + recommended commit-message footer. Unmatched paths classify as `REQUIRES_REBUILD: unmatched-path` (safe default per §11.4.6).

Pre-build gates: `CM-COVENANT-114-51-PROPAGATION` + `CM-AF-CLASSIFY-FIX-HELPER-EXISTS` + `CM-AF-LIVE-ADB-FIRST-COMMIT-MARKER` . Three paired meta-test mutations. No escape hatch — no `--skip-classify` / `--assume-rebuild` / `--no-footer-required` flag.

Classification: universal (§11.4.17). See Constitution §11.4.51 for the full mandate.

Autonomous-Validation (§11.4.52, User mandate 2026-05-18)

Forensic anchor — verbatim user mandate:

"Make sure we have full automation tests which will do all this work in full automation! IMPORTANT: Make sure that all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Every user-facing feature MUST have at least one autonomous validation path: end-to-end via `adb shell` + scripted automation, captured runtime evidence per §11.4.5, PASS/FAIL verdict WITHOUT human presence. Operator-attended tests are SUPPLEMENTARY, never PRIMARY. A feature whose ONLY path is operator-attended is a §11.4.52 violation: the path does not scale to CI, does not run on every commit, does not survive operator unavailability, and produces the exact "tests pass but feature doesn't work for users" failure mode §11.4 forbids.

Acceptable autonomous paths: instrumentation APK (SDK-API exercises + JSON result file), headless intent dispatch + state poll (`am start --es + dumpsys / / proc/<pid>/maps` polling), ADB-driven uiautomator (ONLY if hierarchy has ≥ 1 clickable node — empty hierarchy demands fallback to APK/intent), network-side sink probe (§11.4.13), HelixQA autonomous QA session (§11.4.27).

Coverage ledger (§11.4.25) classifies each feature as `AUTONOMOUS_VERIFIED` / `AUTONOMOUS_DESIGNED` / `OPERATOR_ATTENDED_ONLY` / `NOT_APPLICABLE` . `OPERATOR_ATTENDED_ONLY` blocks release until migrated; cite a tracked work item per §11.4.15 + §11.4.16.

Pre-build gates `CM-COVENANT-114-52-PROPAGATION` + `CM-AF-AUTONOMOUS-PATH-PER-FEATURE` . Paired mutations. No escape hatch — no `--allow-operator-attended-only` / `--skip-autonomous-path` / `--manual-validation-suffices` flag.

Classification: universal (§11.4.17). See Constitution §11.4.52 for the full mandate.

Fixed_Summary parity (§11.4.53, User mandate 2026-05-18)

Forensic anchor — verbatim user mandate (2026-05-18T17:55Z):

"Note: Just like for Issues we have Issues_Summary, for Fixed we MUST HAVE Fixed_Summary - like all other docs: ALWAYS in sync and up to date and ALWAYS exported into the PDF and HTML!"

`docs/Fixed_Summary.md` is the symmetric short-form summary of `docs/Fixed.md`. MUST be regenerated whenever `Fixed.md` changes. HTML + PDF exports MUST travel with the markdown. Stale exports are §11.4.53 violations regardless of whether the underlying `.md` is correct. Same discipline as §11.4.12 Issues_Summary applied to Fixed.md.

Generator: `scripts/testing/generate_fixed_summary.sh` (canonical). Auto-sync wrapper: `scripts/testing/sync_issues_docs.sh` regenerates BOTH Issues_Summary AND Fixed_Summary in one shot, exports HTML + PDF, colorizes per §11.4.23, re-renders PDFs. MUST be invoked after any edit to `Fixed.md`. No `--issues-only` flag exists.

Sort order: closure date DESC (most-recent-Fixed first), §-letter / Fix-# secondary. Documented at top of generated file.

Composes with §11.4.12 (Issues_Summary sibling), §11.4.19 (atomic migration trigger), §11.4.23 (colorizer), §11.4.33 (type-aware closure terminal values), §11.4.44 (revision header), §12.10 (CONTINUATION.md resumption).

Pre-build gates `CM-FIXED-SUMMARY-SYNC` + `CM-COVENANT-114-53-PROPAGATION`. Paired mutations. No escape hatch — no `--skip-fixed-summary-sync`, `--issues-only`, `--summary-not-applicable` flag.

Classification: universal (§11.4.17). See Constitution §11.4.53 for the full mandate.

ATM-NNN ticket identifier (§11.4.54, User mandate 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every workable item in Issues / Issues_Summary / Fixed / Fixed_Summary MUST carry a stable, unique, auto-incremental ATM-NNN ticket identifier."

Every workable item heading in `docs/Issues.md` AND `docs/Fixed.md` MUST carry `[ATM-NNN]` (form `## §X.Y. [ATM-NNN] <title>`, zero-padded ≥ 3 digits). Allocated by `scripts/testing/assign_atm_ticket_ids.sh` and persisted to `scripts/testing/.atm_ticket_state.json` (jsonl). Identifiers are monotonic, never renumbered, never reused, no gaps. `Issues_Summary.md` and `Fixed_Summary.md` MUST carry an `ATM ID` column as the leftmost data column.

Composes with §11.4.15 + §11.4.16 + §11.4.19 + §11.4.33 + §11.4.12 + §11.4.53 + §11.4.55 + §11.4.57.

Pre-build gates `CM-ATM-TICKET-IDS-COMPLETE` + `CM-ATM-TICKET-IDS-UNIQUE` + `CM-ATM-TICKET-IDS-MONOTONIC` + `CM-COVENANT-114-54-PROPAGATION`.

Paired mutations. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.54 for the full mandate.

Reopens-history tracking + per-item Reopens.md (§11.4.55, User mandate 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a Reopens-count column. For any item whose reopens-count > 0 , create `docs/issues/ATM-NNN/Reopens.md` (+ HTML + PDF) with comprehensive reopen history + Fixed cycles."

Every workable item with `reopens_count > 0` MUST have `docs/issues/<ATM-NNN>/Reopens.md` (+ HTML + PDF). Required content: §11.4.44 revision header, item identification, cycle counters, chronological timeline (By AI/User per §11.4.34, On ISO date, Event, Reason, Evidence, Outcome), reasoning chain for each closure, most-recent state-change pointer.

`Issues_Summary.md` and `Fixed_Summary.md` MUST carry a `Reopens` column; count > 0 hyperlinks to per-item `Reopens.md`.

Composes with §11.4.34 (per-event capture), §11.4.54 (ATM-NNN path), §11.4.44 (revision header), §11.4.45 (Status.md analogue), §11.4.53 (Fixed_Summary parity — Reopens column symmetric).

Pre-build gates `CM-REOPENS-DOC-EXISTS-WHEN-COUNT-GT-ZERO` + `CM-REOPENS-DOC-REVISION-HEADER` + `CM-REOPENS-COL-IN-SUMMARIES` + `CM-COVENANT-114-55-PROPAGATION` . Paired mutations. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.55 for the full mandate.

Status_Summary parity + two-audience format (§11.4.56, User mandate 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Every Status.md doc gets a Status_Summary parity companion ALWAYS in sync, exported to HTML + PDF. Two-page format: page 1 = non-developer audience (team-specific), page 2 = software engineer summary."

For every `docs/<domain>/<integration>/Status.md` a companion `Status_Summary.md` MUST exist with: §11.4.44 revision header; **Page 1 — For the** (audience-specific — audio team for `docs/dolby/*` , video team for `docs/video/*` , etc.) plain- language summary, What works, What's broken or pending, Operator actions — NO §-letter jargon, NO captured-evidence file paths; **Page 2 — For software engineers** — §-letter refs, gate names, commit hashes, captured-evidence paths, ATM-NNN cross-refs. HTML + PDF exports travel with the markdown.

Generator:

`scripts/testing/generate_status_summary.sh <Status.md path>` . Invoked from `sync_integration_status.sh` (§11.4.45 wrapper) on every Status.md update.

Composes with §11.4.45 (Status.md — Status_Summary COMPLEMENTS), §11.4.12 + §11.4.53 (parity discipline), §11.4.44 (revision header), §11.4.23 (colorizer), §12.10 (CONTINUATION.md).

Pre-build gates `CM-STATUS-SUMMARY-EXISTS-FOR-EVERY-STATUS` + `CM-STATUS-SUMMARY-TWO-AUDIENCE` + `CM-STATUS-SUMMARY-REVISION-HEADER` + `CM-COVENANT-114-56-PROPAGATION` . Paired mutations. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.56 for the full mandate.

README.md doc-link section + revision metadata (§11.4.57, User mandate 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19):

"Add a doc-link section to README.md — links to Issues + Issues_Summary + Fixed + Fixed_Summary + CONTINUATION + ALL Status docs + their exports. Each link shows revision + last-modified."

Every project's top-level `README.md` MUST contain a section titled `Tracked-Items + Status Documents` (heading MUST contain literal `Tracked-Items`). The section is a markdown table with columns: `Document`, `Last modified`, `Revision`, `Markdown`, `HTML`, `PDF`. Links to: `Issues.md` + `Issues_Summary.md`, `Fixed.md` + `Fixed_Summary.md`, `CONTINUATION.md`, every `docs/**/Status.md` + its `Status_Summary.md` pair. Status docs auto-discovered by the generator via `find docs -name 'Status.md'`.

Generator: `scripts/testing/update_readme_doc_links.sh` extracts each doc's revision + last-modified from its §11.4.44 header, renders the table, replaces the section between explicit markers (`<!-- doc-link-section:begin -->` / `<!-- doc-link-section:end -->`). Invoked from `sync_issues_docs.sh` AND `sync_integration_status.sh`.

Composes with §11.4.12 + §11.4.19 + §11.4.53 (Issues / Fixed / summaries), §11.4.44 (revision-header data source), §11.4.45 + §11.4.56 (Status pairs), §12.10 (CONTINUATION.md).

Pre-build gates `CM-README-DOC-LINK-SECTION-PRESENT` + `CM-README-DOC-LINK-ROWS-COMPLETE` + `CM-README-DOC-LINK-FRESHNESS` + `CM-COVENANT-114-57-PROPAGATION`. Paired mutations strip markers, remove rows, backdate `Last modified`, strip anchor literal. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.57 for the full mandate.

Parallel-development methodology (§11.4.58, User mandate 2026-05-19)

Forensic anchor — verbatim user mandate (2026-05-19T~05:00Z MSK):

"We MUST CREATE adjusted improved version of working methodology where multiple workable items could be done in parallel, all come into the central (main) branch as they are done, then we at particular moment rebuild and reflash the System and in background full testing with validation and verification is done. Each parallel work on one workable (or more) item(s) must use parallel agents as much as possible! ... Writing in depth tests (all supported types of the tests) with Challenges and full HelixQA use is MANDATORY! Every test we execute besides executed with success MUST RESULT in proof that actual functionality being tested REALLY DOES WORK with NO BLUFF of any kind!"

Project work proceeds through the **Parallel Work Unit (PWU) pipeline**. Each PWU is a self-contained workable item with: ATM-NNN identifier per §11.4.54, Issues.md entry per §11.4.15+§11.4.16, file- scope manifest, §11.4.43 RED test, source patch, pre-build gate, post-flash test, paired meta-test mutation per §1.1, HelixQA Challenge bank entry, captured-evidence directory per §11.4.5+§11.4.52.

5-stage pipeline: Stage 1 DEVELOP (parallel PWU agents in isolated worktrees) → Stage 2 MERGE (serial via `commit_all.sh` flock + §11.4.41 4-step merge-first) → Stage 3 REBUILD+FLASH (parallel where hardware allows) → Stage 4 VALIDATE (parallel D3+D4+meta-test+coverage) → Stage 5 SWEEP (parallel HelixQA + Fixed.md migration + README refresh). Stage 1 of round N+1 overlaps with Stages 4-5 of round N — throughput multiplier.

Synchronization: 4-layer lock hierarchy (parent flock / per- submodule git / contention-path advisory locks / per-PWU worktree). Disjoint-scope PWUs fully parallel.

Anti-bluff merge-time enforcement (mandatory, all four): RED-test captured (§11.4.43), paired meta-test mutation (§1.1), 3-iter deterministic-consistency (§11.4.50), captured-evidence per §11.4.5. Metadata-only / configuration-only / absence-of-error PASS REJECTED. HelixQA Challenge MANDATORY for every user-visible PWU.

Pre-build gates `CM-PWU-LOCK-HIERARCHY` + `CM-PWU-ANTI-BLUFF-COVERAGE`

- `CM-PWU-MERGE-QUEUE-DISCIPLINE` + `CM-PWU-PARALLEL-AGENT-LIMIT` + `CM-COVENANT-114-58-PROPAGATION` . Paired mutations cover each gate. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.58 for the full mandate.
Project-specific implementation reference in consumer-side `docs/guides/`
`PARALLEL_DEVELOPMENT_METHODODOLOGY.md` .

README always-sync (§11.4.59, User mandate 2026-05-19)

`README.md` is a §11.4.12-class always-sync document. HTML + PDF exports refresh on every update via `scripts/testing/sync_readme_export.sh` (pandoc + weasyprint); auto-invoked by `sync_issues_docs.sh` so a single doc-sync run refreshes Issues / Issues_Summary / Fixed / Fixed_Summary / CONTINUATION / README (md + html + pdf). README carries a §11.4.44 revision header and a Documentation Map section linking to every Status.md + Status_Summary.md + spec + plan + guide + script-companion doc + changelog + the constitution submodule, plus per-audience navigation (end user / developer / QA / agent). Pre-build gate `CM-README-EXPORT-SYNC` enforces mtime parity (README.html + README.pdf ≥ README.md). Paired meta-test mutation backdates HTML+PDF → gate FAILs. No escape hatch — no `--skip-readme-sync` , `--no-readme-export` , `--readme-stale-OK` flag. Composes with §11.4.12

- §11.4.18 + §11.4.44 + §11.4.45 + §11.4.53 + §11.4.56 + §11.4.57 + §12.10.

Classification: universal (§11.4.17). See Constitution §11.4.59 for the full mandate.

Documentation always-sync composite covenant (§11.4.60, User mandate 2026-05-19)

Eight doc classes — Issues, Issues_Summary, Fixed, Fixed_Summary, CONTINUATION, README, every Status.md (domain-scoped), every Status_Summary.md — MUST be in sync across `.md` + `.html` + `.pdf` . Composite pre-build gate `CM-DOCS-COMPOSITE-SYNC` walks all 8 classes (Status fleet via recursive `docs/** find`), FAILs the build if ANY `.html` or `.pdf` mtime is older than its `.md` . Per-class anchors §11.4.12 / §11.4.44 / §11.4.45 / §11.4.53 /

§11.4.56 / §11.4.57 / §11.4.59 / §12.10 govern individually; the composite catches what they miss when one is silently bypassed. Paired mutation backdates `docs/Issues.html` → gate FAILs. No escape hatch.

Classification: universal (§11.4.17). See Constitution §11.4.60 for the full mandate.

Credentials-handling (§11.4.10)

Forensic anchor — verbatim user mandate (2026-05-12):

"Credentials or any secret and sensitive data MUST NOT leak!"

`.env` patterns git-ignored project-wide. Runtime-load-only — tests load credentials at execution time from operator-populated files under `scripts/testing/secrets/` (or per-project equivalent). Test scripts MUST NEVER print or log credentials.

Host-session safety (§12)

Project procedures MUST NOT use more than **60%** of total system RAM. Heavy work wrapped in bounded execution scopes so the kernel OOM-kills only the scope — `user@<uid>.service` stays alive. Hibernating / suspending / logging out the user is FORBIDDEN (directly OR indirectly).

Continuation document (§12.10)

`docs/CONTINUATION.md` MUST exist at the project root and reflect the live state of the work. Every non-trivial state change updates it in the same commit. Any agent must be able to resume work exactly where the previous session left off by reading this single file.

Data safety (§9)

Hardlinked backup before any destructive operation. Post-op gate MUST pass. Force-push requires explicit per-session human authorization. Every history rewrite gets a `docs/changelogs/` audit entry.

CLI workflow expectations

1. **Read before write.** Use file-reading tools to inspect the current state before editing. Edits in the dark are how bugs sneak in.
2. **Use the project's commit wrapper.** Direct `git add` / `git commit` is forbidden unless explicitly carved out.
3. **Commit messages cite forensic evidence** for any non-trivial fix (logs, captures, /sys readings, dropbox entries, strace). Speculative narratives are §11.4.6 violations.
4. **Update CONTINUATION.md** in the same commit as the work (§12.10).
5. **Run pre-build / post-build / runtime gates** before declaring a change complete. NEVER report "done" without running the relevant gate.
6. **Cite external research sources** for non-trivial fixes (§11.4.8). Either a citation OR the literal "NO external solution found — original work".
7. **Refuse to bluff.** If a step requires information you don't have, say so and STOP. Don't guess.

Hierarchy of project authority

When a project files declares conflicting rules with this base AGENTS.md, resolve in this order:

1. **Project's** `docs/guides/<PROJECT>_CONSTITUTION.md` — most specific, project-tailored.
2. `constitution/Constitution.md` (this submodule) — universal.
3. **Project's root** `CLAUDE.md` / `AGENTS.md` — agent-facing summaries; lossy compared to the Constitution.
4. **This** `constitution/CLAUDE.md` / `AGENTS.md` — universal agent-facing summary; lossy compared to the Constitution.

If a project's own files contradict the Constitution, that is itself a Constitution violation and MUST be reported. Do NOT silently follow the contradicting project file.

When stuck

- Read `constitution/Constitution.md`.
- Cross-reference the project's `CLAUDE.md` / `AGENTS.md`.
- If still unclear, ask the operator. Do NOT guess. Do NOT bluff.

§11.4.65 — Universal Markdown export mandate (User mandate, 2026-05-19)

Every Markdown document inside the project that is NOT part of an application or service's source-code tree MUST have synchronized `.html` and `.pdf` siblings. Includes: project-root `*.md`, `docs/**/*`.md, `scripts/**/*`.md (doc-format companion docs), owned-submodule top-level `README.md` / `CLAUDE.md` / `AGENTS.md` / `CHANGELOG.md` and their `docs/**/*`.md, `constitution/**/*`.md, owned HelixQA submodules' equivalents. Excludes: `external/**`, `prebuilts/**`, `packages/modules/**`, `kernel-5.10/**`, `out/**`, `build/**`, application/service source-code trees, and third-party submodules NOT in the owned set. Every edit triggers regeneration via `scripts/testing/sync_all_markdown_exports.sh` (pandoc HTML + weasyprint PDF, timeout 60 per file, capped at 500 candidates). HTML + PDF mtime MUST be $\geq$ source `.md` mtime at all times.

Pre-build gates `CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC` + `CM-COVENANT-114-65-PROPAGATION`. Paired meta-test mutations. Composes with §11.4.12 / §11.4.18 / §11.4.23 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.59 / §11.4.60 / §11.4.63 / §11.4.64. No escape hatch — no `--skip-md-exports`, `--no-pdf-only`, `--md-export-not-applicable` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.65.

Non-compliance is a release blocker regardless of context.

§11.4.66 — Blocker-resolution interactive-clarification mandate (User mandate, 2026-05-19)

When any task is blocked (operator decision, hardware access, external authorization, ambiguous scope), the agent MUST: (1) research what's doable from the agent side without operator input; (2) calculate minimum-viable operator input; (3) construct 2–4 mutually-exclusive options with one marked "Recommended" and

each stating what the agent does after that answer; (4) present via the platform's interactive question mechanism (`AskUserQuestion` on Claude Code) — NEVER free-text "what would you like?" for closed- set decisions; (5) after the answer, resume work without follow-up round-trips. Composes with §11.4.6 / §11.4.7 / §11.4.40 / §11.4.41 / §11.4.42 / §11.4.52. No silent waiting; no bulk-text questions when interactive options would do.

Pre-build gate `CM-COVENANT-114-66-PROPAGATION` enforces the anchor literal across the 42-file consumer fleet. Paired meta- test mutation strips the literal → gate FAILs. No escape hatch — no `--skip-ask` , `--silent-wait` , `--free-form-only` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.66.

Non-compliance is a release blocker regardless of context.

§11.4.67 — Shell-script target-shell-parseability mandate (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19): "any issue we spot must be fixed, bash scripts as well if they are broken!"

- "Make sure that this is mandatory rule!"

Every shell script that may be invoked under a target shell other than the one in its shebang MUST parse cleanly under that target shell. Forensic incident: `device/rockchip/rk3588/tests/test_all_fixes.sh:114` used bash-only `exec > >(tee -a "$f") 2>&1` on a `sh` script.sh callsite — Android mksh parses the whole script BEFORE executing, so the runtime `[ -n "$ {BASH_VERSION:-}" ]` guard could not save it. Fixed by wrapping in `eval 'exec > >(tee ...) 2>&1'` so the parser sees only a string.

Closed-set scope: every tracked `.sh` under `device/rockchip/rk3588/tests/` , `scripts/` , `scripts/testing/` (and equivalent paths in owned submodules). OUT of scope: `external/` , `prebuilts/` , `packages/modules/` , `kernel-5.10/` , `out/` , `build/` , `scripts/legacy/` . Mandatory invariants: (1) every in-scope script parses under `sh -n` ; (2) bash-only constructs (`>(...)` , `<(...)` , `[[ ]]` , `<<<` , arrays, `${var^^}` , etc.) MUST be wrapped in `eval`

OR guarded by bash-only loading; (3) shebangs honest — `#!/bin/bash` only if bash actually expected; (4) fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51.

Pre-build gate `CM-SCRIPT-TARGET-SHELL-PARSEABLE` runs `sh -n` on every in-scope script. Propagation gate `CM-COVENANT-114-67-PROPAGATION` enforces the anchor literal across the 44-file consumer fleet. Paired mutations: inject bash-only outside `eval` → parse gate FAILs; strip `11.4.67` literal → propagation gate FAILs. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.67.

Non-compliance is a release blocker regardless of context.

§11.4.68 — Positive sink-side / downstream evidence mandate (User mandate, 2026-05-20)

A test asserting audio or video routing PASS MUST capture and verify positive sink-side or downstream evidence — never config-only, never metadata-only, never PCM-open-state-only. Closed enumeration of acceptable evidence: (1) sink-side codec-state showing non-empty codec matching the expected regex during playback; (2) PCM `hw_ptr` delta strictly positive across the playback window; (3) ALSA ELD demonstrating negotiated channel count + format; (4) `ffprobe` non-zero frames matching expected codec for video; (5) recording-analyzer matched event per §11.4.2 / §11.4.5 timeline; (6) `tinycap` RMS amplitude above floor.

Failure modes specifically forbidden: silent SKIP on sink unreachable, empty codec-state treated as evidence, policy-side device field used as proof PCM opened, config-XML-only PASS. All produce the §11.4 PASS-bluff failure mode anchored at D3 forensic 2026-05-20.

Mandatory protections: sink-side libraries expose `*_require_reachable` returning exit code 2 (OPERATOR-BLOCKED); harness propagates 2; audio/video-routing tests capture ≥ 1 positive evidence per enumeration; anti-stickiness post-stop re-probe. No escape hatch.

Composes with §11.4.2 / §11.4.5 / §11.4.13 / §11.4.14 / §11.4.46 / §11.4.49 / §11.4.50 / §11.4.52. Pre-build gates `CM-COVENANT-114-68-PROPAGATION` + `CM-POSITIVE-SINK-EVIDENCE-PER-AUDIO-TEST`

- paired meta-test mutations per §1.1.

Canonical authority: constitution submodule `Constitution.md` §11.4.68.

Non-compliance is a release blocker regardless of context.

§11.4.69 — Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!"

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Closes the PASS-bluff pattern where tests reported green while end users hit broken features (2026-05-19 → 20 D3 audio "82/84 PASS" + empty Arvus Codec-In-Use).

The mandate. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (`audio_output`, `audio_input`, `video_display`, `network_throughput`, `network_connectivity`, `bluetooth_a2dp`, `bluetooth_pair`, `touch_input`, `sensor`, `gpu_render`, `storage_read`, `storage_write`, `mediacodec_decode`, `mediacodec_encode`, `miracast`, `cast`, `boot_service`, `package_install`, `permission_grant`, `wifi_link`, `wifi_throughput`, `ethernet_link`, `display_topology`, `drm_playback`, `subtitle_render` — open to additions). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape.

Helper contracts (additive during grace; mandatory after 2026-06-19):

- `ab_pass_with_evidence <description> <evidence_path>` — the new canonical PASS helper. Verifies path exists AND non-empty; emits `PASS: <description> [evidence: <path>]`.
- `ab_skip_with_reason <description> <closed-set-reason>` — reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`. Forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe.
- Bare `ab_pass` deprecated — WARN pre-grace, FAIL post-grace (2026-06-19).

Mechanical enforcement. Three pre-build gates + three paired §11.1 meta-test mutations:

- `CM-SINK-EVIDENCE-PER-FEATURE` — walks tests for `# §11.4.69 FEATURE: <class>` annotation + verifies taxonomy probe + `ab_pass_with_evidence` use.
- `CM-NO-FAIL-OPEN-SKIP` — audits sink-side probe helpers; FAILs if any code path converts empty/unreachable response to PASS-counting SKIP for a feature class with a sink-side probe.
- `CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE` — pre-grace WARN, post-grace FAIL on bare `ab_pass` calls.

Composes with §11.4.1 (FAIL-bluffs forbidden), §11.4.2 (recorded-evidence), §11.4.5 (audio + video 5-layer quality), §11.4.6 (no-guessing), §11.4.13 (sink-side captured-evidence), §11.4.27 (no-fakes-beyond-unit), §11.4.50 (deterministic consistency), §11.4.52 (autonomous-validation), §11.4.68 (audio-specific sink-side — §11.4.69 is the universal generalisation).

No escape hatch — no `--skip-evidence`, `--config-only-pass`, `--allow-fail-open-skip`, `--legacy-ab-pass-permitted` flag. The discipline exists because the 2026-05-20 forensic incident demonstrated the failure: tests reported audio-routing PASS while the user heard nothing and the Arvus Codec-In-Use field was empty.

Propagation gate `CM-COVENANT-114-69-PROPAGATION` enforces this anchor literal across the ~44-file consumer fleet. Paired mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule `Constitution.md` §11.4.69.

Non-compliance is a release blocker regardless of context.

§11.4.70 — Subagent-driven execution is the default (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

"Always do if possible Subagent-driven! Add this into our root (constitution Submodule) `Constitution.md`, `CLAUDE.md` and `AGENTS.md`. This should be the default choice ALWAYS!"

When executing implementation plans authored via `superpowers:writing-plans` (or any equivalent task-decomposed execution flow), the **default execution model is subagent-driven** per `superpowers:subagent-driven-development`. Inline execution via `superpowers:executing-plans` is permitted ONLY when (a) the task is trivial AND fits in a single sub-300-line edit, OR (b) the operator explicitly requests inline execution at brainstorm-handoff time.

Subagents bring an isolated context window per task (no conductor context bloat), a structurally separated review seam (conductor reviews subagent output, eliminating self-review blind spots), parallel-PWU compatibility (§11.4.58 — subagents ARE the parallel work units), and resumability across operator absence (subagents resume from on-disk plan + spec inputs).

Composes with §11.4.4 (four-layer coverage), §11.4.6 (no-guessing — subagent's captured output IS the evidence), §11.4.42 (iteration discipline), §11.4.43 (TDD-fix), §11.4.50 (deterministic consistency), §11.4.51 (LIVE_ADB_FIRST), §11.4.58 (parallel- development PWU).

No escape hatch — `--inline-execution-required`, `--no-subagents`, `--monolithic-execution` are NOT permitted flags. Skipping subagent-driven for non-trivial work without recorded operator authorisation is itself a §11.4 PASS-bluff. Pre-build gate `CM-COVENANT-114-70-SUBAGENT-DEFAULT-PROPAGATION` enforces this anchor literal across the ~44-file consumer fleet. Paired meta-test mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule `Constitution.md` §11.4.70.

Non-compliance is a release blocker regardless of context.

§11.4.71 — Pre-Push Fetch + Investigate + Integrate Mandate (User mandate, 2026-05-20)

Everyday-push variant of §11.4.41 (force-push merge-first). Before pushing to any upstream for any repository (main repo or submodule), the agent MUST follow the 5-step pre-push cycle: (1) `git fetch --all --prune --tags`, (2) `git pull --no-rebase <remote> <branch>` for each remote whose tip differs from local, (3) investigate the diff vs OUR previous HEAD by reading every foreign commit's body (what changed, why, how it affects OUR project), (4) integrate mandatory changes with full §11.4.4(b) four-layer anti-bluff test coverage producing REAL captured-evidence proofs, (5) THEN push to every configured remote in cascade order.

Composes with §11.4.41 (force-push case) + §11.4.26 / §11.4.32 / §11.4.37 / §11.4.40 / §11.4.42 / §11.4.43 / §11.4.4(b) / §11.4.5 / §11.4.6.

Applies to parent repo + constitution submodule + every owned submodule + every nested submodule + every HelixQA dependency. Audit-trail per push reconstructable from `docs/changelogs/<tag>.md` + per-repo `git log` evidence.

No escape hatch — no `--skip-fetch`, `--no-investigate`, `--fast-push`, `--trust-upstream` flag. Pre-build gate `CM-COVENANT-114-71-PROPAGATION` + paired mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.71.

Non-compliance is a release blocker regardless of context.

§11.4.72 — Audio Top-Priority Mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure all fixes for audio are always top priority in main working stream!" The conductor (main working stream — Claude Code session, AI agent, or human operator) MUST treat audio fixes as the highest-priority class on the serial dispatch queue. Audio scope includes §EU HDMI rejection, §EM multichannel HDMI, §ET Arvus integration, §EV/§EW/§EX D3 audio defects, HiFi (Fix #74, #76), AC3 (Fix #106), ES8388 (Fix #103/§104/#105), multichannel LPCM (Fix #112), and every future audio-stack improvement.

Composes with §11.4.42 (iteration-discipline — audio sits at apex of priority order) + §11.4.58 (parallel-development PWU — background research subagents run concurrently but do NOT preempt audio on the main-stream serial dispatch queue).

Operative rule: any time the conductor faces a choice between dispatching an audio task vs a non-audio task on the SAME serial resource, the audio task wins. No escape hatch — no "but this non-audio task is faster" or "but this research is more interesting" override. Audio-stack regressions are user-perceptible and high-impact (D3 silent post-flash is a release blocker), while research and refactors can wait.

Canonical authority: constitution submodule `Constitution.md` §11.4.72.

Non-compliance is a process violation regardless of context.

§11.4.73 — Main-specification document versioning + revision discipline (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make sure everything we add now in previous and upcoming requests IS ALWAYS applied to the main specification — if we have one. Since all these are not major changes we could increase Specification version per change for secondary version instead of the primary. Primary version MUST BE increased for much bigger levels of changes! Document MUST BE updated ALWAYS to follow the versioning rules we are applying here + revision and other properties we have!"

Applies only when a project recognises a main specification document. Two-axis versioning: **primary** (V1/V2/V3/...) bumps for major rewrites (old primary archived); **secondary** (Revision) bumps for additive operator requirements (matches the §11.4.61 metadata-table Revision integer). Every operator-mandated requirement MUST land in the spec as part of the work that implements it. Cross-doc propagation copies MUST reference the active spec file, not a stale archived version. Composes with §11.4.44, §11.4.61, §11.4.59, §11.4.65.

Canonical authority: constitution submodule `Constitution.md` §11.4.73.

Non-compliance is a release blocker.

§11.4.74 — Submodule-catalogue-first discovery + extend-don't-reimplement (User mandate, 2026-05-20)

Direct user mandate (verbatim): "We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in `vasic-digital` and `HelixDevelopment` organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times. For any missing features we MUST IMPLEMENT them properly and extend those Submodules further!"

Before scaffolding any new module / package / helper / utility, the agent MUST: (1) survey `vasic-digital` + `HelixDevelopment` orgs on GitHub + GitLab — the canonical inventory is `submodules-catalogue.md` (142 repos categorised); (2) reuse an existing Submodule when it covers $\geq 80\%$ of the functionality; (3) extend in-place via upstream PR when 80%+ matches but features are missing — never duplicate; (4) document the survey result in the relevant tracker row with

`Catalogue-Check: reuse|extend|no-match <org/repo>@<sha> .`

Every Submodule in the catalogue is subject to the same development-cycle rules (§11.4 anti-bluff, §1.1 paired mutations, §11.4.10 credentials, §11.4.61 metadata + ToC, §11.4.65 universal export, §11.4.73 spec versioning, §2.1 multi-mirror push, §3 propagation order).

Canonical authority: constitution submodule `Constitution.md` §11.4.74.

Non-compliance is a process violation; severe cases (duplicate implementation landed without catalogue check) are release blockers.

§11.4.75 — Mechanical Enforcement Without Exception (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Why do these violations still happen!? This is a serious problem! We cannot rely on stability nor consistency if we cannot respect our Constitution, mandatory rules and constraints! Is there a way to make this always respected, followed and applied without exception fully and unconditionally!? WE MUST HAVE THIS WORKING FLAWLESSLY!!! Do investigate the root causes of such problems! Once all problems are identified WE MUST apply proper mechanisms for this not to happen NEVER EVER AGAIN!"

The §11.4 covenant historically relied on agent + operator vigilance. Three forensic incidents in 2026-05-19 → 20 (two stalled remediation subagents + post-§11.4.66 propagation drift requiring catch-up commit `f93b25a92eb`) demonstrated that late-

binding enforcement at `pre_build_verification.sh` time fires hours-to-days AFTER the violator commit has already reached every remote. §11.4.75 closes the gap with FIVE independent mechanical enforcement layers — bypassing any single layer does not bypass the discipline:

1. **Local pre-commit git hook** — refuses staged `.md` lacking sibling `.html` + `.pdf` (and other staged-only invariants).
2. **commit_all.sh integration** — the canonical project commit script invokes the same checks + auto-runs `sync_all_markdown_exports.sh` to self-repair before commit (`_constitution_sibling_check` function).
3. **Local pre-push git hook** — re-runs siblings + propagation-gate subset on every commit in the push.
4. **post-commit auto-repair hook** — detects orphan `.md` in just-committed manifest, auto-generates siblings via pandoc + weasyprint, creates a `chore(§11.4.75): auto-export ... follow-up commit`. Idempotent + recursion-guarded.
5. **Local-only equivalent** (Phase 39.GF, User mandate 2026-05-20). Remote CI surfaces (GitHub Actions, GitLab pipelines, Jenkins, CircleCI, etc.) are DISABLED — the workflow file is preserved at `.github/workflows/constitution-compliance.yml.disabled-local-only` (NOT `.yml` ; GitHub Actions ignores it). Layer 5 enforcement migrated to the LOCAL pre-build-verification + meta-test ritual the operator MUST execute before tagging per §11.4.40. Layers 1-4 remain authoritative; Layer 5 is the operator's local final gate. A future re-enable PWU may re-establish remote CI.

Helper contracts (mandatory): `scripts/install_git_hooks.sh` (idempotent installer wired into `scripts/setup.sh`), `scripts/git_hooks/{pre-commit,pre-push,post-commit,commit-msg}` , `_constitution_sibling_check` in `scripts/commit_all.sh` . The `commit-msg` hook enforces a Bypass-rationale: `<reason> footer when --no-verify is detected (touch-file marker .git/ATMO_LAST_BYPASS_ATTEMPT )` ; `docs/audit/bypass_events.md` accumulates the audit trail per §11.4 captured-evidence requirement.

Five pre-build gates with paired §1.1 meta-test mutations: `CM-COVENANT-114-75-PROPAGATION` (anchor literal across canonical files), `CM-GIT-HOOKS-INSTALL-SCRIPT` (installer present + executable), `CM-GIT-HOOKS-SOURCE-DIR` (4 hook

bodies present + executable), `CM-COMMIT-ALL-SIBLING-CHECK`
(`_constitution_sibling_check` in `commit_all.sh`), `CM-CI-WORKFLOW-PRESENT` (CI workflow + ≥ 3 required jobs).

Composes with §1.1 (paired meta-test mutations), §9 (data safety), §11.4 (end-user-quality covenant — Layer 5 CI makes the §11.4 promise mechanical), §11.4.41 (this anchor's renumber from §11.4.74 → §11.4.75 was itself driven by §11.4.41 merge-first discipline), §11.4.65 (universal Markdown export — Layer 1+3+4 are §11.4.65's mechanical seam), §11.4.66 (interactive clarification), §11.4.67 (target-shell-parseability — hooks parse under bash AND mksh), §11.4.71 (pre-push fetch + integrate — Layer 3 + 5 honour the merge-first pipeline), §11.4.72 (audio top-priority — hooks hold no lock themselves, so do not race against in-flight audio commits), §11.4.73 (main-spec versioning — when spec exists, hooks include it), §11.4.74 (submodule-catalogue-first — hooks can be added to the canonical catalogue for cross-project reuse).

No escape hatch — no `--skip-hooks` , `--bypass-enforcement` , `--allow-orphan-md` , `--ci-not-applicable` , `--mechanical-enforcement-not-needed` flag exists. The `--no-verify` route IS the deliberate audit-trail bypass; §11.4.75 makes the audit trail mechanical via the `commit-msg` footer requirement.

Canonical authority: constitution submodule `Constitution.md` §11.4.75.

Non-compliance is a release blocker regardless of context.

§11.4.76 — Containers-submodule mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "For any work or requirements of running services or codebase inside the Containers (Docker / Podman / Qemu / Emulators, and so on) we MUST USE / INCORPORATE the Containers Submodule properly:

`https://github.com/vasic-digital/containers` (`git@github.com:vasic-digital/containers.git`). Containers Submodule contains all means for us to Containerize our code and services! If any feature or Containing System is missing or not supported we MUST EXTEND IT properly like we do all of our projects! No bluff work is allowed of any kind!"

§-slot history note. Originally drafted as §11.4.75, renumbered to §11.4.76 per §11.4.71 fetch-before-push after the concurrent `0a70083` landing of §11.4.75 (Mechanical Enforcement).

For ANY containerized workload (Docker / Podman / Qemu / Kubernetes / container-backed emulators), the agent MUST: (1) install `vasic-digital/containers` (`digital.vasic.containers`) as a Git submodule when scaffolding the consuming project; (2) consume via `replace` directive during development + pinned commit SHAs in production; (3) boot infra on-demand via the Submodule's `pkg/boot` + `pkg/compose` + `pkg/health` APIs so the operator is never required to start `podman machine` / `docker compose up` manually — the boot is part of the test entry point (**on-demand-infra invariant**); (4) extend the Submodule via upstream PR when a runtime / lifecycle primitive is missing — never reimplement in-project (per §11.4.74); (5) anti-bluff: integration tests claiming to exercise containerized components MUST actually boot them via the Submodule. Short-circuit fakes that bypass boot are a §11.4 violation. A passing test MUST imply the infra was up.

Tracker rows touching containerization MUST record `Catalogue-Check: extend vasic-digital/containers@<sha>` (or `reuse`); `no-match` requires demonstrating the Submodule cannot model the workload.

Planned anti-bluff gate `CM-CONTAINERS-USED` scans container-touching PRs for `digital.vasic.containers/...` imports. Paired mutation strips the import + asserts FAIL.

Composes with §11.4.74 (catalogue-first), §11.4.75 (mechanical enforcement), §3 (propagation), §11.4.31 (Submodule-Dependency-Manifest), §11.4.36 (`install_upstreams` on clone), §11.4.28 (Submodules-As-Equal-Codebase), §1.1 (paired-mutation gates).

Canonical authority: constitution submodule `Constitution.md` §11.4.76.

Non-compliance: reinventing compose orchestration in-project is a release blocker.

§11.4.77 — Regeneration-mechanism-required mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "We must be sure that after excluding anything from Git versioning we still have the mechanism which will out of the box obtain or re-generate missing content!"

Forensic anchor. 2026-05-20T15:00Z: a consuming project's audio Tier 1

`commit_all.sh` stalled 4 h on `git add -A` scanning 274 GiB `.git-backup-*` + 159 GiB `RKTools/linux/` + 167 GiB `qa-results/` — all untracked but un-gitignored. Bare `.gitignore` fix would orphan every fresh clone (missing `RKTools`, missing test infra, missing build outputs).

Every `.gitignore` entry excluding (a) >~100 MiB OR (b) any artefact essential to building / running / testing the project MUST carry a documented + automated mechanism to either **re-obtain** (vendor tarball, SDK installer, package registry, git submodule, object store) OR **re-generate** (build pipeline, code-gen, asset render, captured-evidence replay, container build).

Required artefacts per qualifying `.gitignore` entry: (1) `.gitignore-meta/<entry-slug>.yaml` declaring pattern + mechanism-type + script-path + expected-disk-usage + vendor-url-or-source + integrity hash + requires-network + requires-credentials; (2) post-clone bootstrap entry (`scripts/setup.sh` or canonical equivalent) running the mechanism non-interactively; (3) pre-build gate verifying regenerated content present OR stamp `.gitignore-meta/.regenerated/<slug>.ok` recent; (4) README + `docs/guides/*.md` describing the mechanism + manual fallback + time/disk budget + per-§11.4.10 credentials.

No escape hatch: bare `.gitignore` additions without the mechanism are §11.4 PASS-bluff variants — codebase appears complete but fresh clone cannot build / run. No `--skip-regen-mechanism`, `--gitignore-is-enough`, `--operator-already-has-content` flag.

Planned anti-bluff gate `CM-GITIGNORE-REGEN-MECHANISM` scans every `.gitignore` addition for matching `.gitignore-meta/` sibling. Paired §1.1 mutation strips a required YAML key → gate FAILs.

Composes with §11.4.6 (no-guessing — verify mechanism on sandbox), §11.4.65 (HTML/PDF siblings are a `re-generate` instance), §11.4.66 (interactive clarification if mechanism unknown), §11.4.71 (re-validate vendor integrity before push), §11.4.74 (extend a reusable downloader Submodule), §11.4.75 (pre-commit + CI replay refuse bare additions), §11.4.76 (container images regenerated via `vasic-digital/containers`), §9 / §9.2 (pre-test mechanism in sandbox), §3 (consuming submodules inherit).

Canonical authority: constitution submodule `Constitution.md` §11.4.77.

Non-compliance is a release blocker regardless of context.

§11.4.78 — CodeGraph code-intelligence mandate (User mandate, 2026-05-20)

Direct user mandate (verbatim): "Make codegraph MANDATORY CHOICE for this purpose for all of our project ... All project which do not have configured and installed codegraph yet MUST DO IT and MUST USE IT!"

Every consuming project worked on by AI coding agents MUST install, initialize, and use **CodeGraph** (<https://github.com/colbymchenry/codegraph> , npm `@colbymchenry/codegraph`) — a local SQLite semantic code-knowledge-graph exposed to agents over MCP (100% local, no cloud). (1) Install globally via npm — no `sudo` (npm prefix MUST be user-writable). (2) `codegraph init + codegraph index : .codegraph/config.json` tracked, `.codegraph/codegraph.db` gitignored with `codegraph index` as its §11.4.77 regeneration mechanism; the `exclude` list MUST exclude every §11.4.10 credential/secret path. **CRITICAL — CodeGraph filter mechanism:** CodeGraph v1.x is zero-config — it uses built-in skip lists + root `.gitignore` for file filtering, NOT `config.json` include/exclude. Use root-anchored `.gitignore` patterns (`/pattern/`) to exclude only actual rubbish (build outputs, caches, credentials). **ALL SOURCE CODE MUST REMAIN INDEXABLE** — AOSP platform trees (`frameworks/` , `packages/` , `external/` , `art/` , `bionic/` , `kernel-5.10/` , etc.) are tracked source and MUST NOT be blocked. (3) Wire the `codegraph serve --mcp` server into every CLI agent the developers use — project-scoped + committed where supported (Claude Code `.mcp.json` , OpenCode `opencode.json` , Qwen Code `.qwen/settings.json` , Crush `.crush.json`), host-local otherwise (Kimi CLI `~/kimi/mcp.json`); configs reference the bare `codegraph` command on `PATH` . (4) Cover the integration with an anti-bluff verification suite whose per-agent end-to-end layer uses an unforgeable challenge (a fact obtainable only by calling a CodeGraph MCP tool); un-runnable agents are documented SKIP gaps per §11.4.3, never faked PASSes. (5) Document everything in `docs/CODEGRAPH.md` . CodeGraph is consumed as the npm package (§11.4.74) — not a git submodule, adds no Git remote.

Universal mandatory-exclude baseline set (§11.4.78 + §11.4.79 refinement — binds EVERY consuming project). Independently of project, the CodeGraph index MUST NEVER include the following file-classes anywhere — they are indexed-rubbish in every project; a consuming project MAY add its own project-specific rubbish paths on top per §11.4.35 but MUST NOT drop any baseline class:

(a) **build outputs / regenerable artifacts** — `/out/` , `/build/` , `/dist/` , `/target/` , `bin/` , `obj/` , `.compressed/` , and any generator-produced tree; (b) **caches / dependency / tooling dirs** — `**/ccache/` , `**/.ccache/` , `.gradle/` , `__pycache__/` , `.pytest_cache/` , `.mypy_cache/` , `.ruff_cache/` , `node_modules/` , `.next/` , `.cache/` , `.terraform/` , `.venv/` , `logs/` ; (c) **credentials / secrets** — `.env` , `.env.*` , `**/*.env` , `secrets/` , `**/*secret*` , `keystores` / `signing-keys` / `service-account JSON`, and every credential-bearing path per §11.4.10 (a secret reaching the index is a §11.4.10 violation, never negotiable); (d) **QA / recording / large regenerable corpora** — `**/qa-results/` , `**/recordings/` , and prebuilt-binary trees. Realized via root-anchored `.gitignore` patterns (CodeGraph v1.x zero-config filter = built-in skip list + root `.gitignore` , NOT `config.json` include/exclude). Own-org submodule sources stay INCLUDED per §11.4.79; third-party vendored trees EXCLUDED. This baseline is UNIVERSAL (§11.4.17) — dropping any class is a §11.4.78 / §11.4.79 violation, and no source-code tree may be excluded to satisfy it. Composes with §11.4.3, §11.4.10, §11.4.12, §11.4.65, §11.4.30, §11.4.74, §11.4.77, §11.4, §1.1. Planned gate `CM-CODEGRAPH-WIRED` + paired mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.78.

Non-compliance is a process violation; a project worked on by AI agents without CodeGraph installed, wired, and anti-bluff-verified is in breach of this mandate.

§11.4.79 — Own-org submodules MUST be included in the CodeGraph index (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "All Submodules we use in the project and that are part of organizations to which we have the full access via GitHub, GitLab and other CLIs MUST BE included into the codegraph database and initialized / scanned / synced!"

Refines §11.4.78 step 2's exclude-list with a per-submodule-ownership split: **own-org** (full write access via project's CLIs — canonical orgs `vasic-digital` + `HelixDevelopment`) MUST be INCLUDED; **third-party** (§11.4.74 no-match → vendor; e.g. `gopkg.in/telebot.v3`) MUST be EXCLUDED. Operational steps: (1) `git submodule update --remote --merge` before re-indexing — respect load-bearing third-party pins (roll back if `--remote` breaks a load-bearing import-path pin). (2) Adjust `.codegraph/config.json` exclude list. (3) Re-index. (4)

Validate via probe resolving a symbol living ONLY inside an own-org submodule.

(5) Paired §1.1 mutation: temporarily add own-org path to exclude → validate FAILs
→ restore.

Composes with §11.4.74 (ownership classifier), §11.4.78 (CodeGraph parent), §11.4.10 (credentials still excluded), §1.1 (paired mutation), §107 (a lying index is a PASS-bluff).

Canonical authority: constitution submodule `Constitution.md` §11.4.79.

Non-compliance is a process violation; severe cases (own-org submodules silently excluded without an audit trail in `.codegraph/config.json`) are release blockers.

§11.4.80 — CodeGraph regular-update + sync automation mandate (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "We MUST regularly check for the updates and execute codegraph npm updates so the latest version of it is always installed on the host machine! [...] Make sure we have proper full automation bash scripts which will run regularly and that these are part of the constitution Submodule since they MUST BE available to all projects [...]. Make sure all updates, sync processes we do and important codegraph related events are all documented under docs/codegraph in Status and Status_Summary documents [...]"

Three deliverables in this constitution submodule: (1) `scripts/codegraph_update.sh` — npm-installs latest; §107 anti-bluff verifies `codegraph --version` reflects new version (npm exit 0 ≠ working binary). (2) `scripts/codegraph_sync.sh` — runs `codegraph status` → `sync` → `status` → project's validate; appends to both project's + constitution's `docs/codegraph/Status.md`. (3) `docs/codegraph/Status.md` + `Status_Summary.md` ledgers, exported to `.html` + `.pdf` per §11.4.65.

Cadence: weekly floor. Scripts inherited by reference (§3 submodule inheritance) — invoke at `${CONST_DIR}/scripts/codegraph_*.sh`, never copy.

Composes with §11.4.78, §11.4.79, §11.4.10, §11.4.45, §11.4.65, §107, §1.1.

Canonical authority: constitution submodule `Constitution.md` §11.4.80.

Non-compliance is a process violation; severe cases (>2 weeks since last update + open AI-agent work) are release blockers.

§11.4.81 — Cross-platform-parity mandate (User mandate, 2026-05-21)

Direct user mandate (verbatim, 2026-05-21): "Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!"

Every consuming project whose supported-platforms manifest lists more than one OS MUST ship per-OS-equivalent implementations + tests for every feature/gate/challenge/mutation that depends on platform-specific primitives. Runtime dispatch via `uname -s` (or equivalent). Three sub-mandates: **(A)** Per-OS implementation REQUIRED — `cgroup/systemd/ /proc` primitives have documented equivalents (POSIX `setrlimit / ulimit`, `launchd`, `rctl`, Job Object). **(B)** Per-OS tests REQUIRED — every gate test has `case "$(uname -s)"` in branches with positive captured evidence per §11.4.2 + §11.4.5 in each branch; SKIP-with-reason only when the platform genuinely cannot enforce. **(C)** Honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists (canonical: XNU does NOT enforce `RLIMIT_AS` for unprivileged processes → SKIP with exact reproducer + adjacent test of what IS enforced, e.g. `RLIMIT_CPU` + `SIGXCPU`). The adjacent test is itself anti-bluff per §11.4 with a paired §1.1 mutation.

Per-OS equivalence catalogue (canonical): `systemd-run --user --scope` ↔ POSIX `ulimit -t -u / launchd; cgroup MemoryMax` ↔ XNU gap (use `RLIMIT_CPU` adjacent) / `rctl` / Job Object; `cgroup TasksMax` ↔ `RLIMIT_NPROC` ; `/proc/<pid>/oom_score_adj` ↔ no Darwin/BSD equivalent.

Composes with §11.4.1 / §11.4.2 / §11.4.3 (strictened — SKIP only when kernel cannot) / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.20 / §11.4.27 / §11.4.69 / §11.4.70 / §107. Pre-build gate `CM-CROSS-PLATFORM-PARITY` + paired §1.1 mutation. No escape hatch.

Canonical authority: constitution submodule `Constitution.md` §11.4.81.

Non-compliance is a release blocker on multi-platform projects.

§11.4.82 — Iteration-speedup discipline mandate (User mandate, 2026-05-22)

Direct user mandate (verbatim, 2026-05-22): "How can we speed-up this whole development and fixing process? ... all speed optimizations critical rules and mandatory constraints MUST BE all added into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md and QWEN.md!"

Iteration cycle time bounds the project's defect-discovery rate. Slow cycles ship fewer validated features per unit of operator time. Every consuming project's build / test / commit / debug pipeline MUST adopt 9 speedup disciplines AS MANDATORY:

- **(A) Phase 1 forensic before any speculative source patch** — Phase 1 of `superpowers:systematic-debugging` is mandatory before any non-trivial source patch. Speculative patches violate §11.4.6 + §11.4.82.
- **(B) Live-ADB-First before rebuild** — §11.4.51 strengthened to release-blocker. Skipping live-probe for LIVE_ADB_TESTABLE changes = §11.4.82 violation.
- **(C) Pre-flight before rebuild orchestrators** — 30-sec pre-flight: device reachable, sinks reachable, memory budget, no stale locks, no orphan processes.
- **(D) Persistent build caches outside containers** — `ccache` + Soong / `sccache` / Gradle daemon state bind-mounted to host. Saves 25-32 min/cycle from cycle #2.
- **(E) Module-only rebuild for `CONFIG_*=m` driver patches** — `make modules` over single driver dir (~2-5 min vs ~25 min full kernel).
- **(F) Parallel multi-device testing** — every owned validation device runs autonomous cycle concurrently with separate output dirs.
- **(G) Subagent scope ≤30 min + worktree isolation + single-responsibility** — empirical pattern: 8-task subagents stall, 2-3-task subagents complete.
- **(H) Lock-file + stale-process hygiene** — clean `.git/index.lock` / `.lock` / orphan git/build processes on session start. Disable auto `git-gc` in concurrent multi-agent repos.
- **(I) Cycle telemetry per §11.4.24** — log commit hash, per-phase wall-clock, speedup flag set, outcome. Weekly aggregation surfaces empirical ROI per discipline.

Composes with §11.4.4 / §11.4.6 / §11.4.9 / §11.4.20 / §11.4.24 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.51 / §11.4.52 / §11.4.58 / §11.4.70 / §12.7 / §107. Pre-build gate `CM-ITERATION-SPEEDUP-DISCIPLINE` (when implemented) + paired §1.1 mutation. No escape hatch — no `--skip-phase1` , `--no-pre-flight` , `--rebuild-everything` , `--unlimited-subagent-scope` , `--ignore-locks` , `--no-telemetry` flag exists.

Canonical authority: constitution submodule `Constitution.md` §11.4.82.

Non-compliance is a release blocker.

§11.4.83 — docs/qa/ end-user evidence mandate (User mandate, 2026-05-22)

Direct user mandate (verbatim, 2026-05-22): "every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under `docs/qa/<run-id>/` (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation MUST preserve full bidirectional communication threads as proof."

Every feature that ships MUST carry a recorded end-to-end communication transcript plus any attached materials committed under `docs/qa/<run-id>/` . A feature with no QA transcript is itself a §11.4 / §107 PASS-bluff. (1) Transcripts MUST be full bidirectional. (2) Attached materials MUST live in-repo (no external-only links — §11.4.13 sink-side violation). (3) Bot-driven / agent-driven QA MUST preserve the full conversation thread as the proof artefact. (4) CI release gates MUST refuse to tag a version whose feature-shipping commit lacks its matching `docs/qa/<run-id>/` .

Composes with §11.4.2, §11.4.5, §11.4.13, §11.4.65, §11.4.69, §107, §1.1.

Canonical authority: constitution submodule `Constitution.md` §11.4.83.

Non-compliance is a release blocker.

§11.4.84 — Working-tree quiescence rule for subagent commits (User mandate, 2026-05-22)

Short tag: `working-tree quiescence` .

Direct user mandate (verbatim, 2026-05-22): "no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before `git add`, the committing agent MUST `grep` its own working tree for mutation markers (`MUTATED` for paired, `// always pass`, `return json.Marshal` shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT."

No subagent or main-thread commit may proceed while any concurrent mutation gate is in flight in the same checkout. Pre- `git add` : `grep` for mutation markers (`MUTATED` for paired, `// always pass`, `return json.Marshal` shortcuts, `// MUTATION / # MUTATION` annotations, `_mutated_*` filenames). Cross-check `git status --porcelain` against declared scope — unaccounted entries → ABORT.

Lesson (forensic). A consuming project's logo-fix subagent (Herald commit `72e81ab`, 2026-05-21) ran in a checkout where a paired §1.1 mutation had injected an `// always pass` shortcut into a JWT verify path. The subagent's `git add` swept the mutation residue into the commit; the resulting commit was pushed to all four mirrors before any other agent caught it. Fix (Herald `d5bd360`) landed within the hour, but the production-equivalent-binary-with-bypassed-JWT window is a real security-defect window. The lesson is now constitutional.

Operative rule. (1) Pre- `git add` `grep` + scope-match → unaccounted file ABORT. (2) Active mutation gates MUST be serialised — mutate → assert FAIL → restore → assert PASS — and tree clean BEFORE unrelated commits. (3) Concurrent subagents same-checkout MUST use a lockfile (`.git/MUTATION_IN_PROGRESS`) or `git worktree add` per subagent. (4) Pre-push `mutation-residue-scanner` MUST run; commits containing mutation markers → push BLOCKED.

Composes with §1.1, §11.4.20, §11.4.70, §11.4.27, §11.4.10, §11.4.71, §107.

Canonical authority: constitution submodule `Constitution.md` §11.4.84.

Non-compliance is a release blocker. A mutation marker in a tagged commit is a critical defect regardless of how briefly it persisted.

§11.4.85 — Stress + Chaos Test Mandate (User mandate, 2026-05-24)

Short tag: `stress-chaos-mandate` .

Direct user mandate (verbatim, 2026-05-24): "Every fix or improvement you do MUST BE covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!"

Every fix MUST ship with full-automation stress + chaos test suites. Happy-path-only coverage is a §11.4 / §107 PASS-bluff at the resilience layer.

Stress (closed-set): sustained load ($N \geq 100$ iters OR ≥ 30 s, p50/p95/p99 recorded) + concurrent contention ($N \geq 10$ parallel, no deadlock, no leak) + boundary conditions (empty/max/off-by-one, every boundary produces categorised result).

Chaos (closed-set, per fix-class): process-death injection + network-fault injection + input-corruption injection + resource-exhaustion injection (disk/OOM/FD) + state-corruption injection. Recovery deterministic + categorised per §11.4.69.

Anti-bluff. Every stress + chaos PASS cites a captured-evidence artefact (`latency.json` / `categorised_errors.txt` / `state_delta_snapshot.json` / `recovery_trace.log`) per §11.4.5 + §11.4.69. Helper library `stress_chaos.sh` provides `ab_stress_run` / `ab_stress_concurrent` / `ab_chaos_*` primitives composing with `ab_pass_with_evidence` / `ab_skip_with_reason` . Chaos cleanup MUST run in trap EXIT — failure = §11.4.14 violation.

4-layer per §11.4.4(b): pre-build gate + paired meta-test mutation + on-device test (if LIVE_ADB_TESTABLE) + HelixQA Challenge entry. Composes with §11.4 / §11.4.1 / §11.4.5 / §11.4.6 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.83 / §107.

Canonical authority: constitution submodule `Constitution.md` §11.4.85.

Non-compliance is a release blocker regardless of context. No `--skip-stress` , `--no-chaos` , `--happy-path-suffices` , `--stress-test-later` flag exists.

§11.4.86 — Roster/corpus-backed Status-doc auto-sync mandate (User mandate, 2026-05-25)

Forensic anchor (verbatim): "Make sure that assets and players Status docs are ALWAYS regularly updated and in sync like all others Status docs — any time we add or modify the assets content(s) or we change or add new / remove existing pre-installed video and audio player apps! This MUST WORK OUT OF THE BOX!"

Status docs (§11.4.45) backed by a **tracked roster** (installed apps/components) or **asset corpus** (test/media asset directory) MUST resync **out of the box** the moment a member changes (player added/removed/renamed; asset added/modified/removed) — Status doc + Status_Summary + HTML + PDF, mechanically, not by operator vigilance.

Mechanism (all must hold): (1) drift-proof **fingerprint** = sha256 of sorted member list (NOT mtime), persisted in a sidecar beside the Status doc; (2) **sync helper** regenerates fingerprint + re-exports HTML+PDF (via §11.4.65 exporter), wired so sync is automatic; (3) **pre-build gate** FAILs when live fingerprint ≠ persisted

(mirrors §11.4.12 `CM-ISSUES-SUMMARY-SYNC` + §11.4.45 `sync_integration_status`); (4) paired §1.1 mutation corrupts fingerprint → gate FAILs. Composes with §11.4.12 / §11.4.45 / §11.4.53 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.6. Classification universal (§11.4.17) — consuming project supplies specific docs/roster/helper/gate per §11.4.35.

Canonical authority: constitution submodule `Constitution.md` §11.4.86.

Non-compliance is a release blocker regardless of context. No `--skip-roster-sync` , `--allow-status-drift` , `--roster-sync-not-applicable` flag exists.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing mandate (User mandate, 2026-05-26)

When operator instructs an AI agent to "continue in endless loop fully autonomously" (or semantically-equivalent), the agent MUST treat as HARD-CONTRACT covenant covering five obligations: (A) continue until

`docs/Issues.md` non-terminal Status entries = 0 AND `docs/CONTINUATION.md` §3 Active work empty AND no subagent in-flight AND no external dep in-flight; (B) dispatch background subagents for parallelisable work — main + subagents concurrent, "waiting for results" is the ONLY idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence "physical proofs" (tinycap WAV + RMS / screen recording + fprobe / dumphsys + sink-probe / uiautomator dump / sysfs snapshots) — metadata-only / config-only / absence-of-error / grep-without-runtime PASS are critical defects; (D) §11.4 anti-bluff covenant

family operative end-to-end (tests AND HelixQA Challenges bound equally per forensic anchor "tests pass but features don't work"); (E) loop terminates ONLY on all-conditions-met, explicit operator STOP, host-safety demand (§12 family), or scheduled wake on known-future-actionable signal.

Composes with §11.4 / §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.7 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.58 / §11.4.68 / §11.4.69 / §11.4.70 / §11.4.83 / §11.4.85 / §11.4.86 / §12.10. Pre-build gate `CM-COVENANT-114-87-PROPAGATION` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.87.

Non-compliance is a release blocker regardless of context. No escape hatch — `--idle-OK` , `--skip-endless-loop` , `--bluff-permitted-for-this-task` , `--metadata-only-test-suffices` , `--no-physical-proof-required` are FORBIDDEN flags.

§11.4.88 — Background-push mandate: commit-lock release immediately after commit, push runs detached (User mandate, 2026-05-26)

Forensic anchor: 2026-05-26 a single `commit_all.sh` held its flock ~5 hours because `do_push` ran synchronously after the commit landed — every subsequent commit was blocked on a slow mirror push that had nothing to do with the local commit's durability. Implementation seam for §11.4.87(B) zero-idle.

Mandate (5 obligations): (A) `.git/.commit_all.lock` released IMMEDIATELY after `git commit` returns 0; (B) push runs detached via `nohup ./push_all.sh ... > <log> 2>&1 & + disown` ; (C) `push_all.sh` acquires per-remote flock `.git/.push.<remote>.lock` — concurrent same-remote serializes, different-remote parallelises; (D) failures → `qa-results/push_failures/<ts>_<remote>.log` , autonomous loop checks per §11.4.87(A); (E) `--sync-push` flag escape for §11.4.41 force-push merge-first paths only. Composes with §2.1 / §9.2 / §11.4.41 / §11.4.42 / §11.4.71 / §11.4.87. Pre-build gates `CM-COVENANT-114-88-PROPAGATION` + `CM-BACKGROUND-PUSH-WIRED` + paired §1.1 mutations.

Canonical authority: constitution submodule `Constitution.md` §11.4.88.

Non-compliance is a release blocker. No escape hatch beyond `--sync-push` .

§11.4.89 — Background test execution mandate (User mandate, 2026-05-27)

Forensic anchor — verbatim user mandate (2026-05-27): "Any tests we are executing, especially long test cycles, MUST BE performed in background in parallel with main work stream! This MUST NOT block our capabilities to work on queued workable items."

Symmetric anchor to §11.4.88 at the test-execution layer. Mandate (6 obligations):

(A) Long-running tests (>30s) MUST run via `nohup ... > <log> 2>&1 & + disown` with log under known dir; (B) Main stream proceeds to §11.4.42 priority queue immediately — "waiting for results" is the only acceptable idle reason per §11.4.87(B); (C) Hard-dependency gating: poll exit-status file or `pgrep` before dependent steps — surface as §11.4.66 options if test still running; (D) Failures land in log files; (E) Foreground only for <30s OR operator authorisation; (F) Per-script flock for same-script serialisation. Composes with §11.4.42 / §11.4.66 / §11.4.82 / §11.4.84 / §11.4.85 / §11.4.87 / §11.4.88. Pre-build gates `CM-COVENANT-114-89-PROPAGATION` + `CM-BACKGROUND-TEST-EXECUTION-WIRED` + paired §1.1 mutations.

Canonical authority: constitution submodule `Constitution.md` §11.4.89.

Non-compliance is a release blocker. No escape hatch beyond explicit per-invocation operator authorisation.

§11.4.90 — Obsolete status + per-item obsolescence audit (User mandate, 2026-05-27)

§11.4.15 Status closed-set extended with terminal `Obsolete` (→ `Fixed.md`) (orthogonal to Type). Reasons (closed): `superseded-by-design-change` | `superseded-by-later-mandate` | `feature-removed` | `duplicate-of` | `unsupported-topology` | `not-reproducible` (`not-reproducible` = a reported defect that does NOT reproduce on the canonical tree/baseline — an environment/isolated-worktree artifact, not a real product defect). Every Obsolete heading carries `**Obsolete-Details:**` (Since + Reason + Superseding-item + Triple-check evidence). Colorizer `cell-status-obsolete` = `#E0E0E0` background + strikethrough. Audit cadence per §11.4.40 + §11.4.42. Triple-check non-negotiable. Pre-build gates `CM-COVENANT-114-90-PROPAGATION` + `CM-ITEM-OBSOLETE-DETAILS` + `CM-OBSOLETE-COLORIZER-WIRED` + paired §1.1 mutations.

Canonical authority: constitution submodule `Constitution.md` §11.4.90. Non-compliance is a release blocker.

§11.4.91 — Summary-doc clarity mandate (User mandate, 2026-05-27)

Every summary entry MUST contain self-contained meaningful description ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden anti-patterns: section labels (`Composes with` , etc.); bare metadata fragments; §-letter alone. Generators MUST extract from H1/H2 heading line per §11.4.54 ATM-NNN convention. Pre-build gate `CM-SUMMARY-CLARITY-DESCRIPTIONS` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.91. Non-compliance is a release blocker.

§11.4.92 — Multi-pass change-evaluation discipline (User mandate, 2026-05-27)

Every non-trivial change MUST pass 5-pass evaluation BEFORE commit-ready: (1) Main-task verification, (2) Regression-blast-radius, (3) Cross-feature interaction, (4) Deep-research validation per §11.4.8, (5) Anti-bluff confirmation. Documentation per pass. Only after all 5 passes complete may commit/push/test/release proceed. Composes §11.4.4 / §11.4.5 / §11.4.6 / §11.4.8 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.78 / §11.4.82 / §11.4.83 / §11.4.85 / §11.4.87 / §11.4.89. Pre-build gates `CM-COVENANT-114-92-PROPAGATION` + `CM-MULTI-PASS-EVALUATION-EVIDENCE` .

Canonical authority: constitution submodule `Constitution.md` §11.4.92. Non-compliance is a release blocker.

§11.4.93 — SQLite-backed single-source-of-truth for workable items (User mandate, 2026-05-27)

Text-based Issues/Fixed/Summary trackers converted to SQLite DB at `docs/.workable_items.db` (gitignored + §11.4.77 regen mechanism). Schema: items (atm_id PK + Type + Status incl. Obsolete + Severity + title + description ≥ 40 chars + composes_with JSON + current_location); item_history (audit log

§11.4.34); `obsolete_details` (§11.4.90); `operator_block_details` (§11.4.21); `firebase_metadata` (§11.4.47); `meta`. Go binary `cmd/workable-items/` : sync md-to-db / db-to-md / diff / validate / add / close. Bidirectional byte-identical round-trip. `commit_all.sh` refuses on diff. Pre-build gate runs validate. Anti-bluff: unit + integration + stress + chaos per §11.4.85 + paired §1.1 mutation + HelixQA Challenge CME-WORKABLE-ITEMS-001. 6-phase migration §LA. Go binary in constitution submodule per §11.4.74. Composes §11.4 / §11.4.12-93 covenant family.

Canonical authority: constitution submodule `Constitution.md` §11.4.93. Non-compliance is a release blocker.

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode (User mandate, 2026-05-27)

Binds

§11.4.20+§11.4.42+§11.4.58+§11.4.70+§11.4.72+§11.4.82+§11.4.87+§11.4.88+§11.4.89 into single always-on enforcement: idle only when externally blocked / operator STOP / §12 host-safety; before any wake/sleep survey parallel-work feasibility; priority order MANDATORY (audio-first per §11.4.72); subagent-driven default non-trivial; background default >30s; stability-preserving per §11.4.92 + §11.4.84 + §12.6-§12.9; progress updates surfaced at milestones. Anti-pattern: wake without queue survey, "nothing to do" with non-trivial progressable, serial parallelisable work, lower priority over higher. Pre-build gates `CM-COVENANT-114-94-PROPAGATION` + `CM-PARALLEL-WORK-AUDIT` .

Canonical authority: constitution submodule `Constitution.md` §11.4.94. Non-compliance is a release blocker.

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored (User mandate, 2026-05-27)

§11.4.93 amendment: DB at `docs/workable_items.db` is TRACKED, NEVER gitignored. Authoritative source data, NOT build artefact. Every state change → stage+commit+push DB alongside MD regen per §11.4.19+§2.1. WAL-checkpoint before commit. Destructive DB ops require §9.2+authorization. Pre-build gates `CM-COVENANT-114-95-PROPAGATION` + `CM-WORKABLE-ITEMS-DB-TRACKED` .

Canonical authority: constitution submodule `Constitution.md` §11.4.95. Non-compliance is a release blocker.

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate (User mandate, 2026-05-27)

Operational catalogue for 5-7h AOSP containerised build:

SAFE: (A) docs/MD work; (B) host-side scripts/; (C) pre-build + meta-test gate authoring; (D) on-device test scripts authoring; (E) constitution edits + push; (F) any submodule commit + push per §11.4.88; (G) read-only live-ADB probes; (H) subagent dispatch per §11.4.20/§11.4.70 + §11.4.84; (I) web research + APIs; (J) workable-items DB ops; (K) pre-build + meta-test execution backgrounded per §11.4.89.

UNSAFE: (α) git checkout/reset on source tree; (β) mass deletes/renames under built source; (γ) submodule pointer updates affecting built APKs; (δ) out/ mutations; (ε) make clean / m clobber; (ζ) container destruction; (η) disk-filling; (θ) §12 host-safety breaches.

Conductor MUST dispatch ≥ 1 (A)-(K) item per pause point. "Build running, nothing else to do" NEVER true. Pre-build gates `CM-COVENANT-114-96-PROPAGATION` + `CM-PARALLEL-WORK-DURING-BUILD-AUDIT` .

Canonical authority: constitution submodule `Constitution.md` §11.4.96. Non-compliance is a release blocker.

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence (User mandate, 2026-05-27)

Operating-mode capstone: (A) every minute of progressable idle time = §11.4.97 violation; dispatch CONTINUOUSLY; (B) emit 1-line operator update at every commit/subagent-return/anchor-landed/evidence-captured/migration-closure — no prompt required; (C) continuous physical-proof gathering per §11.4.5+§11.4.6+§11.4.69; (D) composes with §11.4.5/6/13/20/27/42/50/52/69/70/72/83/85/87/88/89/94/96; (E) idle-only-when-blocked closed-set unchanged. Pre-build gates `CM-COVENANT-114-97-PROPAGATION` + `CM-IDLE-TIME-AUDIT` .

Canonical authority: constitution submodule `Constitution.md` §11.4.97. Non-compliance is a release blocker.

§11.4.98 — Full-Automation Anti-Bluff Mandate — Live tests MUST be re-runnable end-to-end without manual intervention (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28): "Make sure we have full automation testing of all scenarios with real bot, main group and users without any manual intervention or contribution of real user! ... No bluff is allowed!"

Closes the manual-intervention gap §11.4 + §11.4.85 + §11.4.87 + §11.4.89 + §11.4.94 did not explicitly forbid: a test requiring human action during execution is a §11.4 PASS-bluff at the automation layer. (A) Binding rule: every test MUST be self-driving end-to-end; PASS/FAIL/SKIP-with-reason without further human action after startup. (B) Single exception: one-time credential bootstrap OUTSIDE test execution (`.env` , `~/ .bashrc` , OAuth, MTPROTO-session-activation) — configuration, not test driving. (C) Concrete live-messenger requirements: (1) no "operator MUST type" prompts — drive programmatically (MTPROTO/real-user-API/webhook-fixture/in-process loopback); (2) no UUIDs colliding with active dev session (Herald 2026-05-28 lesson: same-UUID `claude --resume` returns silent exit -1); (3) no 60s human-response windows (§11.4.50 determinism violation); (4) re-runnability proof at `-count=3` consecutive automated runs with self-cleaning state; (5) §11.4.98 audit — every existing test classified COMPLIANT vs NON-COMPLIANT; (6) no false-positive PASS — silent-skip-as-PASS forbidden, stale-evidence forbidden, SKIP-with-reason per §11.4.3. (D) Composes with §11.4.85+89+87+94 = continuously-validated fully-automated non-flake anti-bluff regime. (E) Inheritance per §11.4.35 — restate literal anchor `11.4.98` in every consumer's CLAUDE/AGENTS/QWEN; pre-build gate `CM-COVENANT-114-98-PROPAGATION` ; paired §1.1 mutations strip → gates FAIL. (F) Enforcement: manual-action commit BLOCKED at release-gate; NON-COMPLIANT test not rewritten in 30 days → §11.4.90 Obsolete citing §11.4.98.

Canonical authority: constitution submodule `Constitution.md` §11.4.98. Non-compliance is a release blocker.

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate — instructions/guides/manuals MUST be verified against latest official online sources BEFORE publication (User mandate, 2026-05-28)

Forensic anchor: "Make sure we ALWAYS check against latest versions of services we use web/online docs before creating instructions! This situation is illustration of how we can misguide ourselves or get banned!"

Case study (Herald 2026-05-28): first-draft MTPROTO guide recommended VoIP + omitted `recover@telegram.org` pre-login email; both contradicted official Telegram + gotd/td docs and could have caused permanent account ban. Misguidance-by-stale-docs = §11.4 PASS-bluff at documentation layer.

(A) Pre-commit cross-reference: agent MUST fetch LATEST official online docs (WebFetch/MCP/direct browsing — NEVER training data) for every operator-facing instruction doc; cross-reference each instruction; seek secondary authoritative sources (maintainer SUPPORT.md, changelogs, vetted FAQs) when official is sparse; cite source URL + date in "## Sources verified" footer; cite in commit-message footer (`Sources verified <date>: <urls>`). (B) Negative findings documented (gaps/silences = no authoritative agreement). (C) Re-verification cadence — STALE after 6 months (90 days for risk-classified services); re-verify before operator-authority citation, at vN.0.0 release boundary, on breaking-change announcements, on operator-error report. (D) Risk-classified services (messaging/cloud/payment/AI/code-hosting/package-managers) — 90-day staleness limit + explicit safety warnings vs latest policies. (E) Composes with §11.4.92 Pass 4 but INDEPENDENT — cannot substitute. (F) Inheritance per §11.4.35 — restate literal `11.4.99` in every consumer; pre-build gate `CM-COVENANT-114-99-PROPAGATION` . (G) Enforcement: missing footer → BLOCKED at release-gate; stale-beyond-grace → §11.4.90 Obsolete `Reason=stale-documentation` .

Canonical authority: constitution submodule `Constitution.md` §11.4.99. Non-compliance is a release blocker.

§11.4.100 — RETIRED. Demoted to consumer project (a consuming project's video-color/visual-quality fidelity) per §11.4.17/§11.4.35 — project-specific (RK3588/MPV/Arvus), not universal. See the consuming project's Constitution/CLAUDE/AGENTS/QWEN.

§11.4.101 — Autonomous-decision-over-blocking mandate (User mandate, 2026-05-28)

Forensic anchor — verbatim user mandate (2026-05-28): "when working in endless working loop fully autonomously try to decide most properly about points which would block execution and wait for us. If we haven't answered now work would be blocked whole night! If possible and if that will not cause any issues make proper and most reliable and safe decision so we achieve maximal efficiency and work gets fully done!"

When operating in autonomous / endless-loop mode (per §11.4.87), the agent MUST minimize operator-blocking and instead make the safe, reliable, reversible decision itself — so work is NOT stalled (e.g. overnight) waiting for input. §11.4.87 says keep working; §11.4.101 says HOW to clear the decision points.

Decision rule (closed-set — proceed autonomously when ALL hold): (a) action reversible OR captured pre-op backup per §9.2; (b) safe choice determinable from captured evidence per §11.4.6 (no guessing — `LIKELY` is not a determination); (c) wrong-choice blast radius bounded AND recoverable; (d) composes with anti-bluff §11.4 + host-safety §12 + data-safety §9.

Block-only-when rule (BLOCK via §11.4.66 ONLY when ALL hold): action irreversible AND high-blast-radius AND safe choice undeterminable from evidence — e.g. external-account state the agent cannot inspect, hardware it cannot access, destructive ops without backup, force-push (also §9.2 + §11.4.41), spending / sending to third parties. `operator-blocked` per §11.4.21 reached only after this rule fires AND self-resolution-exhaustion audit completes.

Maximize-progress-while-blocked: an unavoidable block parks one work unit, not the loop — keep progressing every NON-blocked item in parallel per §11.4.87 + §11.4.94. Posing the question and going idle is a §11.4.94 + §11.4.97 violation.

Composes with §11.4.6 / §11.4.21 / §11.4.40 / §11.4.41 / §11.4.66 / §11.4.87 / §11.4.94 / §9.2 / §12. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-101-PROPAGATION` (literal `11.4.101` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--always-block-on-decision`, `--never-decide-autonomously`, `--skip-decision-rule`, `--block-without-self-resolution` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.101. Non-compliance is a release blocker.

§11.4.102 — Mandatory systematic-debugging activation + always-loaded skill-discovery + plugin-dependency availability (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29): "Make sure that we ALWAYS trigger / start the `"/superpowers:systematic-debugging"` skills when any issues happen! If this is possible to activate and use in this situations out of the box when we spot problems / issues / bugs / misalignments / inconsistencies we MUST activate the skill(s) and make strongest efforts in full in depth analysis / debugging and determine root causes of all problem or obtain relevant data and information we need! ... we MUST make sure that `"/using-superpowers"` skill is ALWAYS loaded, applied and used! All dependencies (plugins) that Claude Code or other market places are offering MUST BE installed if these are not already available for loading and use!"

Three cooperating invariants — the difference between guess-and-retry and investigate-to-root-cause-first.

(A) Mandatory systematic-debugging activation. On ANY spotted issue / bug / test failure / gate failure / regression / misalignment / inconsistency / unexpected behaviour, the agent MUST activate `superpowers:systematic-debugging` (or platform-equivalent) BEFORE proposing/writing/applying any fix — Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST. Full four-phase arc: root-cause (read real failure, reproduce, gather facts) → pattern (classify defect, search same pattern elsewhere) → hypothesis (falsifiable cause, prove/disprove with captured evidence) → implementation (fix only against proven root cause). Guess-and-retry, symptom-patching, re-running a failed test hoping it passes ("probably transient/flaky") WITHOUT a completed investigation are §11.4.102 violations. Real-world signal: calling a failure `transient / flaky / intermittent / probably-timing` without captured forensic evidence is simultaneously a §11.4.6 (no-guessing) + §11.4.7 (demotion-evidence) violation — §11.4.102 makes the corrective response mechanical.

(B) Mandatory always-loaded using-superpowers . `superpowers:using-superpowers` (or platform-equivalent skill-discovery discipline) MUST be loaded + applied at session start and consulted before any task: survey available skills before acting; if ANY skill could apply — even at 1% relevance — it MUST be invoked rather than improvised. Skipping skill-discovery, or improvising a task a loaded skill exists for, is a §11.4.102 violation.

(C) Mandatory plugin / dependency availability. Every skill plugin / marketplace package / capability dependency the project relies on (Claude Code, its marketplaces, or any other runtime's plugin ecosystem) MUST be installed + loadable BEFORE dependent work proceeds. A missing plugin blocking a mandated skill is a release-blocker until installed + confirmed loadable. Install mechanism: the runtime's own plugin/marketplace path — for Claude Code the in-session `/plugin` flow (add marketplace → install plugin → confirm skill appears in available-skills list); for other runtimes the documented package/extension installer. Anti-bluff: confirm by observing the skill in the live capability list (install exit 0 ≠ skill loadable, per §11.4.80 lesson).

Composes with §11.4.4 / §11.4.6 / §11.4.7 / §11.4.8 / §11.4.43 / §11.4.70 / §11.4.82(A) / §11.4.92. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-102-PROPAGATION` (literal `11.4.102` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--skip-systematic-debugging` , `--guess-and-retry-OK` , `--symptom-patch-permitted` , `--skip-skill-discovery` , `--plugin-optional` , `--missing-plugin-is-warning` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.102. Non-compliance is a release blocker.

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Forensic anchor — verbatim user mandate (2026-05-29): "Do this working approach continuously and make it part of regular working routine and add it to the Constitution Submodule documented fully."

Promotes the proven multi-stream operating pattern into the project's standing default working routine (not a per-request opt-in). Adds the load-bearing invariant: the main work stream MUST always stay FREE. (A) Main stream FREE — ALL

commit AND push run detached (`nohup ... & + disown` per §11.4.88), never blocking on a push or slow mirror. (B) ≥ 3 parallel background streams at all times + auto-backfill (User mandate 2026-05-29; raised from ≥ 2) — run at least three subagent-driven background streams (per §11.4.70/§11.4.20, isolated per §11.4.58 + §11.4.84) alongside the main stream whenever three-plus non-contending actionable items exist; the moment any one stream is FULLY done, a new stream MUST immediately start and take its place (claim next-highest-priority non-contending item), so the active-stream count NEVER drops below 3 while actionable items remain. Idle below 3 only when no remaining non-contending actionable items OR all externally blocked (§11.4.94/§11.4.97/§11.4.101). Standing band 3–6, bounded above by §12.6 60% memory + §11.4.58 6-agent cap. (C) Most-critical + most-visible first; audio always top per §11.4.72 + §11.4.42. (D) Safe-during-build scope only — while a heavy `gradle / m -j /42` GB containerised AOSP build (§12.9) runs, streams restrict to the §11.4.96 SAFE catalogue; NEVER a second concurrent heavy build (§12.8). (E) Heavy anti-bluff on every closure (§11.4.5/§11.4.6/§11.4.50/§11.4.69/§11.4.102). (F) Idle ONLY when genuinely externally blocked (hardware/network/in-flight build-test-push) OR operator STOP OR §12 host-safety per §11.4.94(A)+§11.4.97; a block parks one work unit, not the loop (§11.4.101).

Composes with §11.4.58 / §11.4.70 / §11.4.72 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.94 / §11.4.96 / §11.4.97 / §11.4.101 / §11.4.102 / §11.4.42 / §11.4.84 / §12.6-§12.9 / §9.2. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-103-PROPAGATION` (literal `11.4.103` across consumer fleet) + paired §1.1 meta-test mutation (strip literal $\rightarrow$ gate FAILs; gate-code = separate work item). No escape hatch — no `--block-main-stream`, `--synchronous-commit`, `--synchronous-push`, `--single-stream-only`, `--skip-parallel-streams`, `--serialise-actionable-work`, `--idle-without-queue-survey` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.103. Non-compliance is a release blocker.

§11.4.104 — Participant identity, attribution & notification-tagging (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31): "Every supported messenger must relate messages to participants (Subscribers/Users); the same logical person may have a different username on every messenger. Workable items must carry who created them and who they are assigned to. Notifications must @-tag the right participant — but never the operator (who drives the system) and never the system agent."

MANDATORY for every consumer that ships a messenger/notification surface (Herald + flavor binaries = reference impl; others inherit per §11.4.35). Detailed spec: Herald `docs/design/PARTICIPANT_ATTRIBUTION.md` — restated, not redefined. (A) Participant identity — every messenger MUST relate messages to a **Participant** (logical Subscriber/User); the SAME person MAY have a DIFFERENT username per messenger (logical subscriber: canonical messenger-neutral `handle`, `kind` ∈ {human, agent, service} ; + per-channel aliases `channel / channel_user_id` / the `@username` used for tagging). Canonical handle closed set: `Claude` (reserved system-agent sentinel; never tagged) OR a subscriber `@username`. (B) Operator = env var, not a DB flag — the one human who drives via the agent CLI, designated by `HERALD_<CHANNEL>_OPERATOR_USERNAME` ; a normal Participant whose handle equals that value. (C) Workable items MUST carry `created_by` + `assigned_to` (canonical handles): CLI-prompt → Operator; system/agent-detected → `Claude` ; received-message → sender's resolved `@username` ; `assigned_to` defaults to Operator, overridable; legacy items carry `""` and MUST still parse + validate. (D) Tagging matrix: tag `assigned_to` if human ≠ Operator; tag `created_by` if human ≠ Operator ≠ `Claude` ; NEVER tag `Claude` or the Operator; de-dup; resolve to the channel `@username`, skip if not on that channel. (E) Anti-bluff (composes §11.4): real SQLite round-trip with the new columns byte-identical (incl. legacy fixtures WITHOUT the fields); tagging matrix proven by a truth-table + a cell-flip mutation forcing FAIL; E2E real-event → real-message asserting the exact `@username` s + a NEGATIVE case proving the Operator is NOT tagged; evidence under `docs/qa/<run-id>/`.

Composes with §11.4 + §11.4.1..§11.4.16 / §11.4.5 / §11.4.69 / §11.4.50 / §11.4.91 / §11.4.93 / §11.4.95 / §1.1. Classification: universal (§11.4.17); projects with no messenger surface inherit it latently (binds the moment they ship one, §11.4.96 pattern). Propagation gate `CM-COVENANT-114-104-PROPAGATION` (literal

11.4.104 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--skip-attribution` , `--no-participant-tagging` , `--tag-operator-anyway` , `--attribution-later` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.104; detailed spec Herald `docs/design/PARTICIPANT_ATTRIBUTION.md` . Non-compliance is a release blocker.

§11.4.105 — Natural-language intent recognition & clarification (User mandate, 2026-05-31)

Forensic anchor — verbatim user mandate (2026-05-31): "Users must NOT need to know command syntax (no `COMMAND: ...` prefix). They send a clear natural-language message; the System determines the intent. The System recognizes the commands it has; if none matches it infers the exact intent; if it is totally unable it replies, tags the user (`@user ...`), and asks to clarify precisely. We MUST always do our best to determine exact intent so we never annoy end users. This is a CORE part of the System."

MANDATORY for every consumer that ships a messenger/command surface (Herald + flavor binaries = reference impl; others inherit per §11.4.35). Detailed spec: Herald `docs/design/INTENT_RECOGNITION.md` — restated, not redefined. (A) No required command syntax — users MUST NOT need to know any syntax (no `COMMAND: prefix`); they send plain natural language and the System determines the intent. (B) Three-tier resolution, first that succeeds wins: TIER 1 recognize the System's existing command set from natural language → action (confident deterministic match); TIER 2 when no command matches, infer the exact intent (LLM maps language → action), NEVER guessing; TIER 3 when neither a command nor a confident intent can be determined, REPLY to the message, TAG the sender (`@username` , via the §11.4.104 IdentityResolver) and ask a PRECISE clarifying question NAMING the candidate intents — no guessing, no silent drop. (C) Never guess, never drop — a wrong action is worse than a clarifying question (composes §11.4.6); a message is NEVER silently dropped; only genuine ambiguity reaches Tier 3, which always replies-tags-and-asks. (D) Anti-bluff (composes §11.4): every tier ships unit + integration + E2E + full-automation tests with real captured evidence — Tier 1 truth-table (natural-language → action+fields + conservative negatives that MUST fall through to "no match"); Tier 3 E2E whose

recorded reply body is EXACTLY `@<sender> <specific question> + a` NEGATIVE proving a clear command does NOT trigger clarify; a paired §1.1 mutation breaking the confidence guard (false-match) OR dropping the clarify tag MUST FAIL a test; evidence under `docs/qa/<run-id>/` .

Composes with §11.4 + §11.4.1..§11.4.16 / §11.4.6 / §11.4.104 / §11.4.5 / §11.4.69 / §11.4.98 / §1.1. Classification: universal (§11.4.17); projects with no messenger surface inherit it latently (binds the moment they ship one, §11.4.96 pattern).

Propagation gate `CM-COVENANT-114-105-PROPAGATION` (literal `11.4.105` across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no `--require-command-syntax` , `--guess-intent-ok` , `--skip-clarify` , `--drop-on-ambiguous` flag.

Canonical authority: constitution submodule `Constitution.md` §11.4.105; detailed spec Herald `docs/design/INTENT_RECOGNITION.md` . Non-compliance is a release blocker.

§11.4.106 — Docs Chain — mechanical documentation/DB sync engine (Operator mandate, 2026-05-31)

Forensic anchor — operator mandate (2026-05-31): Docs Chain is the canonical mechanical enforcer of the documentation-sync mandates; consumers MUST use the engine instead of ad-hoc per-project doc-sync scripts, register their chains via per-context YAML, and never accept a faked transform.

MANDATORY for every consumer. Docs Chain (the `vasic-digital/docs_chain` engine — a universal Go bidirectional document-and-database dependency-propagation engine) is the canonical mechanical enforcer of the documentation-sync mandates. Detailed spec: the engine docs `~/Projects/docs_chain` → `docs/CONSTITUTION_INTEGRATION.md` (distribution + inheritance + anchor mapping table) + `docs/USE_CASE_CATALOGUE.md` (chain recipes) — restated, not redefined. (A) Use the engine, never ad-hoc scripts — consumed by reference (the flat-layout sibling `~/Projects/docs_chain` / the constitution-exposed path), inherited like §11.4.80's `codegraph_*` scripts: referenced, NEVER copied; ad-hoc `sync_*` / `generate_*` `summary` / `update_readme_doc_links` scripts are superseded and retired per registered context. (B) Consumer-owned contexts — the engine is project-agnostic; the consumer registers its chains as data via `.docs_chain/contexts/*.yaml` (§11.4.28 decoupling); `state.json` +

*.docs_chain.tmp are gitignored. (C) Anchors it mechanizes (per the CONSTITUTION_INTEGRATION mapping table): §11.4.12 / §11.4.53 / §11.4.45 / §11.4.56 / §11.4.57 / §11.4.59 / §11.4.60 / §11.4.65 / §11.4.86 / §11.4.93 / §11.4.95 / §12.10 / §11.4.44 — with content-hash change detection (NOT mtime, §11.4.86), atomic-rename + SQLite-txn commit + rollback (§9.2), both-dirty sync → conflict-not-silent-merge (§11.4.6), verify as the deterministic CI/pre-build gate (§11.4.50), per-run captured evidence to qa-results/docs_chain/<run-id>/ (§11.4.69). (D) NOT a replacement for authoring discipline — the source author still writes the §11.4.44 revision header; the engine only keeps exports in sync. (E) Anti-bluff (composes §11.4): the engine NEVER fakes a transform — a missing pandoc/weasyprint surfaces a typed ToolAbsentError + honest §11.4.3 SKIP-with-reason, never a fake PASS / partial write; every sync / verify carries real captured evidence.

Composes with §11.4 + §11.4.1..§11.4.16 / §11.4.6 / §11.4.28 / §11.4.80 / §9.2 / §11.4.50 / §11.4.69 / §11.4.5 / §1.1, plus the sync anchors it mechanizes (§11.4.12/.53/.45/.56/.57/.59/.60/.65/.86/.93/.95/.44, §12.10). Classification: universal (§11.4.17); projects with no derived-export/DB-sync surface inherit it latently (binds the moment they ship one, §11.4.96 pattern). Status (§11.4.6): engine Phases 1–3 AND the CLI/YAML loader (Phase 4) IMPLEMENTED+tested — cmd/docs_chain/main.go registers sync / verify / diff (alias)/ doctor , go test -count=1 ./... GREEN 7/7 packages; this CLI/loader is in the working tree but UNTRACKED and the engine is NOT yet a registered git submodule (in-tree at ./docs_chain , HEAD = parent HEAD); only submodule distribution (Phase 6) remains PLANNED + OPERATOR-GATED — wire as phases land, claim no unshipped behaviour. Propagation gate CM-COVENANT-114-106-PROPAGATION (literal 11.4.106 across consumer fleet) + paired §1.1 meta-test mutation (strip literal → gate FAILs; gate-code = separate work item). No escape hatch — no --ad-hoc-sync-ok , --skip-docs-chain , --fake-transform , --sync-evidence-optional flag.

Canonical authority: constitution submodule Constitution.md §11.4.106; detailed spec the Docs Chain engine docs docs/CONSTITUTION_INTEGRATION.md + docs/USE_CASE_CATALOGUE.md (vasic-digital/docs_chain). Non-compliance is a release blocker.

§11.4.107 — Anti-bluff AV/test-validation techniques mandate (User-driven research, 2026-06-02)

Forensic (genericised, 2026-06-02): a test PASSEd on a SINGLE captured frame showing "a picture" — but it was a FROZEN/STALE frame from previously-played content (stale-producer/stuck-decoder), feature broken for the user while green; a sibling FLASHED media on the WRONG output for ~1 s before routing with no test sampling that window; a third shipped a comparator that PASSEd its own deliberately-degraded fixture (the analyzer, not the feature, was the bluff). §11.4.5 mandates captured evidence + a presence pass; §11.4.107 raises it to liveness + correct-routing + self-validated-analyzer.

Every test asserting audio/video output is genuinely playing/advancing MUST satisfy ALL of: (1) a single captured frame is NOT proof — prove LIVE, ADVANCING frames over a window via a freeze-detection oracle (near-duplicate/ `freezedetect` -class OR perceptual-hash adjacent-frame distance, NOT byte-identical — byte-identity is only a pre-filter); (2) an independent frame-advance counter from the platform's compositor/decoder telemetry must increase (a different-domain oracle — flat $\Rightarrow$ stuck decoder $\Rightarrow$ FAIL even if pixels appear to move); (3) loading/buffering is a distinct state — wait for genuine playback-start, never false-FAIL a still-loading stream nor false-PASS a spinner (timeout+unreachable $\Rightarrow$ SKIP per §11.4.3, timeout+reachable $\Rightarrow$ FAIL); (4) not-stale-from-previous cross-check (new first frame $\neq$ previous last frame); (5) measured FPS / no-lost-frames within tolerance; (6) no-flash-on-the-wrong-output — high-frequency sampling of the non-target output during routing transitions, any content frame there $\Rightarrow$ FAIL (content-protection regime classified, never guessed); (7) drive through the realistic feed/UI path, not deep-link shortcuts (UI-not-introspectable $\Rightarrow$ operator-attended per §11.4.52, never fake-PASS); (8) metamorphic relations solve the oracle problem when there is no golden source (same content output-A vs output-B must match; paused $\Rightarrow$ counter stops; 2 $\times$ speed $\Rightarrow$ ~2 $\times$ advance); (9) full-reference quality metrics (SSIM/VMAF/ ΔE_{2000}) vs a golden source for owned content; (10) mutation-test every analyzer with a golden-good + golden-bad fixture pair so the analyzer itself cannot bluff (PASS on the golden-bad fixture = bluff gate — §1.1 applied to analyzers); (11) per-channel audio RMS/loudness (EBU R128) + XRUN/underrun census (a single aggregate RMS misses a dead channel); (12) OCR overlay/subtitle detection needs a per-word confidence floor + ROI (avoids both false-FAIL and false-PASS); (13) thresholds

calibrated on the project's own fixtures, not hardcoded from literature (§11.4.6). Honest gap: true photon FPS at the sink has no clean software oracle — flag it, never claim it.

4-layer per §11.4.4(b): pre-build gate (`CM-AV-LIVENESS-NO-FROZEN-FRAME` -class — every output-is-playing test references the liveness battery not a single frame + analyzer-self-validation gate wiring golden-good/golden-bad fixtures into meta-test) + runtime/on-device test + paired §1.1 mutation (single-frame-only → gate FAILs; analyzer PASSing its golden-bad fixture → self-validation FAILs) + HelixQA Challenge. Every PASS via `ab_pass_with_evidence` citing motion / not-stale / frame-advance / fps / per-channel-loudness / metamorphic / self-validation artefacts (`video_display` / `audio_output` / `subtitle_render` per §11.4.69) — never a single screenshot. Classification: universal (§11.4.17) — platform-neutral, reusable by ANY project validating media playback or any pixel/audio output; the project supplies its capture mechanism + calibrated thresholds per §11.4.35. Composes §11.4.5/.6/.50/.68/.69/.85 + §11.4.2/.3/.13/.48/.52/§1.1. Propagation gate `CM-COVENANT-114-107-PROPAGATION` (literal `11.4.107`) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.107. Non-compliance is a release blocker. No escape hatch — no `--single-frame-proves-playback` , `--skip-liveness` , `--byte-identical-freeze-OK` , `--no-frame-advance-counter` , `--allow-wrong-output-flash` , `--deep-link-shortcut-OK` , `--unvalidated-analyzer-OK` , `--aggregate-rms-suffices` , `--hardcoded-thresholds-OK` flag.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate (systematic-debugging Phase 4.5, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): across one batch multiple fixes were "green" at every gate yet NOT working for the end user — a source-committed change passed by both the source pre-build gate AND the post-build gate never reached the boot-time command-line embedded in the deployed boot artifact (feature dead despite all-green); sibling fixes correctly built into the system image were masked by stale per-user overlay copies from a prior deployment (running code = stale shadow, not the fresh build); deployment did not wipe the mutable overlay so the shadow survived; validation ran against the stale shadow and

reported green on code never exercised. Per `superpowers:systematic-debugging` Phase 4.5: each fix revealing a fresh "fixed-but-not-working" in a DIFFERENT place is ONE architectural VERIFICATION flaw, not independent bugs — patching symptoms is thrashing. A fix crosses FOUR distinct layers, and "fixed" at one does NOT imply fixed at the next: (1) SOURCE (committed — grep-the-source gate), (2) ARTIFACT (the change's BYTES landed in the produced artifact — image / bundle / installer / embedded command-line), (3) RUNTIME-ON-CLEAN-TARGET (active on a CLEAN/fresh deployment, the layer the end user experiences, no stale overlay shadowing it), (4) USER-VISIBLE (works for the end user — §11.4.5/§11.4.69 captured-evidence). The mandate (ALL hold): (1) **runtime-signature-as-definition-of-done** — a fix is DONE only when its declared runtime signature verifies on a CLEAN deployment; source-committed ≠ artifact-contains-it ≠ active-on-clean-target ≠ user-visible-working; (2) every fix declares ONE machine-checkable runtime signature (running-system property / sink-side report per §11.4.13 / §11.4.5/§11.4.69 captured-evidence / counter delta — NEVER a re-grep of the source); this registry of per-fix runtime signatures is the SINGLE SOURCE OF TRUTH for "fixed", REPLACING "a gate greps the source"; (3) gates span all four layers — source (pre-build), artifact (post-build — assert the BYTES landed, not merely that the source still contains the change), runtime-on-clean-target (post-deploy — assert the runtime signature), user-visible (§11.4.5/§11.4.69); (4) eliminate the stale-deployment / shadow layer by construction — deployment MUST yield a CLEAN state (wipe the mutable overlay) OR a pre-validation assertion MUST prove `running-artifact == built-artifact` before any validation; validation against possibly-stale state is INVALID and its PASS is a §11.4 PASS-bluff; (5) meta-rule — ≥3 "fixed-but-not-working" discoveries in one cycle signal an architectural VERIFICATION flaw, not three independent bugs; on the 3rd STOP patching symptoms (§11.4.4), fix the VERIFICATION pipeline, re-certify EVERY item through it on a clean target; (6) a batch is "validated" only after COMPREHENSIVE per-item runtime-signature verification on a clean baseline, NOT after the touched items' own tests pass. Classification: universal (§11.4.17) — platform-neutral verification-pipeline disciplines reusable by ANY project that builds → deploys → validates an artifact; the project supplies its artifact format, clean-deployment / equal-artifact mechanism, and per-fix runtime-signature observables per §11.4.35. Composes §11.4.1/.2/.4/.5/.6/.27/.40/.46/.50/.52/.69/.102. Propagation gate `CM-COVENANT-114-108-PROPAGATION` (literal `11.4.108`) + recommended per-fix gate `CM-RUNTIME-SIGNATURE-REGISTRY` + paired §1.1 mutations (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.108. Non-compliance is a release blocker. No escape hatch — no

`--source-green-is-done` , `--skip-artifact-byte-check` , `--validate-against-running-state` , `--no-clean-deployment` , `--skip-runtime-signature` , `--spot-validate-touched-only` flag.

§11.4.109 — Mandatory Anti-Forgetting Enforcement: PreToolUse Guard Hook + Subagent Constitutional Preamble + Orchestrator Pre-Action Checklist (Operator mandate)

UNCONFIRMED forensic anchor (pending operator's verbatim mandate quote).

Background: emulator subagents ran raw host-direct `emulator / adb` because the orchestrator forgot to inject the Containers-submodule rule at dispatch time. A rule forgotten at dispatch is not enforcement. Two-layer fix: (A) `constitution/scripts/hooks/guard-forbidden-commands.sh` — a portable bash/awk

`PreToolUse` hook (no `jq` dependency, no project-specific paths) that blocks host-direct `emulator / force-push bypass / sudo / host-power` at the tool-call boundary regardless of agent memory (hook = the floor); (B) `constitution/docs/AGENT_GUARDRAILS.md` — canonical preamble the orchestrator pastes verbatim into every subagent dispatch + Orchestrator Pre-Action Checklist (preamble = the ceiling). Consuming projects wire the hook in `.claude/settings.json` (or equivalent) and maintain a `docs/AGENT_GUARDRAILS.md` containing the `SUBAGENT CONSTITUTIONAL PREAMBLE` , `ORCHESTRATOR PRE-ACTION CHECKLIST` , and anchor literal `11.4.109` . The hook is inherited by reference — NEVER copied locally.

Gates: `CM-ANTI-FORGETTING-ENFORCEMENT` (hook + doc + test) + `CM-COVENANT-114-109-PROPAGATION` (literal `11.4.109` across consumer fleet).

Paired §1.1 mutations (gate-code = separate work item). Classification: universal (§11.4.17). Composes §11.4.6/.10/.75/.76/.78/.79/.80/.81/.84/.98/.102/§12.

Canonical authority: `Constitution.md` §11.4.109; ref impl `constitution/scripts/hooks/guard-forbidden-commands.sh` + `constitution/docs/AGENT_GUARDRAILS.md` . Release blocker. No `--skip-pretooluse-hook` / `--no-guardrails-doc` / `--anti-forgetting-optional` / `--single-layer-sufficient` escape.

§11.4.110 — Pre-build build-readiness verdict + change-impact clash detection mandate (operator mandate, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a fix shipped a new system-property *read* but introduced no matching security-policy grant + no property-context type entry — the read was silently denied at runtime + the feature was dead, while every pre-build gate stayed green because the gate only grepped the source file.

Generalises: a change introduces a new dependency on a *second* artifact (security policy, context file, service registry, interface freeze-snapshot, symbol table, build-graph node) that the pre-build never cross-checks. This is §11.4.108's

SOURCE → ARTIFACT gap shifted LEFT to pre-build — most such clashes are statically catchable from the change diff itself. The mandate (ALL hold): (1) a single deterministic **READY-FOR-BUILD verdict** gates the rebuild (orchestrator refuses to start unless READY); (2) a **diff-driven change-impact + clash detector** cross-checks every newly-introduced second-artifact dependency (new property read ⇔ property-context type + read-grant; new service ⇔ service-context; new init service ⇔ security label; new/changed stable interface ⇔ freeze-snapshot updated; new native-lib dep ⇔ module/prebuilt resolves; new policy rule ⇔ every type/attribute defined; two batch changes on the same seam ⇔ collision acknowledged); (3) **coverage-completeness is a gate** — every changed file maps to ≥1 gate + ≥1 deployed-target test + ≥1 paired §1.1 mutation, baseline ratchets per §11.4.50; (4) **two-speed honesty** — grep-speed always-on gates vs REQUIRES_BUILD heavy gates (build-graph parse-only dry-run, full neverallow compilation, ABI diff) as diff-gated opt-in stages bounded per §12.6/§12.7; (5) every gate + wired analyzer is anti-bluff by paired §1.1 mutation; (6) **honest boundary** — a READY verdict proves static internal-consistency + ready-to-build, NOT that the feature works on the deployed target; the regime empties the *preventable* class, not the *all-defects* class (runtime/USER-VISIBLE remains §11.4.108's job). Classification: universal (§11.4.17) — platform-neutral pre-build-rigor disciplines reusable by ANY artifact-building project; the project supplies its property/service/policy registries, build-graph parse-only command, interface-freeze mechanism + changed-file → gate mapping per §11.4.35. Composes §11.4.1/.4/.6/.9/.27/.50/.67/.75/.92/.108.

Propagation gate CM-COVENANT-114-110-PROPAGATION (literal 11.4.110) + recommended CM-READY-FOR-BUILD-VERDICT / CM-CHANGE-IMPACT-CLASH-DETECTOR / CM-COVERAGE-COMPLETENESS-GATE /

CM-BUILDGRAPH-DRYRUN-WIRED / CM-SEPOLICY-NEVERALLOW-WIRED + paired §1.1 mutations (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.110. Release blocker. No --source-green-is-ready / --skip-clash-detector / --skip-coverage-gate / --no-ready-verdict / --grep-proves-neverallow / --skip-buildgraph-dryrun / --build-without-ready-verdict escape.

§11.4.111 — Resolve-by-stable-name-not-by-enumeration-index mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): a platform bound an audio output to a kernel-enumerated index (`card=0`); a second device of a different class enumerated FIRST at boot + took slot 0, shifting the intended device to slot 1 — the static index binding now pointed at the wrong device (policy mis-attached the sink, mis-assigned its TYPE, the output switcher labelled the AV receiver "Wired Headphone" + routing collapsed to stereo). The lower layer of the SAME stack already resolved by **name** (controller-name registry) — proving the brittleness was the *index*, not the resolution capability. Generalises to every enumerated resource whose ordinal is assigned at discovery/boot/hotplug (non-deterministic across reboots, additions, topology changes). The mandate: any binding to a hardware device / resource handle / enumerated entity (audio cards, display connectors, network interfaces, storage devices, GPU render nodes, input/camera devices, container/process slots) MUST resolve by a **stable identifier** (name / UUID / serial / label / controller-name / content-hash / sink-reported identity), NOT by enumeration index / ordinal / slot, UNLESS the platform documents the ordinal as deterministically pinned AND the pin is captured + asserted in the binding. Where a stable identifier exists at one layer, every layer binding the same resource MUST use it (mixed by-name-here / by-index-there is the forbidden weak link). Honest boundary (§11.4.6): when only an ordinal exists, pin deterministically via the platform's mechanism, capture the pin in the §11.4.108 runtime signature, document residual fragility as UNCONFIRMED: -class — never silently trust an unpinned ordinal. Classification: universal (§11.4.17) — platform-neutral binding-robustness discipline; the project supplies its stable-identifier mechanism (ALSA card name, DRM connector name, NIC predictable name, block-device UUID, etc.) + the layers that must agree per §11.4.35. Composes §11.4.6/.8/.69/.108/.110. Propagation gate CM-COVENANT-114-111-PROPAGATION (literal 11.4.111) + recommended CM-RESOLVE-BY-NAME-NOT-INDEX + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.111. Release blocker. No --allow-index-binding / --ordinal-is-stable-enough / --skip-name-resolution / --trust-unpinned-index escape.

§11.4.112 — Structural-impossibility won't-fix classification mandate (research-derived, 2026-06-03)

Forensic anchor (genericised, 2026-06-03): deep research (§11.4.8) into relocating protected (content-protection / secure-surface) video to a secondary display PROVED — from authoritative platform/HDCP docs + reproducible captured behaviour (a secure surface is blanked on any output lacking the secure flag; mirror/screenshot of a secure layer returns black) — that the goal is **structurally impossible by platform design**, not a missing feature. Without a durable classification it is re-investigated every cycle (re-read the same sources, re-run the same probes, re-derive the same impossibility — compounding wasted effort). The mandate: when deep research per §11.4.8 PROVES (cited authoritative sources AND, where applicable, reproducible captured evidence) a goal is structurally impossible on the target platform (forbidden by platform design / hardware-protocol constraint / documented kernel-or-API limitation — NOT merely unimplemented or hard), the goal MUST be: (1) classified `won't-fix` + closed per §11.4.90 with reason `structurally-impossible` ; (2) documented with the impossibility evidence — cited source URLs (per §11.4.99) + reproducible probe/captured evidence — in the tracker entry + relevant `docs/` guide; (3) NOT re-attempted in future cycles — a reopen MUST cite NEW evidence the constraint changed (per §11.4.34 + §11.4.7), never re-derive the same impossibility; (4) paired with the correct posture — the entry states what the project DOES instead, so "impossible" is never confused with "broken/unhandled". Honest boundary (§11.4.6): reserved for *proven* impossibility — "could not find a way" / "very hard" / "no time" are `operator-blocked` (§11.4.21) or open work, NOT `won't-fix`; mislabelling them is a §11.4 planning-layer bluff. A future platform change can make the impossible possible (durable, not eternal). Classification: universal (§11.4.17) — platform-neutral effort-conservation + honesty discipline; the project supplies the impossible goal, its constraint + cited evidence per §11.4.35. Composes §11.4.6/.7/.8/.34/.90/.99. Propagation gate `CM-COVENANT-114-112-PROPAGATION` (literal `11.4.112`) + recommended `CM-WONT-FIX-STRUCTURAL-IMPOSSIBILITY` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.112. Release blocker. No
`--wont-fix-without-proof` / `--reattempt-closed-impossible` / `--skip-impossibility-evidence` / `--impossible-equals-broken` escape.

§11.4.113 — Absolute no-force-push + merge-onto-latest-main mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03): "Any force-push is strictly forbidden! We must for every Submodule take as a base latest commit on Submodule's main (or master) branch, then on top of it carefully to merge all changes that have to be pushed! Once all merging is carefully done we perform commit and push to all Submodule's upstreams!" Force-push is STRICTLY FORBIDDEN with NO exception — `git push --force` , `--force-with-lease` , `+<ref>` , any history-rewriting overwrite of a remote ref — against EVERY repository this Constitution governs (main repo, this constitution submodule, every owned + nested submodule, every upstream). No operator-approval path, no "after merge-first audit" path. Mandated 6-step integration for any repo/submodule with commits to publish OR diverged mirrors: (1) `git fetch --all --prune --tags` all remotes; (2) base = LATEST commit on canonical `main` / `master` (most-advanced mirror tip); (3) carefully MERGE every change on top — union, preserve BOTH sides, NEVER `-s ours` / rebase / reset that drops commits (§9 no-commit-loss); (4) resolve conflicts carefully — no markers, no file dropped, gates/tests still pass; (5) commit the merge (stage only intended files, NEVER `git add -A` in a submodule per §11.4.30); (6) push to ALL upstreams — each is a fast-forward because the merge commit descends from every mirror tip, so NO force is needed (an upstream still rejecting → return to step 1 for it, merge its new tip, re-validate, re-push). TIGHTENS §11.4.41 / §11.4.71 / §9.2 / CONST-043 — force-push escape hatch REMOVED: even WITH operator approval, even after a clean merge-first audit, force-push is forbidden, because the merge-onto-latest-main path is always available so force is never necessary; those clauses' merge-first/fetch-first machinery stays, only their terminal "...then force-push" step is struck.

Classification: universal (§11.4.17). Composes §2.1 / §9 / §9.2 / §11.4.4 / §11.4.6 / §11.4.26 / §11.4.37 / §11.4.40 / §11.4.41 / §11.4.71 / §11.4.88 / CONST-043.

Propagation gate `CM-COVENANT-114-113-PROPAGATION` (literal `11.4.113`) + recommended `CM-NO-FORCE-PUSH-ABSOLUTE` (scan tracked scripts/hooks for

`push --force / --force-with-lease / push +<ref> + reject`; §11.4.109-class PreToolUse guard blocks the class) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.113. Release blocker. No `--force / --force-with-lease / --allow-force-push / --force-push-
authorised / --skip-merge-onto-main` escape.

§11.4.114 — Last-known-good-tag regression isolation mandate (1.1.8-dev remediation, 2026-06-03)

When a previously-working feature/behaviour is observed broken, the FIRST diagnostic action MUST be to identify the last release tag (or commit/build/deploy) at which it was KNOWN-GOOD and diff/bisect the broken state against it — BEFORE any open-ended root-cause hunt or speculative fix. The known-good revision is the regression oracle: bounds the search to commits between good and now (`git diff <good-tag>..HEAD --stat` of the feature's files = captured evidence), marks it a regression REPAIR not from-scratch design, gives each suspect file a behavioural oracle. An operator-volunteered known-good tag is load-bearing → act on it first. Default to SURGICAL forward-fix (keep post-good-tag features, revert ONLY the broken sub-part) over wholesale revert (which loses the batch's other working features) unless operator prefers wholesale. Honest boundary (§11.4.6): "it worked before" is a HYPOTHESIS until the tag is identified AND the feature confirmed working there; "probably regressed last batch" without the diff is a guess; a never-worked feature is not a regression. Classification: universal (§11.4.17). Composes §11.4.4 / §11.4.6 / §11.4.7 / §11.4.40 / §11.4.43 / §11.4.102 / §11.4.108. Propagation gate `CM-COVENANT-114-114-PROPAGATION` (literal `11.4.114`) + recommended `CM-REGRESSION-ISOLATED-AGAINST-KNOWN-GOOD` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.114. Release blocker. No `--skip-known-good-diff / --root-cause-from-scratch / --assume-regression-source / --wholesale-revert-without-isolation` escape.

§11.4.115 — RED-baseline-on-the-broken-artifact + polarity-switch mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.43's RED step. Every RED test MUST REPRODUCE the defect on the CURRENT pre-fix artifact (the actual broken build/deploy), capturing positive evidence per §11.4.5 / §11.4.69 / §11.4.107 that the defect is genuinely present — never a synthetic failure the fix is then written to agree with. The SAME source MUST carry a single polarity switch (env flag / parameter, canonical `RED_MODE` , default `1` = reproduce-and-assert-defect-present) that flipped to `0` post-fix becomes the GREEN regression-guard asserting the defect is ABSENT. One source, two roles: the bug-catcher IS the regression-guard; no separate happy-path test is the primary guard (that only proves the test agrees with the fix — the §11.4.43 PASS-bluff). RED-on-broken then GREEN-on-fixed (clean target per §11.4.108) both captured. Honest boundary (§11.4.6): a RED run that does NOT fail on the broken artifact is a finding (close per §11.4.7 with negative evidence, or fix the blind test). Classification: universal (§11.4.17). Composes §11.4.1 / §11.4.2 / §11.4.5 / §11.4.69 / §11.4.107 / §11.4.4 / §11.4.7 / §11.4.43 / §11.4.50 / §11.4.108 / §11.4.114. Propagation gate `CM-COVENANT-114-115-PROPAGATION` (literal `11.4.115`) + recommended `CM-RED-POLARITY-SWITCH-PRESENT` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.115. Release blocker. No `--green-only-test` / `--skip-red-reproduction` / `--separate-happy-path-suffices` / `--synthetic-red-OK` escape.

§11.4.116 — Real-time conductor↔autonomous-test-framework sync channel mandate (1.1.8-dev remediation, 2026-06-03)

Any autonomous, long-running test/QA/validation framework an external orchestrator (conductor agent / operator) depends on for real-time decisions MUST expose a real-time sync channel: (1) a structured append-only event stream (JSONL or equivalent, one event per line, never rewritten — session-start / phase-transition / per-test-or-challenge-start / captured-evidence-path / external-call (LLM/vision/sink-probe) / error / per-item-verdict); (2) an atomically-rewritten status snapshot (single small file, write-temp-then-rename so no torn read) with current session/phase/item + counters + last verdict. Verdicts use the closed PASS / FAIL / SKIP / OPERATOR-BLOCKED vocabulary (§11.4.45). The conductor tails it live → real-time sync, §11.4.4 interrupt on a fresh defect, never idles blindly (§11.4.94 /

§11.4.97). Anti-bluff: every verdict event MUST carry the evidence path that backs it (§11.4.69) — a PASS event with no evidence path is a channel-layer bluff; a snapshot PASS contradicted by a stream with no evidence event → treat as FAIL; no start-event → no verdict-event. Owned project-agnostic submodule (§11.4.28): channel stays project-neutral, the consumer registers its data via the public API at runtime, never hardcoded. Classification: universal (§11.4.17). Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.27 / §11.4.28 / §11.4.45 / §11.4.52 / §11.4.89 / §11.4.94 / §11.4.97. Propagation gate `CM-COVENANT-114-116-PROPAGATION` (literal `11.4.116`) + recommended `CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.116. Release blocker. No `--no-sync-channel` / `--end-of-session-report-suffices` / `--verdict-without-evidence-path` / `--torn-status-write-OK` escape.

§11.4.117 — Computer-vision / OCR pixel-oracle fallback for non-introspectable UIs mandate (1.1.8-dev remediation, 2026-06-03)

Any test that drives a UI control OR asserts on-screen content MUST NOT assume the accessibility/semantic/DOM hierarchy is the source of truth for what the user sees. When the hierarchy is blank/partial/known-unreliable (TV-Compose, leanback, canvas/ `SurfaceView` /GL, games, custom-rendered UIs), the test MUST fall back to a PIXEL ORACLE: (1) DRIVE input by computer-vision template-match (locate a control by rendered appearance → tap coordinates), not a node id; (2) ASSERT content by ROI OCR (read the rendered text the user reads) with a per-word confidence floor + ROI per §11.4.107(12), not a hierarchy text attribute that may be absent/stale. The tool MUST both drive input AND read pixels — a hierarchy-only tool is NOT a content oracle. Makes §11.4.52's "near-empty hierarchy → INFEASIBLE" constructive: pixel-drive, not only SKIP. Anti-bluff (§11.4.107(10)): the CV/OCR analyzer is self-validated (golden-good PASS, golden-bad FAIL, wired into meta-test); thresholds calibrated on the project's own frames, not hardcoded (§11.4.6). Honest boundary: BOTH hierarchy blank AND pixel oracle infeasible (secure surface black-captures §11.4.112, geo-unreachable) → SKIP-with-reason §11.4.3 + tracked operator-attended migration §11.4.52, never fake PASS. Classification: universal (§11.4.17). Composes §11.4.5 / §11.4.6 / §11.4.48 / §11.4.49 / §11.4.52 / §11.4.107 / §11.4.112. Propagation gate `CM-`

COVENANT-114-117-PROPAGATION (literal 11.4.117) + recommended
CM-CV-OCR-PIXEL-ORACLE-FALLBACK + paired §1.1 mutation (gate-code =
separate work item).

Canonical authority: Constitution.md §11.4.117. Release blocker. No --
hierarchy-is-content-oracle / --skip-pixel-fallback / --unvalidated-
ocr-OK / --hardcoded-ocr-threshold-OK escape.

§11.4.118 — Discovery-pressure to confirm known-issue-set completeness mandate (1.1.8-dev remediation, 2026-06-03)

A remediation/release cycle MUST NOT treat "every reported defect is fixed" as "the build is good" — the reported set is biased by what the operator happened to test ("we only see what we test"). After/alongside fixing the reported set, run a discovery + stress pass across ALL target devices/environments that deliberately exercises subsystems/journeys/edge-cases BEYOND the reported defects — to CONFIRM the reported set is the COMPLETE critical set + surface unreported defects before the user does. The pass MUST produce PROVABLE coverage: an enumerated list of subsystems/journeys/stress-scenarios exercised, each with outcome (no-new-issue, or a new tracker entry per §11.4.15 / §11.4.16). "We found no other issues" is a §11.4 bluff unless paired with "here is the enumerated set we exercised" — absence of evidence of looking is not evidence of absence. New findings trigger §11.4.4 interrupt + §11.4.114/§11.4.115 isolation → RED → fix. Honest boundary (§11.4.6): discovery reduces the unknown-unknown surface, does not prove zero remaining defects — earned claim is "reported set + enumerated discovery set addressed," un-exercised subsystems stated as honest coverage gaps (§11.4.3), never silently implied clean. Classification: universal (§11.4.17). Composes §11.4.4 / §11.4.5 / §11.4.69 / §11.4.25 / §11.4.40 / §11.4.42 / §11.4.52 / §11.4.85 / §11.4.114 / §11.4.115 / §11.4.119. Propagation gate CM-
COVENANT-114-118-PROPAGATION (literal 11.4.118) + recommended CM-
DISCOVERY-COVERAGE-ENUMERATED + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.118. Release blocker. No --
reported-set-suffices / --skip-discovery-pass / --no-issues-without-
coverage-list / --assume-complete escape.

§11.4.119 — Single-resource-owner partitioning for parallel hardware testing mandate (1.1.8-dev remediation, 2026-06-03)

Strict refinement of §11.4.58 / §11.4.103 for hardware contention. When multiple parallel streams exercise SHARED hardware / any exclusive-access resource (a media-playback path, a single HDMI/audio sink, an exclusive device handle, a serial/JTAG line, a GPU under exclusive capture), exactly ONE stream MUST own each such resource at a time. The owner drives it (playback, input injection, capture); every other concurrent stream targeting it MUST be READ-ONLY (passive probes — `dumpsys` / `/proc` / `/sys` reads / sink-side network probes / log tails). Parallelism partitioned by resource: distinct devices/sinks/handles fully concurrent (stream-per-device), but the same device's exclusive resource is single-owner. Ownership enforced by an advisory lock/token (§11.4.58 L3 + the hardware analogue of §11.4.84 quiescence), event-driven (claim when the resource frees, release the moment work completes so the next queued stream claims it per §11.4.103 auto-backfill). Why: concurrent drivers of one exclusive resource produce CROSS-CONTAMINATED evidence (sink reports the wrong stream's audio, foreground belongs to whichever `am start` landed last, input events interleave) — a PASS under contention is a §11.4 evidence-integrity bluff. Honest boundary (§11.4.6): a "read-only" probe stream MUST issue NO state-changing command against a device it does not own; a non-partitionable resource → streams serialize (single-owner over time), not run concurrently and hope. Classification: universal (§11.4.17). Composes §11.4.5 / §11.4.69 / §11.4.13 / §11.4.50 / §11.4.58 / §11.4.82 / §11.4.84 / §11.4.103 / §11.4.118. Propagation gate `CM-COVENANT-114-119-PROPAGATION` (literal `11.4.119`) + recommended `CM-SINGLE-RESOURCE-OWNER-PARTITION` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.119. Release blocker. No `--allow-concurrent-resource-drivers` / `--skip-ownership-lock` / `--read-only-may-mutate` / `--contended-evidence-OK` escape.

§11.4.120 — Fix-breaks-its-own-gate reconciliation mandate (1.1.8-dev remediation, 2026-06-03)

When a correct fix causes a pre-existing gate/test to FAIL because that gate asserted the OLD (now-removed-or-changed) behaviour, the gate FAIL is the CORRECT signal the fix landed — it MUST NOT be suppressed by either forbidden

response: (1) FAKE-PASSING the gate (edit to `always pass`, weaken to a tautology, or delete — a §11.4 gate-layer bluff + a §11.4.84 mutation-residue risk); (2) REVERTING the correct fix to satisfy the stale gate (re-introducing the defect). Required response = RECONCILIATION: rewrite the gate to assert the NEW mechanism the fix introduced, backed by captured evidence of the new correct behaviour, AND update its paired §1.1 mutation so the mutation breaks the NEW invariant. Discriminator vs bluffing: after reconciliation gate + mutation still form a valid §1.1 pair (mutate the new invariant → gate FAILs); a bluffed gate's mutation no longer makes it FAIL. The reconciliation MUST be a visible evidence-cited change, never a silent assertion-weakening. Honest boundary (§11.4.6): a post-fix gate FAIL is NOT automatically "stale gate" — investigate per §11.4.102 first; the FAIL may be the gate correctly catching a REGRESSION the fix introduced (then the FIX is wrong). Reconcile ONLY when investigation PROVES the gate asserted old-correct-now-removed behaviour AND the new behaviour is intended + evidence-confirmed. Classification: universal (§11.4.17). Composes §11.4.1 / §11.4.4 / §11.4.6 / §11.4.84 / §11.4.102 / §11.4.108 / §1.1. Propagation gate `CM-COVENANT-114-120-PROPAGATION` (literal `11.4.120`) + recommended `CM-GATE-RECONCILED-NOT-FAKE-PASSED` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.120. Release blocker. No `--fake-pass-stale-gate` / `--revert-fix-for-gate` / `--weaken-assertion-to-pass` / `--delete-failing-gate` escape.

§11.4.121 — No-commit-while-build-writes-tracked-artifacts mandate (1.1.8-dev remediation, 2026-06-03)

A commit (especially `git add -A` / any broad stage) MUST NOT run while a build/packaging/generation step is actively writing artifacts INTO tracked (version-controlled) directories — it races the writer and stages a PARTIAL or stale artifact (a §11.4.108 SOURCE → ARTIFACT integrity failure landed in version control). The commit MUST be deferred until the build step that writes tracked artifacts has COMPLETED, so the tree is quiescent at the artifact layer AND the committed artifacts are the FRESH, whole outputs (no stale pre-rebuild artifact, no half-written file). Before committing tracked build outputs, verify the writing step finished (process exit / completion marker / per-artifact mtime ≥ build-start). Build-output analogue of §11.4.84 (no commit while a mutation gate is in flight): both close

"commit captures transient non-final tree state" at two write-sources. Gitignored build outputs (`out/` / `dist/`) are exempt — the race doesn't apply. Honest boundary (§11.4.6): "the build probably finished" is not "the build finished" — verify with a completion signal; source changes NOT touching the build's tracked-artifact dirs are fine mid-build. PROJECT-SPECIFIC instance (which tracked dir + which build steps write it) recorded in the consuming project's own governance per §11.4.35. Classification: universal (§11.4.17). Composes §11.4.6 / §11.4.30 / §11.4.58 / §11.4.84 / §11.4.88 / §11.4.96 / §11.4.103 / §11.4.108. Propagation gate `CM-COVENANT-114-121-PROPAGATION` (literal `11.4.121`) + recommended `CM-NO-COMMIT-DURING-ARTIFACT-WRITE` + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.121. Release blocker. No `--commit-during-build` / `--stage-partial-artifact-OK` / `--assume-build-finished` / `--skip-build-completion-check` escape.

§11.4.122 — No-silent-removal-of-existing-components-without-operator-confirmation mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03): "Never ever remove any application, system component or service from already existing codebase / System without interactively asked question to us! THIS IS MANDATORY RULE / CONSTRAINT!" Case study (FACT): in the 1.1.8-dev burn-down, F2 (an Apple-TV-class app) + F4 (a Huawei HMS component) were removed WITHOUT asking; the operator reversed both.

No application / system component / service / package / feature / driver / module / library / prebuilt asset — any already-existing end-user capability — may be removed (deleted, dropped from the package set, disabled-into-non-shipping, unbundled, de-listed) from the existing codebase / shipped System WITHOUT FIRST interactively asking the operator (via the §11.4.66 interactive mechanism — `AskUserQuestion` on Claude Code, never free-text, never silent, never an autonomous decision) and receiving an EXPLICIT keep-or-remove decision. A removal is: a deletion from the package/service/module manifest set whose NET EFFECT is "a capability that shipped before no longer ships." Adding / replacing-with-operator-approved-equivalent / fixing is NOT a removal; when uncertain, treat AS a removal and ask (§11.4.6 no-guessing). A silent removal is a release blocker regardless of rationale (dedup / "broken anyway" / geo-restricted / incompatible /

superseded). The tracked DROP path: ask → operator approves → mark item Obsolete (→ Fixed.md) with reason feature-removed + operator-approval citation (§11.4.90) → then remove. Classification: universal (§11.4.17). Composes §11.4.66 / §11.4.101 (removal of a live capability is exactly the high-blast-radius class §11.4.101 reserves for an operator question, never an autonomous decision) / §11.4.90 / §11.4.112 / §11.4.6 / §11.4.40 / §11.4.42. Propagation gate CM-COVENANT-114-122-PROPAGATION (literal 11.4.122) + recommended gate CM-NO-SILENT-COMPONENT-REMOVAL + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.122. Release blocker. No --remove-without-asking / --silent-removal / --autonomous-removal-OK / --dedup-removal-exempt / --it-was-broken-anyway escape.

§11.4.123 — Rock-solid-proof-or-deep-research mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03): "Every single reported issue MUST BE fully and 100% validated with rock solid proofs! Nothing can be considered fixed or completed without hard evidence! No false results or bluff(s) of any kind is allowed! If we are not sure on how to achieve full testing, validation and verification of something we MUST ALWAYS perform deep web research for all possible data (articles, documentation, guides, and other resources) and opensourced codebases which we can use to solve our problems and perform testing with validation and verification which produces rock-solid evidence(s) and leaves no space for false results or any kind of bluff!" Case study (FACT): in the 1.1.8-dev remediation the validation method for FLAG_SECURE secondary-display capture (black frames) + non-introspectable streaming-app UIs (blank accessibility tree) was unclear; deep web research (docs/research/testing_frameworks_20260603/) yielded the CV/OCR/liveness/sink-probe oracle stack (now §11.4.107/§11.4.112/§11.4.117), making rock-solid evidence possible — "unclear how to validate" is a research trigger, never a bluff licence.

Every reported issue / fix / claimed completion MUST be 100% validated with rock-solid CAPTURED proof (§11.4.5/§11.4.69/§11.4.107) before it is marked fixed/implemented/completed (§11.4.33). Nothing is fixed/complete without hard captured evidence — metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden (§11.4/§11.4.1); no false results, no bluff of any kind. Operative

addition: when UNSURE how to fully test/validate/verify something (no obvious evidence-producing method, OR the candidate yields only metadata/config/absence-of-error evidence), the agent MUST ALWAYS FIRST perform deep web research per §11.4.8+§11.4.99 (official docs, articles, guides, vendor refs, standards, issue trackers, reusable open-source codebases) to DISCOVER or BUILD an evidence-producing validation method that leaves no room for a false result. Declaring "untestable"/"not automatable" or accepting a metadata-only PASS WITHOUT first exhausting this deep-research path is itself a §11.4.123 violation (same severity as a PASS-bluff); the cited research (URLs + the method, OR "NO external solution found — original work" per §11.4.8) is the proof the path was exhausted. Only after that research genuinely fails may the item be

PENDING_FORENSICS: / Operator-blocked (§11.4.21)/ structurally-impossible won't-fix (§11.4.112) with the cited research as earned evidence. Classification: universal (§11.4.17). Composes §11.4.5/§11.4.6/§11.4.8/§11.4.52/§11.4.69/§11.4.99/§11.4.107/§11.4.118/§11.4.21/§11.4.112. Propagation gate CM-COVENANT-114-123-PROPAGATION (literal 11.4.123) + recommended gate CM-ROCK-SOLID-PROOF-OR-RESEARCH + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.123. Release blocker. No --metadata-pass-suffices / --skip-proof / --untestable-without-research / --config-only-closure-OK / --bluff-when-unsure escape.

§11.4.124 — Dead/unwired-code investigate-before-remove mandate (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04): "Before removing any seemingly-dead (zero-importer / unwired) codebase, we MUST investigate via git history where/how it was originally used and how it became dead. Removal is permitted ONLY when we have captured PROOF it is genuinely no longer needed — and that removal MUST be its own separate commit with a proper descriptive message. If there is no such proof, the code MUST be investigated for where/how it should be wired in properly, and any missing or unwired tests MUST be added. We MUST ALWAYS be extra careful with any codebase removal."

"Zero importers / never called / unwired $\Rightarrow$ dead $\Rightarrow$ delete" is a GUESS (§11.4.6), never a finding — a "no references" result proves only current non-reference, not genuinely-unneeded. Before removing ANY seemingly-dead element (zero-importer / never-called / unwired function / method / type / file / module / package / asset / config / build target) the agent MUST FIRST investigate via git history (`git log --follow` , `git log -S / -G` pickaxe, blame on the deleted call-site) and capture as FACT: (1) WHERE/HOW it was originally wired in, (2) WHEN/HOW it became dead (call-site deleted deliberately / by mistake-regression / never-completed / refactored-unreachable), (3) whether "no references" is real OR a hidden reference the static tool cannot see (reflection / dynamic dispatch / build-tags / codegen / DI / plugin registry / FFI / config-driven). Removal is permitted ONLY with captured PROOF the element is genuinely no longer needed; that removal MUST be its OWN SEPARATE COMMIT (independently reviewable + revertible per §11.4.84/§11.4.92) with a descriptive message citing the git-history evidence (+ §11.4.122 operator-confirmation if it is an end-user capability; tracked via §11.4.90 `obsolete` ($\rightarrow$ `Fixed.md`)). With NO such proof the element MUST NOT be removed — instead investigate WHERE/HOW to wire it in properly (restore a mistakenly-deleted call-site per §11.4.114; finish never-completed wiring) AND add any missing / unwired tests (§11.4.27/§11.4.43/§11.4.115). ALWAYS be extra careful with removal: when uncertain, default to NOT removing (investigate + wire + test) per §11.4.6 + §11.4.101 + §11.4.122; "probably dead" is never sufficient — the bar is captured proof. Classification: universal (§11.4.17). Composes §11.4.6 / §11.4.8 / §11.4.84 / §11.4.90 / §11.4.92 / §11.4.101 / §11.4.114 / §11.4.122 / §11.4.27 / §11.4.43 / §11.4.115. Propagation gate `CM-COVENANT-114-124-PROPAGATION` (literal `11.4.124`) + recommended gate `CM-DEAD-CODE-INVESTIGATE-BEFORE-REMOVE` (a net-deletion commit must be removal-only + cite the git-history investigation OR be part of a tracked `Obsolete` item) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.124. Release blocker. No

```
--zero-importers-means-dead / --delete-unwired-on-sight /  
--skip-git-history-investigation / --remove-without-proof /  
--bundle-removal-with-other-work escape.
```

§11.4.125 — Code-review-agent gate before pre-build + main build (mandatory multi-layer review) (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04): "After all fixes/ changes/implementations are done, BEFORE running pre-build tests and the main build, dispatch code-review agent(s) that analyze all work done + all existing data/ facts + the existing codebase + current git history to determine quality, safety, and whether the fixes/changes will REALLY work; they MUST validate and verify that every test covering the fixes/changes genuinely validates the work with NO chance of false results or bluff of any kind. Any finding MUST be fixed, polished, improved, and covered with additional tests before the build proceeds. Multiple strong layers of checks."

After all fixes / changes / implementations in a batch are done, and BEFORE running the pre-build test sweep AND the main (artifact) build (for ANY project), the agent MUST dispatch one or more dedicated code-review agent(s) (subagent-driven by default per §11.4.70/§11.4.20) performing a multi-layer review that: (1) analyzes ALL work done in the batch (every fix/change + its source diff + stated intent); (2) analyzes ALL existing data + facts (captured evidence per §11.4.5/ §11.4.69/§11.4.107, tracker entries, prior findings, the §11.4.108 runtime-signature registry); (3) analyzes the existing codebase (blast radius per §11.4.92, cross-feature interaction, contract integrity of every dependency); (4) analyzes current git history (what each change touched, how it composes with concurrent/recent work, whether it reproduces a known-broken pattern per §11.4.114/§11.4.124); (5) determines quality + safety + will-it-REALLY-work (robust + not error-prone — no solve-A-create-B; no host/data/security regression; genuinely delivers the end-user-visible behaviour per §11.4/§107); (6) validates + verifies the tests covering the work — every covering test genuinely exercises the work-under-test and catches its negation, with ZERO chance of a false result or bluff (a test that PASSES on broken-for-the-user work, a metadata-only/config-only/absence-of-error/grep-without-runtime assertion, or a gate whose paired §1.1 mutation does not make it FAIL is a finding). Any finding (defect / error-prone change / safety risk / will-not-really-work / bluff-or-false-result-capable test / missing-coverage gap) MUST be fixed, polished, improved, and covered with additional tests (four-layer per §11.4.4(b), TDD-RED-first per §11.4.43/§11.4.115) BEFORE the pre-build sweep + main build proceed; the review iterates (re-review after each remediation) until no blocking findings remain. The review is itself anti-bluff (its conclusions are captured evidence per §11.4.5/§11.4.69; a rubber-stamp review of a defective batch = PASS-

bluff). It is one of MULTIPLE STRONG LAYERS — complementing, never replacing, the §1 pre-build sweep, §11.4.92 multi-pass (author-side self-review; §11.4.125 adds the structurally-separated reviewer seam per §11.4.70), §11.4.108 four-layer fix-verification, §11.4.110 build-readiness verdict, and the post-build / runtime-on-clean-target / user-visible layers. Composes §11.4 / §11.4.1 / §11.4.4 / §11.4.6 / §11.4.40 / §11.4.43 / §11.4.50 / §11.4.70 / §11.4.20 / §11.4.92 / §11.4.102 / §11.4.107 / §11.4.108 / §11.4.110. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-125-PROPAGATION` (literal `11.4.125`) + recommended gate `CM-CODE-REVIEW-GATE-BEFORE-BUILD` (build starts only with a fresh code-review-completed marker for the current batch, produced after the last fix + before the pre-build sweep + main build) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: `Constitution.md` §11.4.125. Release blocker. No `--skip-code-review` / `--build-without-review` / `--no-review-gate` / `--review-optional` / `--trust-the-author` escape.

§11.4.126 — Default autonomous-loop working mode from first prompt (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04): "Make sure that you continue work in endless fully autonomous loop, do not stop until new fully validated and verified version (tag) is created and published (all submodules and main repo) or IN A CASE OF some other main stream work until it is fully completed with all side work streams and nothing else is left in our working queue! THIS MUST BE ALWAYS the default working mode without us asking you! We tend to achieve ABSOLUTE EFFICIENCY, with this and all other projects which will incorporate this MANDATORY RULE / CONSTRAINT!!! This way of (your) working will be ALWAYS applied / followed / executed / fully respected, as soon as we assign / send first request (prompt) in the session! This stops only if we explicitly say so or nothing is left to be done in current working scope (release that will come / upcoming version)!!! Any mimicking (imitation) of this behavior / rules / mandatory constraints, false results or any kind of bluff(s) is ABSOLUTELY FORBIDDEN!!!"

The endless fully-autonomous loop is the **DEFAULT working mode**, engaged automatically the moment the operator sends the FIRST request / prompt of a session — the operator MUST NOT have to ask for it, request it, restate it, or re-enable it per session. §11.4.87 framed the endless-loop covenant as an explicit-

instruction opt-in; §11.4.126 is the **capstone** that promotes the same covenant to always-on: from the first prompt onward, every agent operates in the §11.4.87 loop discipline as the standing default, with §11.4.94 zero-idle, §11.4.97 maximum-idle-use, §11.4.101 autonomous-decision-over-blocking, and §11.4.103 continuous-parallel-stream all engaged by default — no per-session activation handshake. The continuation contract: the loop continues until ONE of two terminal conditions holds — (A) **Release scope** — a new, fully-validated-and-verified version (tag) is created AND published across all owned submodules AND the main repo to all configured remotes (per §2.1 + §11.4.40 + §11.4.113); OR (B) **Non-release main-stream scope** — the main-stream goal is fully completed AND every side work stream is done AND the working queue holds nothing left for the current scope. Until (A) or (B) holds, the agent MUST keep working per §11.4.42 / §11.4.72 / §11.4.94 / §11.4.103. The loop STOPS ONLY on: (1) the operator explicitly saying so (STOP / pause / end); (2) nothing left to do in the current working scope with the queue genuinely empty per the (A)/(B) terminal conditions; (3) a §12 host-session-safety demand. Idle-while-blocked parks one work unit, it does not stop the loop (§11.4.101 + §11.4.94 + §11.4.97). Goal — ABSOLUTE EFFICIENCY; applies to this project AND every project that incorporates this Constitution. Anti-bluff: mimicking / imitating this loop behaviour, narrating continuation without performing it, fabricating progress, or emitting false / bluff results of ANY kind is ABSOLUTELY FORBIDDEN (composes §11.4 / §11.4.1 / §11.4.2 / §11.4.5 / §11.4.6 / §11.4.50 / §11.4.69 / §11.4.107); a report claiming the loop ran while no real work / no captured evidence was produced is a §11.4 PASS-bluff at the operating-mode layer. Classification: universal (§11.4.17). Composes with §11.4.87 / §11.4.94 / §11.4.97 / §11.4.101 / §11.4.103 / §11.4.66 / §11.4.6 / §11.4.40 / §11.4.42 / §11.4.72 / §11.4.113 / §2.1 / §12. Propagation gate CM-COVENANT-114-126-PROPAGATION (literal 11.4.126) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: Constitution.md §11.4.126. Release blocker. No --ask-before-continuing / --single-turn-only / --not-default-loop / --mimic-OK escape.

§11.4.127 — Session-handoff resumption-prompt mandate (User mandate, 2026-06-06)

Forensic anchor — verbatim user mandate (2026-06-06): "make sure that in situations like this now when new session is needed you ALWAYS prepare such sentence - which will be valid for particular moment and the phase of the project and enough for work to continue."

When a fresh session is needed (context limits / degradation) OR the operator asks whether a new session is needed / requests a handoff, the agent MUST ALWAYS prepare + proactively provide a ready-to-paste **resumption prompt valid for that EXACT moment + project phase** — self-contained for ZERO-loss resume. Two variants: SHORT first-sentence ("Read <handoff docs> , then continue <goal> ...") + FULL block on demand. MUST (1) point to live handoff docs (`.remember/remember.md` if present + `docs/CONTINUATION.md` §12.10, read FIRST + `git fetch --all`); (2) state PHASE + NEXT action + terminal goal; (3) embed exact live-state anchors (build IDs/MD5, device/target serials, commit HEAD, in-flight PIDs+log paths, captured-evidence paths); (4) restate binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware/target gotchas); (5) be MOMENT-VALID never generic. Handoff docs current BEFORE the prompt (§12.10). Missing/stale/generic = §11.4.127 violation. Composes §12.10 / §11.4.6 / §11.4.66 / §11.4.87 / §11.4.103 / §11.4.126. Classification: universal (§11.4.17). Gate `CM-COVENANT-114-127-PROPAGATION` (literal `11.4.127`) + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.127. Release blocker. No `--skip-handoff-prompt` / `--generic-prompt-OK` / `--no-resumption-sentence` / `--handoff-without-state` escape.

§11.4.128 — Always-on device-recording mandate (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): ALWAYS live-record all available data from all test/manual-testing devices, EXTRA carefully (never harm device / performance / no side effects); raw NOT processed without need (token-conscious); raw ALWAYS git-ignored + code-intelligence-excluded; only curated evidence committed, only at release prep.

For EVERY test/debug device + every device under known manual testing, across EVERY transport (USB / wireless ADB / SSH / serial / network introspection API), ALWAYS live-record all analysable data: activities, all logs, perf metrics (CPU/mem/I/O/thermal/load), every sink-side report (§11.4.13), any live-changeable param. (1) **Side-effect-free** — non-invasive read-only probes, bounded sampling + write-volume, observer-effect budget; a perturbing recorder = §11.4.128 violation. (2) **Background + parallel + subagent-driven** (§11.4.103 + §11.4.70) — never blocks main stream. (3) **Token-conscious record-now-analyse-later** — raw NOT processed without need; only analyse-trigger = release-tag prep (§11.4.40/§11.4.42) OR explicit operator ask. (4) **Raw git-ignored (w/ §11.4.77 regen declaration) + code-intelligence-excluded (§11.4.78/79)** — only curated evidence committed, only at release prep under `docs/qa/<run-id>/` (§11.4.83). (5) **Layout**

`<recording-root>/YYYY-MM-DD/<combined main+submodules hash>/<DEVICE>_<SERIAL>/recording_NNN/<files>` . (6) **Anti-bluff** — recorder-running-but-no-growing-corpus = §11.4 bluff; curated findings trace to real raw paths; recorder health = captured evidence (§11.4.5/69). Composes §11.4.2/.5/.13/.69/.40/.42/.70/.77/.78/.79/.83/.103/.119. Classification: universal (§11.4.17). Gate `CM-COVENANT-114-128-PROPAGATION` (literal `11.4.128`) + rec. gate `CM-DEVICE-RECORDING-ALWAYS-ON` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.128. Release blocker. No

`--skip-recording` / `--record-without-layout` / `--commit-raw-corpus` / `--index-raw-corpus` / `--analyse-corpus-always` / `--invasive-probe-OK` escape.

§11.4.129 — Huge-blocker release protocol (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): on a huge blocker during release validation: STOP all testing → fix ALL discovered issues → process all recorded data from last session → land rock-solid fixes → author NEW validation+verification tests of ALL supported test types → rebuild → reflash → RESTART full validation+verification of every fix/change from the last release tag to now — both devices in parallel, recorded, real physical proofs, no bluff.

On a HUGE BLOCKER (release-blocking-severity: core capability broken / regression invalidating the cycle / blast radius reaching the batch's other fixes), in order, NO shortcut: (1) STOP all testing every device (§11.4.4 STOP at release granularity). (2) Fix ALL discovered issues (not just the blocker) — root-cause §11.4.102 + isolate against last known-good tag §11.4.114. (3) Process all recorded data from last session (the §11.4.128(3) release-prep analyse-trigger). (4) Land rock-solid fixes §11.4.123 + §11.4.43/.115 + §11.4.9. (5) Author NEW tests of ALL supported types §11.4.27 + §11.4.85, anti-bluff + paired §1.1 mutation. (6) Rebuild (full, not module-only) + reflash CLEAN target §11.4.108. (7) RESTART full validation from last tag to now §11.4.40 — RESTART not resume — both/all devices parallel §11.4.103/.119, recorded §11.4.128, real physical proofs §11.4.5/.69/.107, no bluff. BINDS the existing release anchors for the huge-blocker case (STOP → fix-all → process-recordings → new-tests-all-types → rebuild → reflash → full-restart + restart-not-resume), citing not duplicating. Composes §11.4.4/.40/.42/.9/.27/.85/.102/.108/.114/.115/.123/.128/.103/.119. Classification: universal (§11.4.17). Gate `CM-COVENANT-114-129-PROPAGATION` (literal `11.4.129`) + rec. gate `CM-HUGE-BLOCKER-FULL-RESTART` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.129. Release blocker. No `--resume-after-blocker` / `--spot-validate-after-fix` / `--skip-recording-analysis` / `--skip-new-tests` / `--module-only-after-blocker` / `--single-device-restart` escape.

§11.4.130 — Post-remediation validate-the-fix-FIRST-after-redeploy (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): when a blocker found during release validation is fixed and a new artifact (rebuild / new flashing image / redeploy) is produced + the target reflashed, FIRST re-test the SPECIFIC last-failing features + validate the just-incorporated fixes BEFORE the broader / full validation.

When a blocker / critical failure found during release validation is FIXED and a new artifact is produced + the target reflashed / redistributed / updated, in order, NO shortcut: (1) re-test the SPECIFIC last-failing features FIRST (targeted guard tests for exactly the defects this fix addressed) BEFORE any broader / full-suite validation; (2) validate the just-incorporated fixes with real captured evidence — the

§11.4.115 RED test flips GREEN at `RED_MODE=0` on the new artifact AND the §11.4.108 runtime-signature verifies on the CLEAN target the redeploy produced (metadata-only/config-only/absence-of-error/grep-without-runtime PASS forbidden §11.4/.1; proof §11.4.5/.69/.107/.123); (3) ONLY after the targeted fix is CONFIRMED working proceed to the §11.4.40 full retest from last tag to now. Rationale: a first fix attempt may not work / be incomplete / regress again under the new artifact — confirming FIRST catches fix-did-not-take immediately instead of hours later at the END of a full cycle (then restarting §11.4.129); cheap-confirmation-first is §11.4.82 applied to the post-blocker reflash. This is §11.4.46 specialised for the post-blocker-reflash case + the targeted-confirmation phase that GATES §11.4.129's step-7 full-restart. Honest boundary §11.4.6: "probably took" ≠ "took"; a still-FAILing targeted re-test re-enters the §11.4.114/.115 isolate → RED → fix loop, never proceeds to the full cycle on a still-broken fix. Composes §11.4.4/.40/.46/.108/.114/.115/.123/.129/.82. Classification: universal (§11.4.17). Gate `CM-COVENANT-114-130-PROPAGATION` (literal `11.4.130`) + rec. gate `CM-FIX-FIRST-AFTER-REDEPLOY` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.130. Release blocker. No `--skip-targeted-retest` / `--full-cycle-first` / `--assume-fix-took` / `--validate-fix-at-end` / `--skip-red-green-flip-on-new-artifact` escape.

§11.4.131 — Standing session-resumption file mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): "Make this markdown a standard file which will be written EVERY TIME when we need fresh session out of the box! It MUST BE always up to date and in sync so whenever new session is created all we have to do is just point to it!"

Every project MUST maintain a SINGLE canonical, always-current **session-resumption file** at a fixed, project-declared standard path (declared once per §11.4.35, never moved without a §11.4.66 operator decision) — the OUT-OF-THE-BOX entry point for any fresh session (creating a new session = ONLY point the new agent at this one file). §11.4.131 promotes §11.4.127 (PREPARE-on-demand) into a STANDING, version-controlled ARTIFACT, ALWAYS present + ALWAYS in sync. (A) Exists at the declared path at all times (literal path per §11.4.35, never silently moved). (B) (Re)written whenever a fresh session is needed OR the live state materially changes (new HEAD / build-artifact id / phase / device state / in-

flight job / blocking decision) — the §12.10 trigger set; a stale file = §11.4.131 violation (same class as a §12.10 stale-CONTINUATION). (C) Carries SHORT + FULL §11.4.127 variants; points to `.remember/remember.md` + `docs/CONTINUATION.md` read FIRST + `git fetch` ; embeds exact live-state anchors (HEAD, artifact checksums, device serials, in-flight PIDs+logs, evidence paths); states PHASE + NEXT + terminal goal; restates binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware gotchas); MOMENT-VALID never generic (§11.4.6). (D) §11.4.65 export (`.html` / `.pdf` siblings) + §11.4.44 revision header. (E) Given ONLY this file's path a fresh session fully resumes with zero additional context. Composes §12.10/.127/.65/.44/.6/.66/.126. Classification: universal (§11.4.17). Gate `CM-COVENANT-114-131-PROPAGATION` (literal `11.4.131`) + rec. gate `CM-SESSION-RESUMPTION-FILE-PRESENT` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.131. Release blocker. No `--skip-resumption-file` / `--ephemeral-prompt-only` / `--stale-resumption-OK` / `--generic-template-OK` escape.

§11.4.132 — Risk-ordered validation priority mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): "We MUST ALWAYS first test and validate features, functionalities and fixes/changes that have been worked most recently, the ones which were most problematic, which have the most chance to crash or break again, the ones which have been re-opened the most times! Then, after we validate and verify all this with real (physical) proofs and hard evidence, with no false results and bluffs of any kind, we continue with all other existing tests in the test suites! This IS MANDATORY."

Tests / validations / verifications MUST run in RISK-DESCENDING order — the highest-risk set FIRST, and ONLY AFTER that set is fully GREEN with real (physical) captured evidence does the remainder of the suite run. Closed risk factors, highest-first: (a) most-recently-worked; (b) historically most-problematic (longest defect history, most prior fixes/failures); (c) highest crash/break/regress likelihood (greatest blast radius/complexity/dependency surface); (d) most-reopened per §11.4.55 reopens-count. Each highest-risk item MUST pass with real (physical) captured evidence per §11.4.5/§11.4.69/§11.4.107 — no metadata-only / config-only / absence-of-error / grep-without-runtime PASS (§11.4/§11.4.1), no false

results, no bluff (§11.4.6). ONLY AFTER the entire highest-risk set is GREEN with captured proof does the rest of the suite run; arbitrary-order or lower-risk-before-highest-risk-GREEN = §11.4.132 violation. REFINES/STRENGTHENS §11.4.130 (generalises "validate-just-fixed-first" to the full risk-ordered set) + §11.4.46 (adds explicit risk-ordering) + §11.4.42 (applies implementation-priority discipline to VALIDATION ordering). Classification: universal (§11.4.17) — consuming project supplies its recency/problematic-history/reopen-count sources (e.g. §11.4.93 DB `reopens_count + last_modified`) per §11.4.35. Composes §11.4.4/.5/.6/.7/.40/.42/.46/.50/.55/.69/.107/.130. Gate `CM-COVENANT-114-132-PROPAGATION` (literal `11.4.132`) + rec. gate `CM-RISK-ORDERED-VALIDATION-PRIORITY` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.132. Release blocker. No `--skip-risk-ordering` / `--any-order-OK` / `--suite-order-fixed` escape.

§11.4.133 — Target-System + hardware safety mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): "Make sure that all changes we do to the System are ALWAYS safe for the System itself and for the hardware the system runs on! This is MANDATORY."

Every change to the TARGET system (firmware, kernel, init/boot scripts, drivers, sysfs/devfreq/voltage/clock/thermal/regulator register writes, partition/bootloader/U-Boot, HAL, framework, prebuilts, device config) MUST ALWAYS be safe for BOTH (a) the target System itself — MUST NOT brick, boot-loop, corrupt data, or render the device unrecoverable — AND (b) the hardware it runs on — MUST NOT exceed safe electrical/thermal/voltage/clock limits or damage panels/storage/radios/regulators. Concrete obligations: (1) reversible-first — verify irreversible high-blast-radius changes (bootloader/U-Boot MD5, partition layout) against known-good values + capture a pre-op backup (§9.2) BEFORE applying; (2) NO unverified hardware-control writes — never write an unverified value to a voltage/clock/regulator/thermal-throttle/current-limit sysfs node or register that could exceed datasheet limits, the safe range established as FACT (§11.4.6), never guessed; (3) thermal/perf changes (forcing a performance governor, pinning the top OPP, disabling thermal management) MUST respect the device's cooling design, validated by captured thermal evidence; (4) flashing MUST use the sanctioned tool + a freshly-built integrity-verified image — never an ad-hoc partition write or stale/

unverified artifact; (5) unprovable-safety $\Rightarrow$ blocked — a change whose target/hardware safety cannot be established from captured evidence is treated as UNSAFE and blocked (§11.4.6 + §11.4.101 reversible-first + §11.4.123 rock-solid-proof). DISTINCT from §12 host-session safety: §12 protects the DEVELOPER's HOST + session; §11.4.133 protects the TARGET device + its hardware — both apply, neither weakens the other. Classification: universal (§11.4.17) — consuming project supplies its concrete hardware-control surfaces, datasheet-safe ranges, known-good bootloader/image hashes, and sanctioned flashing tool per §11.4.35. Composes §12 / §11.4.6 / §11.4.101 / §11.4.108 / §11.4.123. Gate CM-COVENANT-114-133-PROPAGATION (literal 11.4.133) + rec. gate CM-TARGET-HARDWARE-SAFETY + paired §1.1 mutation.

Canonical authority: Constitution.md §11.4.133. Release blocker. No --unsafe-hardware-write / --skip-system-safety / --brick-risk-accepted escape.

§11.4.134 — Code-review iterate-until-GO + rock-solid-evidence mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): "For any fixes/changes given back to us for re-work by the code-review process, once we fix/improve everything per the code-review's requests, we MUST RE-RUN code-review AGAIN until we get a GO from it with NO new issues reported or warnings of any kind! All results produced by this whole process MUST ALWAYS give us rock-solid PHYSICAL evidence that the fixed/improved codebase really works now as expected, with no false results and no bluff(s) of any kind."

When the §11.4.125 code-review returns ANY finding (BLOCKING, nit, or warning) and the author fixes/improves per that review, the code review MUST BE RE-RUN — and KEEP being re-run after each remediation round — until a clean GO with ZERO new issues AND ZERO warnings. A single "addressed the findings" pass is NOT sufficient: the corrected batch MUST pass a FRESH adversarial review (a re-review can surface NEW findings introduced by the fixes — the §11.4.1 fix-A-creates-B mode). The loop terminates ONLY on a clean GO; a residual warning is itself a finding that re-arms the loop. Every round's verdict AND every fix's validation MUST carry rock-solid PHYSICAL captured evidence per §11.4.5 / §11.4.69 / §11.4.107 (captured audio / video / sysfs / dumphsys / sink-side / runtime-signature) proving the fixed codebase REALLY works — never metadata-only / config-only /

absence-of-error / grep-without-runtime; no false results, no bluff; a GO unbacked by physical evidence is a §11.4 PASS-bluff at the review-loop layer. REFINES / STRENGTHENS §11.4.125 (iterate-until-no-blocking): makes the loop EXPLICIT, raises termination to ZERO findings AND ZERO warnings, BINDS physical evidence to every round. Classification: universal (§11.4.17). Composes §11.4.125 / §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.69 / §11.4.107 / §11.4.50 / §11.4.108 / §11.4.123. Gate `CM-COVENANT-114-134-PROPAGATION` (literal `11.4.134`) + rec. gate `CM-CODE-REVIEW-ITERATE-UNTIL-GO` + paired §1.1 mutation.

Canonical authority: `Constitution.md` §11.4.134. Release blocker. No `--skip-rereview` / `--single-review-pass` / `--warnings-ok` / `--evidence-optional` escape.

§11.4.135 — Standing regression-guard suite + every-fixed-defect-gets-a-permanent-regression-test (User mandate, 2026-06-08). Every project MUST maintain a STANDING regression-guard suite that runs on EVERY build+deploy and BLOCKS the release tag on any failure. Every closed defect (stable ticket id, e.g. ATM-NNN) MUST, in the SAME commit as its fix (extending the §11.4.43 DOCUMENT step), register a permanent §11.4.115 RED-on-broken-artifact regression test into the suite — `RED_MODE=1` capturing the historical defect on a pre-fix artifact (the proof the guard is real), `RED_MODE=0` the standing GREEN guard asserting the defect is ABSENT. A closure without a registered guard is a §11.4.123 violation. The suite runs FIRST in the post-deploy cycle (highest-risk set per §11.4.132) and is a §11.4.40 release-gate blocker. Forensic anchor (FACT): the wrong-subtitle-on-2nd-display defect was "fixed" via a source-side `CONTROL_MENU_LABEL_DENYLIST` that NO test mirrored or re-ran, so the NEXT chrome class recurred silently while the GREEN suite passed. Industry-standard bug-driven testing (Google content-driven testing; AOSP CTS/Tradefed) made mechanical + enforced. Composes §11.4.4 / §11.4.40 / §11.4.43 / §11.4.46 / §11.4.50 / §11.4.107 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.124 / §11.4.130 / §11.4.132. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-135-PROPAGATION` (literal `11.4.135`) + recommended gates `CM-REGRESSION-GUARD-REGISTERED` / `CM-REGRESSION-GUARD-SUITE-WIRED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.135. Non-compliance is a release blocker. No escape hatch — no `--skip-regression-guard`, `--no-guard-on-close`, `--guard-optional` flag.

§11.4.136 — Real-content end-to-end playback-test mandate (User mandate, 2026-06-08). Refines/strengthens §11.4.107. Any test asserting media playback works MUST drive REAL content (catalog stream or offline reference clip) through the user's path (§11.4.48 UI-driven → §11.4.117 CV/OCR fallback) and assert it genuinely PLAYS via the §11.4.107 liveness battery PLUS a decoder-health census — a numeric drop-buffer budget, no buffer-timestamp re-order/discard, no codec-reject (cite Android/Media3 ExoPlayer OEM pre-OTA playback-test mandate: "too many dropped buffers" >25, "unexpected presentation timestamp", "test timed out"). Metadata-only / launch-only / registration-only / single-frame / config-only PASS is forbidden (§11.4 / §11.4.1). A golden/reference clip corpus (BBC ExoPlayer testing samples) is the offline ground-truth. Composes §11.4.5 / §11.4.48 / §11.4.50 / §11.4.107 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-136-PROPAGATION` (literal `11.4.136`) + recommended gate `CM-REAL-CONTENT-PLAYBACK-TEST` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.136. Non-compliance is a release blocker. No escape hatch — no `--launch-proves-playback`, `--skip-decoder-health`, `--metadata-playback-pass-suffices` flag.

§11.4.137 — Subtitle/caption content-correctness oracle + secure-display-proxy-honesty mandate (User mandate, 2026-06-08). Refines §11.4.117 + §11.4.107 + §11.4.112. Forensic anchor (FACT): tests tasked to "physically verify the 2nd-display subtitle" PASSEd GREEN while subtitles did NOT show / showed WRONG — the streaming player surface is FLAG_SECURE so `screencap -d <secondary>` returns BLACK (autonomous PIXEL verification structurally impossible per §11.4.112), so the test fell back to the accessibility-scraped/`persist.atmosphere.subdebug` proxy, and the proxy accepted a chrome/menu LABEL (`Аудио и субтитры`) as a valid subtitle because the prose floor accepted any multibyte prose and NO menu-label denylist + NO position/cadence check existed. The mandate: a subtitle-correctness test MUST classify the cue's *content class* — a present cue is NOT a correct cue. CHROME (FAIL) if a known control/menu label (closed multilingual deny-list MIRRORED from source, case-folded incl. non-ASCII), time/numeric chrome, not prose, OUTSIDE the lower safe-title band (CEA-708 9-anchor grid), OR STATIC across the window (real subtitle changes → ≥2 distinct prose cues, a metamorphic relation). DIALOGUE (PASS) only when prose + not-denied + not-chrome + position-ok + cadence ≥2 OR fuzzy-matches the SOURCE-extracted cue via normalized edit distance (§11.4.123 host ground truth).

The oracle MUST be self-validated golden-good/golden-bad (§11.4.107(10)) and the deny-list MUST be verified present in the SHIPPED artifact (§11.4.108) — a source-green denylist with no test mirror + no artifact check is the exact recurrence pattern forbidden here. Secure-display honesty (§11.4.112): where FLAG_SECURE makes pixel verification impossible, the rock-solid autonomous proof is the player's caption telemetry + source-track presence + content-class oracle — NEVER a faked pixel "physical" pass; human-eye pixel confirmation is `operator_attended` (§11.4.52) with a tracked migration item. App-agnostic (keys off content class). Composes §11.4.3 / §11.4.5 / §11.4.6 / §11.4.107 / §11.4.108 / §11.4.112 / §11.4.115 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-137-PROPAGATION` (literal `11.4.137`) + recommended gate `CM-SUBTITLE-CONTENT-CORRECTNESS-ORACLE` + paired §1.1 mutation (strip the denylist/position/cadence check → golden-bad `Аудио и субтитры` `PASSes` → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.137. Non-compliance is a release blocker. No escape hatch — no `--present-cue-is-correct`, `--skip-chrome-oracle`, `--length-heuristic-suffices`, `--pixel-pass-on-secure-display`, `--skip-position-check`, `--skip-cadence-check` flag.

§11.4.138 — Operator-escape => mandatory bluff-audit + permanent guard (User mandate, 2026-06-08). When the operator (or any out-of-band channel) finds a defect that the GREEN test suite passed, this is by definition a §11.4 PASS-bluff — it MUST trigger, before the fix is closed: (1) a §11.4.102 systematic-debugging pass to FACT-root-cause; (2) a bluff-audit identifying the EXACT assertion that should have caught it but didn't, cited to `file:line` (canonical example: `lib/subtitle_content_validation.sh:sub_is_prose()` returning TRUE for `Аудио и субтитры`); (3) a permanent §11.4.135 regression guard registered in the SAME commit as the fix, with its §11.4.115 RED capturing the operator-found defect; (4) the bluff-audit committed under `docs/research/<scope>/<defect>_bluff_audit/`. Closing an operator-found defect WITHOUT the bluff-audit + permanent guard is itself a §11.4 violation (the bluff that let it through is still live and the defect will recur). Composes §11.4 / §11.4.1 / §11.4.102 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.135. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-138-PROPAGATION` (literal `11.4.138`) + recommended gate `CM-OPERATOR-ESCAPE-BLUFF-AUDIT` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md`

§11.4.138. Non-compliance is a release blocker. No escape hatch — no `--close-without-bluff-audit` , `--operator-find-is-just-a-bug` , `--skip-permanent-guard` flag.

§11.4.139 — Fresh-process clean-artifact runtime-signature mandate (User mandate, 2026-06-08). Refines §11.4.108. Before any post-deploy validation — ESPECIALLY a non-pixel proxy verification (the subdebug/accessibility-cue channel used for FLAG_SECURE displays) — the harness MUST assert `running-artifact == built-artifact`: the deploy yielded a CLEAN target (mutable-overlay/userdata wiped) OR a pre-validation check proves no stale overlay shadows the deployed code (e.g. every guarded package — incl. the Presenter that emits the subtitle cue — resolves to the system partition, no per-user override). A stale shadow of the cue-emitting component (e.g. a Presenter APK predating the denylist) makes the proxy report on code that was never deployed — any PASS is a §11.4 PASS-bluff. Each fix declares ONE machine-checkable runtime signature verified on the clean target (the §11.4.108 registry IS the definition of done); for the subtitle class the signature is "the shipped Presenter APK contains the denylist literal (case-insensitive) AND the subdebug channel emits `candidate REJECTED reason=chrome-label` for a menu label." Composes §11.4.46 / §11.4.108 / §11.4.130 / §11.4.135 / §11.4.137. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-139-PROPAGATION` (literal `11.4.139`) + recommended gate `CM-CLEAN-ARTIFACT-RUNTIME-SIGNATURE` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.139. Non-compliance is a release blocker. No escape hatch — no `--validate-against-running-state` , `--skip-clean-precondition` , `--shadow-OK` flag.

§11.4.140 — Universal action-prefix system (ACTION_NAME ::) (User mandate, 2026-06-09; GRAMMAR_ADDENDUM 2026-06-09; arrow form 2026-07-02). When a user prompt's FIRST non-blank line starts with a recognised action prefix, you MUST: (1) look the action token up in the action registry `constitution/actions/registry.yaml` (or `$HELIX_ACTION_REGISTRY`); (2) if it is a registered action, REPLACE the prefix with that action's `expansion` text and apply its `rules` ; (3) execute the remainder of the prompt under the expanded instruction. **Five EQUIVALENT forms** — same action, same expansion, same execution: (1) `ACTION_NAME :: <rest>` (bare `::`), (2) `PREFIX::ACTION_NAME :: <rest>` (namespaced `::`), (3) `/ACTION_NAME <rest>` (bare slash), (4) `/PREFIX::ACTION_NAME <rest>` (namespaced slash), (5) `ACTION_NAME ---> <rest>` (bare arrow) $\equiv$ `PREFIX::ACTION_NAME ---> <rest>` (namespaced arrow). Thus `BACKGROUND :: x` $\equiv$ `DEFAULT::BACKGROUND :: x` $\equiv$ `/BACKGROUND x` $\equiv$ `/DEFAULT::BACKGROUND x` $\equiv$ `BACKGROUND ---> x` $\equiv$ `DEFAULT::BACKGROUND ---> x` . `PREFIX` is an action NAMESPACE; the reserved default namespace is **DEFAULT** , and an action runs WITH or WITHOUT the prefix. Grammar (all hold): anchored at the FIRST non-blank line only (mid-prose tokens never match); the action token AND the namespace are UPPERCASE-only `[A-Z][A-Z0-9_]*` (lowercase never matches); the namespace separator `::` inside the token carries NO surrounding spaces (`PREFIX::ACTION_NAME`), DISTINCT from the action-body separator `" :: "` (one ASCII space on each side of `::` — avoids `C+ + Foo::Bar` , YAML `key: value` , URLs) in forms 1/2, the arrow-body separator `" ---> "` (one ASCII space on each side) in form 5, and the slash-body separator (one space) in forms 3/4; stacked prefixes (`A :: B :: rest`) apply outer-to-inner, left-to-right (expand `A` , re-scan, expand `B` , then the residual is the task); a leading `\` escapes the prefix for the `::` , arrow, and slash forms (`\BACKGROUND :: x` , `\BACKGROUND ---> x` , `\ /BACKGROUND x` — treat literally, strip the backslash, NO expansion) so action names can be discussed. **Conflict rule (slash form):** `/ACTION_NAME` (form 3) is honored as the action ONLY when `ACTION_NAME` (case-folded) does not collide with a built-in/host slash command (registry `slash_bare: auto` + `slash_conflicts: [..]`); form 4 (`/PREFIX::ACTION_NAME`) is ALWAYS unambiguous and always honored. An unknown token that matches the grammar shape (any of the 5 forms) but is NOT registered is NEVER silently

expanded or silently dropped — ask which registered action was meant (§11.4.66 / §11.4.105) or treat it as a literal prompt, NEVER invent an expansion (§11.4.6); any prompt not satisfying the grammar is an ordinary prompt and the system is a no-op. The registered action **BACKGROUND** expands to: *"The following prompt that we will provide MUST BE executed in background in parallel with all main work streams using the subagents-driven development approach! All work done MUST PRODUCE rock solid evidence covered with hard physical proof(s) that all done is working as expected and as specified without any false results and without any bluff!"* (composes §11.4.20 / §11.4.70 subagent-driven, §11.4.58 / §11.4.103 parallel streams, §11.4.89 background execution, §11.4.5 / §11.4.69 / §11.4.107 captured physical evidence, §11.4 anti-bluff). A second registered action **REMINDER** re-surfaces previously-scheduled, CRITICAL, status-UNCERTAIN work: FIRST verify the ACTUAL current status from captured evidence (never assume done or not-done — §11.4.6 no-guessing), THEN act on the delta — report the captured proof if genuinely complete, resume from the exact point if partial, action it NOW if not started, or surface the block per §11.4.66/ §11.4.101 — always producing a status verdict (composes §11.4.6 / §11.4.87 / §11.4.94 / §11.4.97 / §11.4.103 / §11.4.108 / §11.4.130 / §11.4.147). The system is UNIVERSAL (every CLI agent reads this block via its context carrier per §11.4.35), extensible (new action = new registry row), decoupled + reusable (§11.4.28), and loads out-of-the-box. Classification: universal (§11.4.17). **Canonical authority:** constitution submodule `Constitution.md` §11.4.140. Non-compliance is a release blocker. No escape hatch — no `--skip-action-prefix` , `--ignore-prefix` , `--no-registry` , `--invent-expansion-OK` , `--single-layer-only` flag.

§11.4.141 — Token-efficiency mandate (research-derived + operator mandate, 2026-06-09). Every project worked on by AI coding agents MUST cut token spend (input AND output) toward **30–40% of current (a 60–70% reduction)** WITHOUT degrading quality/performance/safety or breaking any existing mechanism, via a composable, safety-ranked measure set: (1) **prompt-cache the static governance prefix** — the always-loaded governance forms a byte-stable cache-breakpointed prefix with no volatile bytes ahead of it; cache reads cost $\sim 0.1 \times$ base input (the dominant cost driver — measured $\sim 170K$ tokens of governance re-sent every turn, externally corroborated by Claude Code issue #24147); caching is transparent so it

removes no rule, weakens no gate, changes no verdict — only billing (PRIMARY, biggest + safest lever); (2) **subagent model-tiering + output-to-file** — mechanical non-judgment work (search/grep/status/doc-export/read-only probes) to a Haiku-class model, the strong model RESERVED for all reasoning/verdicts/fix-design (§11.4.102)/code-review (§11.4.125)/demotion (§11.4.7), large output persisted to a file not an inline 350–520K-token transcript; the cheap model never emits a PASS so §11.4.50 + anti-bluff are untouched; (3) **thin always-loaded INDEX + on-demand detail** — concise index (one line per fix/anchor, EACH carrying the literal 11.4.N token so propagation gates pass) with the canonical full text kept gate-scanned in constitution/Constitution.md and reachable in one hop — a deduplication realising §11.4.35, never a deletion; (4) **CodeGraph/retrieval-first over full-file loading** (§11.4.78/§11.4.79); (5) **output-token reduction** — terse status + effort:"low" on the mechanical allowlist only; (6) **tool-call batching + no re-reads**; (7) **compaction/context-editing for long sessions. Mandatory measured proof:** a token-accounting harness measures tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload from the authoritative usage object (input/cache_read/cache_creation/output split; NEVER tiktoken, NEVER the client-side cost estimate), reproduced N times (§11.4.50), pass = AFTER ≤ 40% of BEFORE OR the measured best-safe reduction with a cited cold-cache reason; the AFTER run MUST show ZERO regression on the pre-build sweep + meta-test mutation sweep + propagation gates + a strong-model reasoning probe + a cache-warm proof (cache_read_input_tokens > 0) — cost reduction with quality regression is a §11.4 FAIL. The headline number is the *measured* reduction, never the design estimate (§11.4.6/§11.4.123). No measure may break/degrade any existing mechanism, and the rule is structured so none can. Composes

§11.4.5/.6/.20/.40/.50/.58/.69/.70/.78/.79/.80/.103/.106/.123/.125/.128/§12.6/§1.1. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-141-PROPAGATION (literal 11.4.141) + recommended gate CM-TOKEN-EFFICIENCY + paired §1.1 mutation (inject a pre-breakpoint volatile token → cache collapses → measured reduction falls below bar → gate FAILs). **Canonical authority:** constitution submodule Constitution.md §11.4.141. **Project instantiation (§11.4.35):** [the consuming project fills in: confirm the Claude Code governance-prefix cache is warm + free of pre-breakpoint volatiles; tier the §11.4.96-SAFE mechanical subagents (search/grep/status/doc-export) to Haiku with output-to-file under qa-results/; replace the consumer CLAUDE.md 112-row Applied-Fixes inline table + verbatim anchor restatements with a literal-anchor index pointing at

constitution/Constitution.md ; keep CodeGraph/lumen-first navigation; run the harness BEFORE/AFTER to prove the measured reduction.] Non-compliance is a release blocker. No escape hatch — no `--skip-token-efficiency` , `--no-cache-governance` , `--assert-reduction-without-measuring` , `--tier-down-reasoning` , `--inline-all-governance` , `--tiktoken-estimate-OK` flag.

§11.4.142 — Universal code-review mandate — every change reviewed, always, no exception (User mandate, 2026-06-09). Verbatim operator mandate: "ALL changes we do MUST pass through the code review step!!! ALWAYS!!!" EVERY change made to ANY repository this Constitution governs — without exception — MUST pass through an INDEPENDENT code-review step BEFORE it is accepted, committed, or built. NO change class is exempt: source code, fixes, tests, gates, meta-test mutations, documentation, doc-tooling, build/CI scripts, configuration, governance files (Constitution / CLAUDE / AGENTS / QWEN), conductor main-stream edits, sub-agent output, refactors, single-line edits — if a diff exists, it gets an independent review. This is the ABSOLUTE form of §11.4.125 (code-review agent gate after a batch, before pre-build sweep + main build): §11.4.142 strips every implicit scoping §11.4.125's "after all fixes/changes/ implementations are done" phrasing could leave open — no "just a doc edit", no "just a one-liner", no "the author already self-reviewed (§11.4.92)", no "trivial change" carve-out. **Independence is load-bearing** — the reviewer MUST be structurally separated from the author (a dedicated code-review agent, subagent-driven by default per §11.4.70 / §11.4.20, or a distinct human), NEVER the author re-reading their own work; §11.4.92's multi-pass self-evaluation is the AUTHOR-side discipline and PRECEDES (never satisfies) §11.4.142. **The review is itself anti-bluff** (§11.4 / §11.4.1) — a rubber-stamp "looks good" is not a review; it MUST genuinely analyse correctness, safety (no host §12 / data §9 / target-hardware §11.4.133 regression), will-it-really-work (no solve-A-create-B), end-user behaviour (§11.4 / §107), and test-genuineness (§1.1), its conclusions captured evidence per §11.4.5 / §11.4.69, and **it iterates to a clean GO per §11.4.134** — any finding (BLOCKING / nit / warning) re-arms the loop, acceptance only on ZERO new findings + ZERO warnings with rock-solid physical evidence. Honest boundary (§11.4.6): a passing review does NOT replace §11.4.108 four-layer runtime-signature verification nor the §11.4.40 full-suite retest — it is one of MULTIPLE STRONG LAYERS, and the FIRST one every change crosses. Composes §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.20 / §11.4.40 / §11.4.69 / §11.4.70 / §11.4.92 / §11.4.108 / §11.4.110 / §11.4.125 / §11.4.134 / §107 / §1.1. Classification: universal

(§11.4.17) — the consuming project supplies its reviewer-dispatch mechanism + change-acceptance seam (commit wrapper / merge queue / PR gate) per §11.4.35. Propagation gate `CM-COVENANT-114-142-PROPAGATION` (literal `11.4.142`) + recommended gate `CM-EVERY-CHANGE-REVIEWED` (every accepted change carries a fresh independent-review-completed marker for its diff, produced by a reviewer distinct from the author, before acceptance/commit/build) + paired §1.1 mutation (strip the literal → propagation gate FAILs; accept a diff with no independent-review marker → `CM-EVERY-CHANGE-REVIEWED` FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.142. Non-compliance is a release blocker. No escape hatch — no `--skip-review`, `--no-review`, `--trivial-change-exempt`, `--doc-edit-exempt`, `--self-review-suffices`, `--review-after-commit` flag.

§11.4.143 — Real-user-journey mandate for video-streaming-app full-automation tests (User mandate, 2026-06-10). Verbatim operator mandate: "All video streaming apps ... require to choose some title and to press proper UI button to start or resume playing! Proper UI interaction to play exact show with proper content and subtitles is MANDATORY! Without it we just eventually play on 2nd display sample, and that's it mostly! THIS MUST BE ADDED as MANDATORY RULE regarding testing of any video streaming app in general with full automation tests! Universal in root constitution + a consuming project's extensions." Any full-automation test that asserts a video player / streaming application plays content MUST drive the REAL end-user journey through the app's OWN UI — launch → BROWSE the actual catalog → choose a SPECIFIC title → press the real Play/Resume button → confirm THAT chosen content is genuinely playing, with its correct subtitles, on the intended routing target. A test that bypasses the journey with a sample / demo / built-in-loop clip, a deep-link / `am start -a VIEW` / intent shortcut, a synthetic or pre-staged stream, or any path that does NOT exercise the app's own browse-select-play UI is a §11.4 PASS-bluff at the user-journey layer: it validates ROUTING (that *something* reaches the display) while leaving the user-visible behaviour the operator cares about — "the show I picked actually plays" — unproven (the operator's "we just eventually play on 2nd display sample, and that's it mostly" gap). The mandate (ALL hold): (1) **Real journey, not a shortcut** — launch → catalog browse → specific-title selection → real Play/Resume press through the app's own UI (§11.4.48 UI-driven), NEVER a deep-link / intent / sample / loop-clip shortcut; (2) **Chosen-content confirmation** — the PASS proves the SPECIFIC selected title is playing (not merely that pixels move on the target)

via the §11.4.107 liveness battery on the §11.4.136 real-content path + the §11.4.137 subtitle content-correctness oracle for the chosen title's captions; (3) **Non-introspectable UIs use the pixel oracle** — when the app's accessibility hierarchy is blank/unreliable (TV-Compose / leanback / canvas / GL), DRIVE input + ASSERT content via the §11.4.117 CV/OCR pixel oracle, never a hierarchy-only tool; (4) **Login via the credential single-source** (§11.4.10), never hardcoded, never logged; (5) **Honest SKIP, never a faked PASS** — where the autonomous journey is genuinely infeasible (hard human-only login / CAPTCHA, geo-block per §11.4.3, secure-surface pixel-blanking per §11.4.112) the test is

`operator_attended SKIP-with-reason` per §11.4.52 + §11.4.3 with a tracked migration item — NEVER a metadata-only / sample-played / routing-only PASS. Honest boundary (§11.4.6): "the routing fired so the title is playing" is a guess — only the chosen-content liveness + subtitle oracle on the real journey proves it. Composes §11.4.48 / §11.4.107 / §11.4.117 / §11.4.136 / §11.4.137 / §11.4.52 / §11.4.3 / §107 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its concrete app roster, login-credential source, routing target, and UI-driving / pixel-oracle harness per §11.4.35. Propagation gate `CM-COVENANT-114-143-PROPAGATION` (literal `11.4.143`) + recommended gate `CM-VIDEO-REAL-JOURNEY-TEST` (every video-streaming-app playback test drives the real browse-select-play journey + confirms the chosen content + subtitles, or SKIPS-with-reason) + paired §1.1 mutation (replace a real-journey test's browse-select-play path with a deep-link / sample shortcut → gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.143. Non-compliance is a release blocker. No escape hatch — no `--sample-playback-ok`, `--skip-real-journey`, `--deep-link-suffices`, `--routing-only-pass`, `--no-title-selection` flag.

§11.4.144 — Tracked/recorded-device availability-following mandate (User mandate, 2026-06-10). Direct operator mandate: every device the project is tracking / following / recording (test / debug / manual-testing device, across every reachable transport — USB / wireless ADB / SSH / serial / network introspection API) MUST be availability-FOLLOWED — its connection state continuously monitored, any drop handled, never silently abandoned and never presented as a continuous recording. The §11.4.128 always-on recorder KNOWS a tracked device is absent (its per-device loop guards on the reachable state) but, lacking a following discipline, merely spins idle — no data captured, no offline event logged, no

resume, no escalation — so the recording corpus presents a continuous timeline with a silent hole, a §11.4 PASS-bluff at the recording-integrity layer (it claims continuous capture while no data exists for the gap). On a tracked device leaving its reachable state the system **MUST**, automatically: (a) **DETECT** the drop and **log an honest offline event** into the recording corpus (§11.4.6 — a silent gap presented as continuous capture is a fabricated-continuity bluff; §11.4.128 — a silent recording gap defeats always-on recording); (b) **WAIT** for the device to return using the project's ALREADY-DEFINED reconnection timings — never invented numbers (§11.4.6), the SAME grace / reconnect / poll budgets the project's recovery path already uses; (c) **RE-ATTACH** — resume recording / tracking the moment the device returns and log an honest online / resume event; (d) **ESCALATE** to the project's sanctioned device-recovery path (per §11.4.69 feature class `device_recovery`) if the device does not return within the defined timeout — through the sanctioned, authorization-gated recovery entry point **ONLY**, never bypassing its gate and never performing a destructive recovery (e.g. a power-cycle) autonomously without that authorization (§11.4.21 / §11.4.101 — high-blast-radius recovery is gated; while blocked the system keeps following the device, logging the blocked-escalation honestly). A tracked-device drop that produces a silent corpus hole, a never-resumed recording, or a never-escalated permanent absence is the bluff this anchor forbids. Composes §11.4.128 (always-on recording — §11.4.144 closes its drop-handling gap) / §11.4.69 (`device_recovery` sink-side positive evidence) / §11.4.6 (honest offline / online events, reused-not-invented timings) / §11.4.14 (watchdog children reaped on stop) / §11.4.21 + §11.4.101 (gated, non-autonomous escalation). Classification: universal (§11.4.17) — the consuming project supplies its concrete tracking transport, device set, already-defined reconnection timings, and sanctioned recovery entry point per §11.4.35.

Propagation gate `CM-COVENANT-114-144-PROPAGATION` (literal `11.4.144`) + recommended gate `CM-DEVICE-AVAILABILITY-FOLLOWED` + paired §1.1 meta-test mutation (strip the watchdog wiring / let an absence go unlogged → gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.144. Non-compliance is a release blocker. No escape hatch — no `--skip-availability-following` , `--silent-recording-gap-OK` , `--no-reconnect-wait` , `--invent-reconnect-timing` , `--skip-recovery-escalation` , `--autonomous-power-cycle-OK` flag.

§11.4.145 — Independent multi-angle impact-research per change (User mandate, 2026-06-10). For EVERY fix / change / new feature, INDEPENDENT impact-research agents (subagent-driven §11.4.70/§11.4.20, structurally separate from the author — §11.4.92 self-eval PRECEDES, never satisfies — adversarial "refute-the-change" stance to defeat the documented LLM confirmation-bias failure mode) MUST research the change AND its connected/dependent features across a CLOSED SET OF EIGHT ANGLES — (1) correctness/logic, (2) regression (what existing working feature could break — via call-graph impact enumerating every direct + transitive caller and contract-dependent feature, each mapped to its test), (3) latent/dangerous-code — the recents class (runtime-only failure, interface/ABI/contract mismatch, concurrency/data-race, resource lifecycle), (4) security (taint/injection/secret/unsafe-API), (5) performance (latency/memory/thermal under load vs baseline), (6) host/data/target-hardware safety per §12/§9/§11.4.133, (7) cross-feature interaction (shared state/timing/hardware contention), (8) business-logic conformance (matches the spec AND the connected features' contracts, not merely compiles). Forensic anchor (FACT, project-internal): the recents / 3-button-nav path shipped a latent AIDL-interface-contract mismatch that PASSED code review yet failed at RUNTIME ONLY (ANR on a recents-reaping gesture) because no review angle interrogated the interface contract / concurrency-lifecycle / silently-broken connected feature — the §11.4.108 SOURCE → ARTIFACT → RUNTIME → USER-VISIBLE gap caught only after the fact; §11.4.145 shifts that discovery LEFT. Each angle MUST name the TOOL(S) used + obtain the required DEPTH (the diff, the full changed unit, its declared contract, the spec section, every connected feature — NEVER the diff lines alone) + emit a captured-evidence conclusion per §11.4.5/ §11.4.69 ("looks fine" forbidden, §11.4.1); a genuinely-N/A angle is recorded

NOT_APPLICABLE: <reason> per §11.4.6, never silently skipped. Output = a per-change impact-research REPORT (one section per angle: tool + evidence path + verdict) + a single GO/NO-GO that BLOCKS acceptance/commit/build on ANY unmitigated risk in ANY angle; the change is fixed/mitigated + affected angles re-researched, iterating to a CLEAN GO (zero unmitigated risk + zero warning) per §11.4.134 before the §11.4.125 review gate. Honest boundary (§11.4.6): a GO proves cross-angle internal-consistency + bounded blast-radius — it does NOT replace §11.4.108 runtime-signature verification nor §11.4.40 full-suite retest; it is one of MULTIPLE STRONG LAYERS, the FIRST research pass every change crosses. Composes/strengthens §11.4.92/.125/.108/.110/.134/.78/.79/.107/.85/.133/ §12/§9/.99/.6/§1.1. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-145-PROPAGATION (literal 11.4.145) + recommended gate CM-

IMPACT-RESEARCH-PER-CHANGE + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; accept a change with a report missing an angle or an unmitigated NO-GO → CM-IMPACT-RESEARCH-PER-CHANGE FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule

Constitution.md §11.4.145. Non-compliance is a release blocker. No escape hatch — no --skip-impact-research , --single-angle-suffices , --author-researches-own-change , --diff-skim-OK , --no-go-overridable , --trivial-change-no-research flag.

§11.4.146 — Reproduce-first test + same-test-confirms-fix + mandatory extend-to-all-cases workflow (User mandate, 2026-06-10). Every reported problem MUST be handled by a NAMED three-step test workflow — §11.4.146 does NOT re-author its component disciplines, it NAMES + BINDS them into the operator's exact per-defect sequence and adds ONLY the two emphases the components leave implicit. **(STEP 1 — REPRODUCE-FIRST + INVESTIGATE)** before any fix, author the §11.4.43 RED test as a §11.4.115 RED-baseline-on-the-broken-artifact (reproduce the defect on the CURRENT pre-fix artifact, capture defect-present physical evidence per §11.4.5/§11.4.69/§11.4.107); NEW EMPHASIS (D1) that same RED test is ALSO a deliberate investigation instrument per §11.4.102(A) — it MUST gather ADDITIONAL forensic data characterising the defect (triggers, boundaries, input/topology scope, adjacent failure modes) that feeds BOTH the fix design AND the STEP-3 extend scope; a RED test used only as a binary present/absent gate with no characterisation satisfies §11.4.115 but NOT §11.4.146 STEP 1. **(STEP 2 — SAME-TEST-CONFIRMS-FIX)** after the fix the SAME test source (the §11.4.115 polarity switch flipped RED_MODE=1→0) confirms the defect ABSENT — RED-on-broken then GREEN-on-fixed, both captured, on a clean target per §11.4.108/§11.4.139, validated-first-after-redeploy per §11.4.130, deterministic per §11.4.50; no separate happy-path test substitutes (the §11.4.43/§11.4.115 PASS-bluff). **(STEP 3 — EXTEND-TO-ALL-CASES, mandatory per-fix)** NEW EMPHASIS (D2) immediately after STEP 2 the test MUST be FANNED OUT across the full case-space of the SAME functionality — all flows, valid + invalid + boundary (§11.4.85 empty/max/off-by-one) + concurrent (§11.4.85 contention) + failure-injection (§11.4.85 chaos) + topology variants (§11.4.3) — confirming no issue of any kind, with PROVABLE enumerated coverage per §11.4.118 (a listed case-set with per-case outcome, never "no other issues found"); a REQUIRED step, NOT deferred to a release-cycle discovery sweep and NOT reduced to a single guard; each user-visible case carries rock-solid physical evidence per

§11.4.123 + is registered into the §11.4.135 standing regression-guard suite; a newly-discovered fan-out defect triggers §11.4.4 test-interrupt + re-enters STEP 1. (ANTI-BLUFF, all steps) every PASS is rock-solid CAPTURED physical evidence per §11.4.123 (§11.4.5/§11.4.69/§11.4.107) — metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden (§11.4/§11.4.1); unclear validation method ⇒ deep-research-before-declaring-untestable per §11.4.123/§11.4.8/§11.4.99; an operator-found defect the green suite missed triggers §11.4.138. Honest boundary (§11.4.6): STEP 3 reduces the unknown-unknown surface but does not prove zero remaining defects (§11.4.118) — un-exercised cases stated as honest gaps, never silently implied clean. Composes §11.4.43/.115/.102/.130/.108/.139/.50/.85/.118/.135/.3/.123/.5/.69/.107/.138/.4/§107/§1.1. Classification: universal (§11.4.17). Propagation gate

CM-COVENANT-114-146-PROPAGATION (literal 11.4.146) + recommended gate CM-REPRODUCE-FIRST-THEN-EXTEND (every closed defect's fix carries a §11.4.115 reproduce-first polarity test with captured defect-characterisation [STEP 1] + its RED_MODE=0 GREEN confirmation [STEP 2] + an enumerated per-functionality extend case-set registered into the §11.4.135 suite [STEP 3]; a closure with the reproduce → confirm pair but NO enumerated extend case-set FAILs) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; close a defect with only the reproduce → confirm pair and no extend-case-set → CM-REPRODUCE-FIRST-THEN-EXTEND FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.146. Non-compliance is a release blocker. No escape hatch — no

```
--skip-reproduce-first , --fix-without-red , --skip-extend-to-all-cases , --reproduce-confirm-suffices ,  
--defer-extend-to-release-sweep , --single-guard-suffices , --no-defect-characterisation flag.
```

§11.4.147 — Crashed-agent respawn-until-complete + no-work-loss registry

mandate (User mandate, 2026-06-10). Verbatim operator intent: "any agent that

crashed because of something MUST BE respawned and finish its work at some point! We MUST NOT lose any work, forget about or have it corrupted!" Forensic case study (FACT): dispatched subagents died on TRANSIENT causes (API

Error: Server is temporarily limiting requests rate-limit killed 5 at once,

API Error: socket connection closed unexpectedly killed 2, one left

PARTIAL edits — two source files written, the dependent SystemUI sliders + doc + test NOT done), each re-dispatched by pure conductor vigilance with NO

mechanical guarantee — the moment conductor context is lost / compacted / the conductor itself crashes, an in-flight-but-dead work unit is silently dropped (lost / forgotten / corrupted). Every dispatched agent/subagent **MUST** be tracked through its full lifecycle so a crash **NEVER** loses, forgets, or corrupts its work; a crashed agent is **NOT** a completed agent (§11.4.6 — "it probably finished before dying" is a forbidden guess; abnormal termination is positive evidence the unit is **OPEN**). The mandate (ALL hold): **(a) durable agent REGISTRY** — every dispatched agent has an append-only machine-readable registry entry (stable id, task + §11.4.58 file-scope, declared output path(s) + §11.4.5/§11.4.69 evidence dir, dispatch timestamp, status from the CLOSED SET {dispatched | in-flight | crashed | respawned | complete}), reusing the §11.4.116 sync substrate (JSONL event stream + atomic status snapshot; "an entry with no dispatch-event cannot show complete "); the registry is the **SINGLE SOURCE OF TRUTH** for "is this work owed?" and an unregistered dispatch is itself a violation; **(b) mechanical CRASH-DETECTION + RESPAWN-until-complete** — any abnormal termination (transient rate-limit/socket-close, any non-completion exit, or a terminal error after the runtime's own retries are exhausted) flips the entry to `crashed` + keeps the unit **OPEN**, and the unit is respawned (fresh agent claims the same id + scope + preserved state) and re-respawned until an agent reaches `complete` ; respawn is the safe/reversible/bounded decision taken autonomously per §11.4.101 (never blocks the loop for a human), transient causes use the project's **ALREADY-DEFINED** backoff budgets (never invented, §11.4.6), a deterministically-reproducible non-transient terminal error is investigated per §11.4.102 before further respawn (never a blind retry loop); **(c) PARTIAL-STATE preserve** → **§11.4.84-check** → **resume-or-clean-restart** — the crashed agent's uncommitted edits + output doc are **PRESERVED** (never silently discarded → lost, never blindly committed → corruption), the respawn runs the §11.4.84 quiescence check on the preserved tree (every modified file accounted-for vs scope, no mutation/ // always pass / `_mutated_*` residue, no half-written/torn artifact), then **EITHER RESUMES** idempotently when it passes OR **CLEAN-RESTARTS** from a known-good base (§9.2 pre-op backup, reversible §11.4.101) when inconsistent — nothing lost, nothing corrupted; per-agent `git` `worktree` isolation (§11.4.58 L4 / §11.4.84) keeps the partial tree from contaminating other streams; **(d) "crash ≠ done"** **COMPLETION criterion** — a unit is **DONE** only when an agent reaches `complete` with its required captured evidence/output landed (§11.4.5/§11.4.69 + the §11.4.116 verdict-carries-evidence-path rule); the endless-loop done-condition (§11.4.87/§11.4.94/§11.4.97/§11.4.126) **MUST NOT** read satisfied while any entry is

dispatched / in-flight / crashed / respawned -not-yet- complete , the zero-idle survey (§11.4.94) treats every non- complete entry as an OPEN item to reclaim, and a registry showing complete without landed evidence is a §11.4 PASS-bluff at the agent-lifecycle layer. Honest boundary (§11.4.6): the registry + respawn guarantee work is not lost/corrupted, NOT that it is correct — the respawned output still crosses §11.4.108 + §11.4.125/§11.4.142 review + §11.4.40 retest; §11.4.147 is the durability layer beneath those, the agent-side analogue of §11.4.144 (same detect → wait/backoff → re-attach/respawn → escalate shape). Composes §11.4.6/.58/.84/.87/.94/.97/.101/.116/.102/.108/.125/.142/.40/.128/.144/ §9.2/§1.1. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-147-PROPAGATION (literal 11.4.147) + recommended gate CM-CRASHED-AGENT-RESPAWN-TRACKED + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; mark a crashed entry as the loop's done-condition complete without landed evidence, OR drop a dispatched agent without a registry entry → CM-CRASHED-AGENT-RESPAWN-TRACKED FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule Constitution.md §11.4.147. Non-compliance is a release blocker. No escape hatch — no --skip-agent-registry , --crash-equals-done , --no-respawn , --discard-partial-state , --blind-commit-partial , --forget-dead-agent , --loop-done-with-crashed-entries flag.

§11.4.148 — Workable-item integrity (status+type+id) + comprehensive structured description + bidirectional external-tracker sync + BLOCKED unblock-choices mandate (User mandate, 2026-06-10). BINDS + STRENGTHENS the workable-item discipline (§11.4.15 status / §11.4.16 type / §11.4.54 id / §11.4.21 Operator-blocked / §11.4.91 description-clarity / §11.4.93 + §11.4.95 SQLite SSOT / §11.4.106 docs_chain) into ONE integrity contract spanning DB ↔ docs ↔ external tracker, adding the operator's three emphases: **(D1) no item without a valid status + valid type + stable id — on ALL three surfaces** (an item missing any of the three in the DB, the rendered docs, OR the external tracker FAILs the validator — release-blocker; the id is the cross-surface binding key); **(D2) comprehensive structured description per item** — WHAT it is (§11.4.91 ≥6-word/≥40-char clear meaning) + HOW it manifests + HOW to reproduce (§11.4.115/.146) + ACCEPTANCE CRITERIA (§11.4.123/.69 captured-evidence verdict), in the §11.4.93 DB description column AND the docs AND the tracker, never a stub/§11.4.91-anti-pattern fragment; **(D3) BLOCKED items carry WHY + enumerated unblock CHOICES** — tightens §11.4.21: every

`Operator-blocked` item (incl. `Blocked` / `BLOCKED` documented alias normalised to canonical `Operator-blocked` , never a silent fork §11.4.6) MUST enumerate the closed list of decisions/actions that would unblock it (`[A]...·[B]...·[C]...` , mirroring §11.4.66's 2–4-option shape); a `BLOCKED` item with no enumerated choices FAILs; **(D4) regular never-missed bidirectional DB↔docs↔tracker sync** — the §11.4.93/.95 git-tracked SQLite DB is the SSoT, docs + tracker DERIVED; full sync (`db-to-md` / `md-to-db` byte-identical round-trip §11.4.93 + tracker push) runs regularly + on every change, §11.4.86 drift-proof fingerprint (sha256 of the sorted item keyset, NOT mtime) gates freshness, §11.4.106 docs_chain-bound so drift is mechanically caught not vigilance-dependent; **(D5) generic external-tracker push** carries statuses (collapsed onto the tracker's native set per §11.4.33/.112 when fixed, precise value preserved in a header, never lost), types, assignee (§11.4.104 participant handle, UNSET defaults from a project env var, never hardcoded/logged §11.4.10), and sub-tasks, IDEMPOTENT (dry-run-then-real, match-by-stable-key `[<ID>]` prefix / id custom-field, present⇒UPDATE/absent⇒CREATE, rate-limited, credential-redacted, sink-side `created=N` `updated=M` `failed=0` proof §11.4.69), the MACHINERY project-agnostic §11.4.28 (consumer registers its tracker/list/field-map at runtime). Anti-bluff §11.4: every sync + validator pass carries captured evidence §11.4.5/.69; honest boundary §11.4.6 — guarantees well-formed items + agreeing surfaces, NOT that the underlying work is correct (still crosses §11.4.108/.40/.123). Composes §11.4.15/.16/.21/.33/.34/.54/.66/.86/.91/.93/.95/.104/.106/.112/.123/.10/.28/.69/.6/ §1.1. Classification: universal (§11.4.17) — consumer supplies its DB path, id prefix, tracker service + list/board id + field map + default-assignee env var, docs_chain context per §11.4.35. Propagation gate `CM-COVENANT-114-148-PROPAGATION` (literal `11.4.148`) + recommended gates `CM-ITEM-INTEGRITY-STATUS-TYPE-ID` / `CM-ITEM-COMPREHENSIVE-DESCRIPTION` / `CM-BLOCKED-UNBLOCK-CHOICES` / `CM-TRACKER-SYNC-IDEMPOTENT` + paired §1.1 mutation (gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.148. Non-compliance is a release blocker. No escape hatch — no `--item-without-status` , `--item-without-type` , `--item-without-id` , `--stub-description-OK` , `--blocked-without-choices` , `--skip-tracker-sync` , `--one-way-sync-OK` , `--non-idempotent-tracker-push` , `--hardcode-assignee` flag.

§11.4.149 — Per-workable-item testing-diary mandate (User mandate, 2026-06-10). Every workable item MUST carry an append-only TESTING DIARY — a chronological record of every test run against it — distinct from §11.4.93 `item_history` (lifecycle STATE transitions) + §11.4.55 Reopens (reopen cycles): the diary records TEST EXECUTIONS (which may or may not change status). The mandate (ALL hold): **(a)** a `test_diary` table additive to the §11.4.93/.95 SQLite SSoT, one row per run soft-keyed to the item id, capturing ISO-8601-UTC `date_time` + `tested_by` (closed set `User|Operator|AI-agent|HelixQA`) + `result` (§11.4.45 `PASS|FAIL|SKIP` + detail) + in-depth observations (long markdown, facts or explicit `UNCONFIRMED:` / `PENDING_FORENSICS:` §11.4.6, the background a future fix needs) + `action_taken` + `status_changed` +from/to (did this run change status + WHY) + §11.4.69 `evidence_path` + `feature_class`, with a SCHEMA CONSTRAINT making a `PASS` row WITHOUT a non-empty evidence path impossible (a PASS-bluff rejected by the schema itself); **(b)** BOTH an in-depth diary doc + a derived at-a-glance summary VIEW (total/pass/fail/skip + last-verdict + last-run + status-changes + distinct testers/feature-classes, DERIVED never duplicated §11.4.93) feeding §11.4.132 risk-ordering; **(c)** four-format per-item `Diary / Diary_Summary` exports §11.4.65 + §11.4.86-drift-proof-fingerprinted (sha256 of the sorted diary keyset) + §11.4.106 `docs_chain`-bound so a diary row not exported/pushed FAILs a gate (never-missed); **(d)** external-tracker SUB-TASK model — the item task's description stays the item (§11.4.148 D2, NOT polluted with diary text), each run a CHILD sub-task (type `task`) with a `{TODO|In-progress|Completed}` lifecycle (collapsed per §11.4.33/.112), observations + action + an Evidence: line (path not raw artefact §11.4.10/.13) + a diary-entry idempotency key, reusing the §11.4.148 D5 idempotent rate-limited credential-redacted sink-side-proven push, mapper project-agnostic §11.4.28; **(e)** MINIMAL-LLM deterministic bash/Go tooling (zero LLM in the data path — the `observations` prose is authored by whoever ran the test; tooling only stores/renders/pushes/validates), 100% test-covered §11.4.27 (unit + integration-against-the-real-tracker with an honest §11.4.3 SKIP-when-token-absent never a faked PASS + export + HelixQA Challenge + paired §1.1 mutation) so a diary PASS-bluff is mechanically impossible. Honest boundary §11.4.6 — the diary guarantees a complete auditable per-item test-history, NOT that any single run's verdict is correct (rests on its own §11.4.69/.107/.123 evidence). Composes §11.4.6/.27/.45/.50/.55/.65/.69/.86/.93/.95/.106/.107/.123/.132/.148/.10/.13/.28/.33/.112/ §1.1. Classification: universal (§11.4.17) — consumer supplies its DB path, diary

doc layout, export formats, tracker sub-task field map, docs_chain context per §11.4.35. Propagation gate CM-COVENANT-114-149-PROPAGATION (literal 11.4.149) + recommended gates CM-TEST-DIARY-SYNC / CM-DIARY-PASS-REQUIRES-EVIDENCE + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.149. Non-compliance is a release blocker. No escape hatch — no --skip-testing-diary , --diary-without-evidence , --no-diary-summary , --diary-without-tracker-subtask , --llm-in-diary-data-path , --diary-export-optional flag.

§11.4.150 — Mandatory deep multi-angle web research per change/issue, before declaring fixed or structural (User mandate, 2026-06-11). Verbatim operator mandate: "For every single issue we fix or improvement we make — besides tight systematic-debugging, fixing, code review by independent agents, comprehensive tests — we MUST ALWAYS do deep web research from various angles! No matter how big/small/simple/complex, dig deep the internet for articles, technical documentation, APIs, open-source code — EVERYTHING that can help make the best possible solution OR confirm we don't have a more serious problem we're unaware of! We MUST do everything possible to STOP the constant issue-reopening and finally start closing items as fixed+working! Do this ALWAYS in parallel with the main work stream!" For EVERY fix / improvement / change / closure — no matter how big, small, simple, or complex — the agent MUST, IN ADDITION to tight systematic-debugging (§11.4.102), the fix, independent-agent code review (§11.4.125 / §11.4.142), and comprehensive multi-layer tests (§11.4.4(b) / §11.4.40), perform a DOCUMENTED **deep multi-angle web research pass** digging the internet from VARIOUS angles — articles, official + vendor technical documentation, API references, standards, issue trackers, maintainer guidance, reusable open-source code — to BOTH (i) discover the best possible solution AND (ii) confirm there is NOT a more serious underlying problem the team is unaware of. Forensic case study (FACT, 2026-06-11): a subtitle-on-secondary-display goal verdicted Won't-fix: structurally-impossible (§11.4.112 secure-surface pixel-blanking) was OVERTURNED by a deep multi-angle research pass that surfaced the real OCR-of-the-PRIMARY-display path — a "we can't" became a shipped capability; the canonical anti-pattern of a structural verdict reached for want of research. The mandate (ALL hold): **(A) No closure-as-fixed/ structural WITHOUT a documented deep-research pass** — no item may be marked Fixed / Implemented / Completed (§11.4.33) OR classified

structurally-impossible won't-fix (§11.4.112) until a deep multi-angle pass is documented; §11.4.150 makes §11.4.8's pass UNCONDITIONAL (every item, however trivial, at the closure AND structural-verdict gates). **(B) Multiple angles, not a single lookup** — ≥ 2 genuinely-distinct angles (official-docs / standards / known-bug-trackers / alternative-approach / failure-mode / security / performance / platform-constraint / open-source-precedent) so a single-source confirmation-bias miss (§11.4.145) cannot pass; a one-link drive-by is NOT deep multi-angle. **(C) Confirm-no-bigger-problem, not just find-a-fix** — explicitly seek evidence the fix does not mask a deeper defect AND the closure/verdict is genuinely safe; "we found nothing worse" requires the enumerated search (§11.4.118). **(D) Latest-source + cited** — LATEST authoritative versions per §11.4.99 (never training-data/memory), each cited by URL + access date in the research artefact AND the closure commit footer (`Deep-research <date>: <urls>` OR the literal `NO external solution found – original work` per §11.4.8). **(E) Reopen-breaking is the PURPOSE** — STOP the constant Fixed↔Reopened churn (§11.4.34 / §11.4.55); a reopen whose root cause a deep pass would have surfaced is a §11.4.150 miss; composes §11.4.7 (demotion-evidence) + §11.4.112 (structural verdict now REQUIRES the cited-authorities pass). **(F) ALWAYS in parallel with the main stream** — background subagent-driven (§11.4.70 / §11.4.20 / §11.4.103 / §11.4.89) concurrent with main fix/build/test work, NEVER serialising it or stalling the loop (§11.4.94 / §11.4.97 / §11.4.101 / §11.4.126). **(G) Apply ASAP to every workable item** — retroactively + going forward across the §11.4.93 / §11.4.95 SSoT; items closed without a documented pass are re-audited in the §11.4.40 / §11.4.42 release-gate sweep. Honest boundary (§11.4.6): the pass reduces the unknown-unknown surface + breaks the most common reopen causes — it does NOT prove zero remaining defects (§11.4.118) and does NOT replace §11.4.108 runtime-signature verification, §11.4.125 / §11.4.142 review, or §11.4.40 retest; it is the research layer every fix/closure/structural-verdict additionally crosses, and "we probably don't have a bigger problem" without the enumerated multi-angle search is a guess (§11.4.6), never a finding. Composes §11.4.8 / §11.4.99 / §11.4.123 / §11.4.118 / §11.4.145 / §11.4.125 / §11.4.142 / §11.4.7 / §11.4.112 / §11.4.34 / §11.4.55 / §11.4.70 / §11.4.20 / §11.4.89 / §11.4.103 / §11.4.40 / §11.4.42 / §11.4.93 / §11.4.95 / §11.4.6 / §1.1. Classification: universal (§11.4.17) — the consuming project supplies its concrete research corpora, angle set, item tracker, and closure-commit-footer convention per §11.4.35. Propagation gate `CM-COVENANT-114-150-PROPAGATION` (literal `11.4.150`) + recommended gate `CM-DEEP-RESEARCH-PER-ISSUE` (every closed/structural-verdicted item carries a

documented multi-angle deep-research artefact + cited-source closure footer, run in parallel, before the closure/structural verdict is accepted) + paired §11.1 mutation (strip the literal → propagation gate FAILs; close an item or reach a structural verdict with no documented multi-angle research artefact / cited footer →

CM-DEEP-RESEARCH-PER-ISSUE FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.150. Non-compliance is a release blocker. No escape hatch — no `--skip-deep-research`, `--single-source-suffices`, `--trivial-change-no-research`, `--close-without-research`, `--structural-without-research`, `--serialise-research`, `--research-from-memory-OK` flag.

§11.4.151 — Project-prefixed release-tag/version-naming mandate (User mandate, 2026-06-12). Verbatim operator mandate: "Every release tag and version name we create — on the main repository and on every Submodule we own — MUST be prefixed with the project's release prefix, e.g. `myproject-1.0.0-dev-0.0.1`. The prefix MUST come from `HELIX_RELEASE_PREFIX` in our `.env` if it is set, otherwise from the lowercased project root directory name. The SAME prefix MUST be used across the main repo and all owned Submodules in one release so a release is greppable across every repository." Every release tag AND every version name created on the main repository AND on every owned-by-us submodule (§11.4.28) MUST be prefixed with the project's release prefix, form `<PREFIX>-<version>` (canonical example: `myproject-1.0.0-dev-0.0.1`), so a release is identifiable + greppable across every repository it spans (one `git tag -l '<PREFIX>-*'` enumerates the whole release surface). **Prefix resolution order (closed-set, deterministic — §11.4.6 no-guessing):** (1)

`HELIX_RELEASE_PREFIX` from the project's `.env` — authoritative when set; `.env` is git-ignored per §11.4.30 and the variable is documented in the tracked `.env.example` (a §11.4.77 re-obtain mechanism, never committed); (2) fallback = the lowercased snake_case form of the project root directory name (no spaces) per §11.4.29 — used whenever the env var is unset/empty, so a prefix is ALWAYS resolvable from the checkout with zero operator input. **The prefix MUST be IDENTICAL across the main repo and all owned submodules within a single release** — a release that tags the main repo `<PREFIX>-1.2.0` while tagging an owned submodule with a different/unprefixed value is a §11.4.151 violation (the cross-repo grep no longer enumerates the release). Version codes increment monotonically within the prefix (`<PREFIX>-...-0.0.1` → `<PREFIX>-...-0.0.2` → ...), never reset, never skipped. Honest boundary (§11.4.6): the prefix guarantees a

release is identifiable + uniform across every repository, NOT that its contents are correct — the tag is still created only after the §11.4.40 full-suite retest GREEN and reaches every upstream via the §11.4.113 merge-onto-latest-main path (NEVER a force-push), fanned out per §2.1. Composes §2.1 / §11.4.29 / §11.4.30 / §11.4.40 / §11.4.113 / §11.4.126 (the release-scope terminal condition is a published, prefixed tag). Classification: universal (§11.4.17) — the consuming project supplies its concrete prefix value + the `HELIX_RELEASE_PREFIX` env var per §11.4.35.

Propagation gate `CM-COVENANT-114-151-PROPAGATION` (literal `11.4.151`) + recommended gate `CM-RELEASE-PREFIX-NAMING` (every release tag/version on the main repo + every owned submodule carries the resolved `<PREFIX>-` prefix, identical across the release) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; create an unprefixed or differing-prefix release tag → `CM-RELEASE-PREFIX-NAMING` FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.151. Non-compliance is a release blocker. No escape hatch — no `--no-release-prefix`, `--unprefixed-tag`, `--prefix-optional`, `--differing-submodule-prefix` flag.

§11.4.152 — Crashlytics-recorded-data continuous monitoring + systematic-debug + regression-test-coverage mandate (User mandate, 2026-06-13).

Verbatim operator intent: "For every project that has Firebase Crashlytics enabled / wired, we MUST continuously monitor ALL of the Crashlytics-recorded data — crashes, ANRs, performance traces, and non-fatals — systematically debug each, fix and improve, and cover everything with validation and verification tests. This MUST be checked regularly, with no false results and no bluff of any kind!" Every project that has Firebase Crashlytics enabled/wired (SDK linked into a shipping artifact, crash+non-fatal+ANR reporting active) MUST treat the Crashlytics console as a first-class captured-evidence channel from real end-user devices and continuously process every datum it records — an open Crashlytics issue with a green test suite is a §11.4 PASS-bluff at the field-telemetry layer (a real user hitting a broken feature while the suite reports green). STRENGTHENS+COMPLETES §11.4.47: §11.4.47 owns the periodic REVIEW + dedup + Issue-creation pass that SURFACES Crashlytics/Analytics/Performance findings; §11.4.152 owns what happens to each surfaced item AFTER — the systematic-debug → fix/improve → regression-test-coverage lifecycle that drives it to a proven regression-immune closure (§11.4.47 finds it; §11.4.152 fixes it + proves it stays fixed; both mandatory, neither substitutes). The mandate (ALL hold): (1) **continuous monitoring of ALL four surfaces** — fatal crashes, ANRs, performance traces/regressions, AND non-

fatals (skipping any one is a PASS-bluff; non-fatals are the silent class — every catch/fallback/recovered-error path is a real degraded UX and MUST be triaged, not ignored because the app did not crash); (2) **systematic-debugging of each (reproduce-before-fix)** per §11.4.102 Iron Law + §11.4.115 RED-baseline-on-the-broken-artifact (the recorded stacktrace/ANR-trace/non-fatal-context IS the §11.4.5/.69 captured defect-present evidence) BEFORE the fix — a fix without a prior reproducing test is a §11.4.43/.123 violation; unreproducible ⇒ deep-research-before-declaring-untestable §11.4.123/.150; (3) **a fix/improvement for every confirmed issue** against its proven root cause (§11.4.9/.43/.108), "acknowledged in console" is not a fix, muting a recurring issue without a tracked rationale is a §11.4.6/.90 violation; (4) **validation+verification regression-test coverage per closed issue** — in the SAME commit as the fix register a permanent §11.4.135 regression guard (§11.4.115 polarity test: `RED_MODE=1` captures the recorded defect on the pre-fix artifact, `RED_MODE=0` standing GREEN guard asserting ABSENT) exercising the same user-reachable path the stacktrace identifies, with rock-solid captured evidence §11.4.5/.69/.107/.123 + a closure log citing the console issue id/URL + root-cause + fix commit + the validation/verification test paths; a Crashlytics issue marked resolved WITHOUT a falsifiable regression test is FORBIDDEN (the silent-recurrence vector, §11.4.138 operator-escape class applied to field telemetry); (5) **regular cadence** reusing the §11.4.47 five-trigger set (pre-build/pre-flash/pre-distribute/pre-tag blocking; daily + post-deployment-burn-in non-blocking) — a one-time sweep never repeated is a violation; (6) **no false results, no bluff** (§11.4/.1/.6 — "no new issues" requires the enumerated monitored-surfaces+window §11.4.118; "fixed" requires the RED → GREEN flip §11.4.115/.130; a guard whose §1.1 mutation does not FAIL it is a bluff gate). Honest boundary §11.4.6: processing every recorded issue breaks the silent-recurrence vector — it does NOT prove zero remaining field defects nor replace §11.4.108/.40; an issue is "closed" only when its guard is GREEN on a clean artifact, never when the console mark is flipped. Classification: universal (§11.4.17) — the consuming project supplies its Crashlytics handle, console-access credential (§11.4.10, never logged), severity table, and per-issue closure-log path per §11.4.35; the reference project-level instantiation is a consuming project's §6.O (Crashlytics-Resolved Issue Coverage — per-issue validation test + Challenge test + `.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md` closure log) + §6.AC (Comprehensive Non-Fatal Telemetry — every catch/fallback records a non-fatal with triage context). Composes §11.4/.1/.5/.6/.9/.34/.40/.43/.47/.69/.90/.102/.107/.108/.115/.118/.123/.130/.135/.138/.150/

§1.1. Propagation gate `CM-COVENANT-114-152-PROPAGATION` (literal `11.4.152`) + recommended gate `CM-CRASHLYTICS-ISSUE-FULLY-COVERED` (every closed Crashlytics issue carries a systematic-debug root-cause record + a registered §11.4.135 regression guard + a closure-log entry citing the console issue id/URL + the validation/verification test paths; a console-resolved issue with no falsifiable regression guard FAILs) + paired §1.1 meta-test mutation (gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md`

§11.4.152. Non-compliance is a release blocker. No escape hatch — no `--skip-crashlytics-monitoring`, `--monitor-crashes-only`, `--skip-non-fatals`, `--resolve-without-regression-test`, `--console-mark-is-fixed`, `--monitor-once`, `--mute-without-rationale` flag.

§11.4.153 — Comprehensive per-feature Status + Status_Summary document set with mandatory video-recording confirmation (User mandate, 2026-06-15).

Every project MUST maintain, under `docs/features/`, a comprehensive feature Status document set (`Status.md` + its §11.4.56 `Status_Summary.md` companion) enumerating EVERY system component, EVERY client app/binary/surface (TUI / CLI / Web / desktop / mobile / API / gRPC / library / submodule / infrastructure), and EVERY feature — including features ported from any incorporated CLI-agent / submodule catalogue (§11.4.74) — with NO single feature left out. (1) **Total categorized coverage** — per-feature table organized Component → Category, reconciled against the actual codebase (a code-present feature missing from the table, or a row with no code, is a §11.4.6/§11.4.118 bluff). (2) **Per-feature fields** — Component / Feature / Category / Implementation / Wiring (genuinely end-user-reachable, not merely compiled, §11.4.108) / Real-use / Tests-coverage (four-layer §11.4.4(b)) / Validation (PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED per §11.4.45) / **Video-recording confirmation** (path to the real-use video or honest gap marker). (3) **Mandatory per-feature real-use video** — every user-visible "confirmed" claim backed by a recorded real-use video of a genuine end-user scenario (real prompts → real LLM/service responses → real results), NEVER a frozen/stale frame (§11.4.107), NEVER faked/mocked/demo-loop/bluff response or LLM error passed off as success (§11.4.2/§11.4.5), stored at the project-declared recording path (§11.4.35); a confirmed row with no real video or a bluff video = §11.4 PASS-bluff; autonomous-infeasible ⇒ honest §11.4.3/§11.4.52 SKIP + tracked migration item, NEVER a faked confirmation. (4) **Video-analysis remediation loop** — every video analysed; any defect it surfaces triggers §11.4.102 systematic-debug → fix → §11.4.146

retest → re-record → clean GO per §11.4.134 before "confirmed" (a video exposing a broken feature is a §11.4.4 test-interrupt). (5) **Always-in-sync** — §11.4.45-class roster/corpus-backed, §11.4.106 docs_chain-bound + §11.4.86 drift-proof fingerprint (sha256 of sorted feature-key roster AND sorted video-artefact roster, NOT mtime), re-syncs out-of-the-box; stale = violation. (6) **Four-format export** — HTML + PDF + **DOCX** (this doc class ADDS DOCX to the §11.4.65 HTML+PDF set; other classes unchanged), in sync per §11.4.60. (7) Follows §11.4.44/.45/.56/.57/.59/.60. (8) **MP4 format REQUIRED**. All video confirmations MUST be in .mp4 format (H.264). Window-specific capture ONLY (§11.4.159(A)). Vision validation REQUIRED (§11.4.159(D)). .cast files are supplementary only. Honest boundary §11.4.6 — guarantees a complete video-confirmed always-synced ledger, NOT per-feature correctness (rests on each row's §11.4.69/.107/.123 evidence) and NOT a §11.4.40 retest substitute. Classification: universal (§11.4.17). Composes §11.4.2/.5/.44/.45/.52/.56/.57/.59/.60/.65/.86/.102/.106/.107/.108/.118/.123/.134/.146/ §1.1. Propagation gate CM-COVENANT-114-153-PROPAGATION (literal 11.4.153) + recommended gates CM-FEATURE-STATUS-COMPLETE + CM-FEATURE-STATUS-VIDEO-CONFIRMED + paired §1.1 mutation.

Canonical authority: constitution submodule Constitution.md §11.4.153. Non-compliance is a release blocker. No escape hatch — no --skip-feature-status , --feature-without-video , --frozen-video-OK , --bluff-response-video-OK , --skip-video-analysis , --feature-ledger-incomplete-OK , --no-docx-export , --allow-feature-ledger-drift flag.

§11.4.154 — Window-scoped capture + fresh-corpus rotation for feature/QA recordings (User mandate, 2026-06-15). Verbatim: "recording you perform does record the window containing apps and services, not the whole desktop or monitor screen!" + "All old recording files MUST BE removed when new one starts!" Refines §11.4.2/.5/.107/.153 recording discipline with two capture-hygiene invariants. **(A) Window-scoped, NOT whole-screen** — every feature/QA video MUST capture ONLY the window/surface of the app/service under test (GUI window / CLI-TUI terminal pane / web tab-viewport / device-emulator-simulator frame), NEVER the whole desktop/monitor or unrelated windows; whole-desktop capture leaks operator-private content (§11.4.10/.83), dilutes the §11.4.107 liveness/freeze oracle, and breaks the §11.4.137 OCR/ROI content oracle. Target the window/region by stable identity (window id/title, device serial, browser context, tmux target) per §11.4.111 — never a fixed full-screen device index. Platform genuinely cannot capture below whole-screen ⇒ honest §11.4.3 SKIP + tracked migration item, never

a whole-screen pass-off. **(B) Fresh-corpus rotation** — when a new recording run for a scope begins, the agent's OWN prior in-scope stale recordings at the raw recording path MUST be removed FIRST so the live corpus reflects the current run (§11.4.107 not-stale + §11.4.86 roster-freshness). Honest boundary (§11.4.6 + §9.2): "remove old" = the agent's own prior recordings for the SAME scope/project ONLY — NEVER another project's/scope's/operator-authored files; uncertain ⇒ surface, don't delete (§11.4.122); committed `docs/qa/<run-id>/` evidence (§11.4.83) is the durable record, NOT rotated. Classification: universal (§11.4.17). Composes §11.4.2/.5/.10/.83/.86/.107/.111/.122/.128/.137/.153/§9.2/.6. Propagation gate `CM-COVENANT-114-154-PROPAGATION` (literal `11.4.154`) + recommended gate `CM-WINDOW-SCOPED-FRESH-CORPUS-RECORDING` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.154. Non-compliance is a release blocker. No escape hatch — no `--whole-screen-capture-OK`, `--skip-window-scope`, `--keep-stale-recordings`, `--no-corpus-rotation`, `--full-desktop-recording` flag. **(C) MP4 auto-conversion REQUIRED.** Any `.cast` file produced MUST be auto-converted to `.mp4` via `agg + ffmpeg` immediately after capture. The `.mp4` is the primary evidence; `.cast` is supplementary only.

§11.4.155 — Project-name-prefixed feature/QA recording filenames (User mandate, 2026-06-15). Verbatim: "All recorded videos MUST START with prefix: the PROJECT NAME (ALWAYS USE THE PROJECT NAME). Project name MUST be obtained according to the constitution's own project-name resolution." Every recorded video the project produces — every feature/QA real-use recording (§11.4.153), every window-scoped capture (§11.4.154), every always-on device recording (§11.4.128), every raw or curated artefact at the project-declared recording path (§11.4.35) + the committed `docs/qa/<run-id>/` trail (§11.4.83) — MUST have a filename that STARTS WITH the PROJECT-NAME prefix, ALWAYS; an unprefixed recording is a §11.4.155 violation (a multi-project/scope corpus on one host per §11.4.128/.103 becomes un-greppable + un-attributable — the §11.4.151 identify-and-grep failure on the recording-corpus axis). **Prefix resolution (closed-set, deterministic — §11.4.6, IDENTICAL to §11.4.151):** (1) `HELIX_RELEASE_PREFIX` from `.env` (authoritative, git-ignored §11.4.30, documented in tracked `.env.example` §11.4.77) else (2) lowercased snake_case project-root dir name §11.4.29 — ALWAYS resolvable, zero operator input. SAME prefix for EVERY recording in a checkout so `ls '<PREFIX>--- '*` enumerates the corpus; canonical form `<PREFIX>---<feature-or-scope>---<run-id>.<ext>`

(`---` delimits the prefix unambiguously); MUST equal the §11.4.151-resolved release-tag prefix (divergence is itself a §11.4.155 violation — one project, one name). Honest boundary (§11.4.6): the prefix guarantees attribution + greppability, NOT content validity (still §11.4.107/.137/.153) and does NOT relax §11.4.154's window-scope/rotation (rotation removes the agent's OWN `<PREFIX>---*` only; foreign/operator files surfaced not deleted §11.4.122/§9.2). Classification: universal (§11.4.17). Composes §11.4.151/.128/.153/.154/.111/.83/.6/.29/.30/.35/.77/.86/§1.1. Propagation gate `CM-COVENANT-114-155-PROPAGATION` (literal `11.4.155`) + recommended gate `CM-RECORDING-PROJECT-NAME-PREFIX` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.155. Non-compliance is a release blocker. No escape hatch — no `--no-recording-prefix`, `--recording-without-project-name`, `--unprefixed-recording`, `--prefix-optional-for-recording`, `--differing-recording-prefix` flag.

§11.4.156 — All CI/CD automation (GitHub Actions / GitLab pipelines / equivalents) MUST be disabled (User mandate, 2026-06-15). Verbatim operator mandate: "Any GitHub actions or GitLab pipelines MUST BE disabled! Add this critical mandatory rule / mandatory constraint into the root constitution Submodule, commit and fetch all its changes to all upstreams and make sure we respect and follow this rule (we do apply it) ASAP!!!" Every repository this Constitution governs — main repo, this constitution submodule, every owned + nested submodule we author and push — MUST ship with ALL server-side CI/CD automation DISABLED: no push to any owned upstream may trigger a GitHub Actions run, GitLab pipeline, or equivalent (Jenkins/CircleCI/Travis/Drone/Woodpecker/Bitbucket/Azure, any `on: push / schedule / workflow_dispatch`). GENERALISES + makes ABSOLUTE the §11.4.75 Layer-5 posture (remote CI DISABLED, workflow preserved at a `...disabled-local-only non-.yaml` name a provider ignores) across ALL governed repos; enforcement migrates to the LOCAL §11.4.75 git-hook ritual + §11.4.40 pre-tag sweep, never a remote runner. ALL hold: **(A)** zero active `.github/workflows/*.yaml|yml / .gitlab-ci.yml / .gitlab/** /` equivalent at the ROOT of any governed repo/submodule (the only place a provider executes); **(B)** "disabled" = a push triggers ZERO runs — delete OR rename to a non-trigger name (§11.4.75 `.disabled / .disabled-local-only`); a live- `on: + if:false` workflow still queues runs, NOT compliant; **(C)** scope = repos we author+push — vendored/third-party nested configs below the root (AOSP `external/**`, `prebuilts/**`, vendored submodules) are INERT (a provider

never runs a non-root config), OUT of scope, MUST NOT be mass-edited (§11.4.29 vendor-exempt); test = "does a push to OUR upstream trigger a run?" yes⇒disable, inert⇒document+leave (§11.4.6 verify-not-assume); **(D)** no new CI may be added (release blocker); **(E)** pre-push verify `git ls-files | grep -E '^\.github/workflows/.*\ya?ml$|^\.gitlab-ci\.yaml$'` empty for authored repos, §11.4.109-class PreToolUse guard + gate enforce mechanically. Honest boundary (§11.4.6): file-level disabling stops FILE-triggered runs, NOT provider-side server settings (org-default required workflows, branch-protection required checks, provider scheduled exports) — the operator turns those off; the agent documents what it cannot reach, never claims unachieved completeness. Composes §11.4.75/.29/.6/.40/.42/.109/.113/§2.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-156-PROPAGATION` (literal `11.4.156`) + recommended gate `CM-NO-ACTIVE-CI` + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; add a root `.github/workflows/x.yml` to an authored repo → `CM-NO-ACTIVE-CI` FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.156. Non-compliance is a release blocker. No escape hatch — no `--allow-ci`, `--enable-workflow`, `--keep-pipeline`, `--remote-ci-OK`, `--ci-exempt` flag.

§11.4.157 — GEMINI.md maintained in lockstep with CLAUDE.md /

AGENTS.md / QWEN.md (User mandate, 2026-06-15). Verbatim operator mandate: "Make sure with CLAUDE.md, AGENTS.md, QWEN.md we maintain GEMINI.md too! Add this mandatory fact / rule to root constitution Submodule we are inheriting / extending - CONSTITUTION.md, CLAUDE.md, QWEN.md, AGENTS.md, GEMINI.md and other related relevant files!" Forensic FACT (2026-06-15): when §11.4.156/§11.4.157 were authored, Constitution/CLAUDE/AGENTS/QWEN carried the family through §11.4.155 but GEMINI.md had silently drifted to §11.4.141 — 14 mandates (§11.4.142–155) never propagated — a §11.4 propagation-bluff (a Gemini-CLI agent reads a stale Constitution). GEMINI.md is a FIRST-CLASS governance context carrier EQUAL to CLAUDE.md/AGENTS.md/QWEN.md, never optional/best-effort. ALL hold: **(A)** five-carrier lockstep — no governance change is complete until GEMINI.md carries it alongside the other three mirrors; GEMINI.md is added to the §11.4.26 propagation + cross-reference set explicitly; **(B)** no silent drift — GEMINI.md lagging the other mirrors' highest rule is a §11.4.157 violation (§11.4.65-class), back-fill required; **(C)** equal status — GEMINI.md restates the SAME literal `11.4.N` anchors the propagation gates require (§11.4.35), fleet count INCLUDES GEMINI.md; **(D)** consumer projects' own

CLAUDE/AGENTS/QWEN/GEMINI bind too (§11.4.35). Honest boundary (§11.4.6): the §11.4.142–155 GEMINI.md back-fill is a tracked release-blocking remediation; claiming GEMINI.md "in sync" while the back-fill is incomplete is itself a §11.4.157 violation. Composes §11.4.26/.35/.17/.44/.65/.140/.156/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-157-PROPAGATION` (literal `11.4.157`, GEMINI.md INCLUDED) + recommended gate `CM-GEMINI-MD-LOCKSTEP` + paired §1.1 meta-test mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.157. Non-compliance is a release blocker. No escape hatch — no `--skip-gemini-md`, `--gemini-optional`, `--gemini-lag-OK`, `--four-carrier-suffices` flag.

§11.4.158 — Intensive all-feature/flow/edge-case video-recording + read-the-screen content-verification mandate (User mandate, 2026-06-16). Every project MUST be covered by intensive automated testing that exercises + RECORDS every feature/flow/use-case/edge-case (valid/invalid/boundary/concurrent/failure-injection §11.4.85), each recording showing the feature GENUINELY WORKING with REAL results (real prompts → real responses → real outputs §11.4.153) and NO false/simulated/stale/frozen result (§11.4.2/.5/.107) — a recording showing an error/bluff/non-working feature is a FINDING (→ §11.4.153(4)/§11.4.4 fix → retest → re-record), never a confirmation. The testing System MUST ACTUALLY READ every shown log line / message / UI label / dialog / toast / status text and VERIFY it is a genuine working result (OCR/ROI §11.4.117/.137 + confidence floor for pixel surfaces; direct capture for terminal/log; §11.4.107(10) self-validated golden-good/golden-bad analyzer) — "a video was produced" is NOT evidence, "the System read the screen + confirmed a genuine result" is. HelixQA (§11.4.27) MUST drive this exercise → record → read → score pass, PASS only on a read-confirmed genuine result + captured artefact path (§11.4.69). Default recording save path = `$HOME/Downloads` (host user's home Downloads, resolved at runtime never hardcoded) unless a project declares an override per §11.4.35; §11.4.155 project-prefix + §11.4.154 window-scope/rotation + §11.4.128 git-ignored raw corpus + §11.4.83 curated docs/qa evidence apply. **Vision analysis MANDATORY for every recording** — after every recording, the agent MUST read the terminal output / video content and verify the feature actually works. A recording without a vision-confirmed verdict is a §11.4.158 violation. Composes §11.4.2/.5/.25/.27/.52/.69/.83/.85/.107/.108/.117/.118/.128/.137/.138/.153/.154/.155/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-158-PROPAGATION` (literal `11.4.158`) + recommended gates

CM-INTENSIVE-RECORDING-COVERAGE +

CM-RECORDING-CONTENT-READ-VERIFIED + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.158. Non-compliance is a release blocker. No escape hatch — no `--skip-recording-coverage`, `--video-without-content-read`, `--happy-path-recording-suffices`, `--recording-path-anywhere`, `--unread-recording-OK`, `--skip-helixqa-read` flag.

§11.4.159 — Mandatory window-specific video recording + vision validation

mandate (User mandate, 2026-06-20). Every feature test, validation, verification, challenge, and QA session that produces video evidence MUST comply with ALL of the following: **(A) Window-specific recording ONLY** — every video MUST record ONLY the target application window (Terminal pane, TUI app, browser tab, emulator frame), NEVER the whole desktop/monitor screen; use macOS `screencapture -l<window_id>`, Linux `xdotool + ffmpeg`, or equivalent; whole-desktop capture leaks operator-private content (§11.4.10), dilutes the §11.4.107 liveness oracle, and breaks §11.4.137 OCR/ROI; platform genuinely cannot capture below whole-screen => honest §11.4.3 SKIP + tracked migration item. **(B) MP4 format REQUIRED** —

`.mp4` (H.264, `movflags +faststart`, `pix_fmt yuv420p`); `.cast` files are supplementary only; auto-convert via `agg + ffmpeg` per §11.4.154(C). **(C)**

Project-name prefix REQUIRED — filename MUST start with project name in snake_case from `HELIX_RELEASE_PREFIX` in `.env` (§11.4.151) or lowercased project root dir name (§11.4.29); format `<project_name>-<feature>-`

`<timestamp>.mp4`; unprefixed = violation. **(D) Mandatory vision validation** —

after EVERY recording, the agent MUST read the terminal output / video content and verify: (i) feature ACTUALLY WORKS, (ii) LLM responses are REAL, (iii) all tests show PASS, (iv) no "TODO implement" / "simulate" / "for now" patterns, (v) output demonstrates end-user working feature; MUST produce a verdict (PASS/FAIL) with evidence path (§11.4.69). **(E) Terminal window cleanup** — after each recording, dismiss/close ONLY the Terminal window used for that recording

(window-specific close via `osascript` / `xdotool`); MUST NOT close windows belonging to other processes. **(F) Real results ONLY** — every recording MUST show REAL working features; errors/empty output/simulated responses trigger

fix → retest → re-record. **(G) Re-runnable evidence** — the command shown MUST be re-runnable to produce the same results. **(H) Fresh-corpus rotation** — remove agent's own prior in-scope stale recordings FIRST (§11.4.154). **(I) Content verification MANDATORY — not duration-based** — the value of a recording is

NOT its duration but its CONTENT; a recording MUST demonstrate the ACTUAL feature being used with REAL results; before accepting ANY recording, the agent MUST verify: expected output patterns ARE present (e.g., test PASS lines, API response data, feature-specific output), the feature ACTUALLY WORKS as demonstrated (not just "something ran"), LLM responses are REAL content (not simulated, not placeholder, not empty), every claim of "working" is backed by visible evidence; a 5-second recording showing a feature working correctly is MORE valuable than a 60-second recording of empty terminal; duration is NOT a proxy for quality. **(J) Expected-content specification REQUIRED** — before recording, the agent MUST specify what content SHOULD appear (expected patterns, expected test results, expected API responses); after recording, the agent MUST verify these patterns ARE present; if expected content is MISSING, the recording is REJECTED regardless of duration. **(K) Content-verification recording workflow** — mandated workflow: (1) SPECIFY expected content patterns, (2) RECORD the feature execution, (3) EXTRACT all text from the recording, (4) VERIFY expected patterns are present, (5) CHECK for simulated/placeholder content, (6) ACCEPT only if ALL patterns found AND zero bluffs detected, (7) REJECT and re-record if ANY pattern missing or bluff detected. **(L) Root cause analysis REQUIRED for rejected recordings** — when a recording is rejected (missing expected content, bluff detected, or empty capture), the agent MUST investigate WHY before re-recording per §11.4.102; determine the root cause (timing issue, wrong command, tool failure, etc.) and fix it; simply re-recording without understanding WHY the first attempt failed is a §11.4.159 violation. **(M) Real-time monitoring RECOMMENDED** — for complex features, use real-time monitoring that analyzes output DURING recording (not after); this catches issues immediately and allows corrective action before the recording completes. Classification: universal (§11.4.17). Composes §11.4.2/.3/.5/.10/.29/.69/.83/.107/.111/.128/.137/.151/.153/.154/.155/.158/§1.1. Propagation gate `CM-COVENANT-114-159-PROPAGATION` (literal `11.4.159`) + recommended gate `CM-WINDOW-VIDEO-VALIDATED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.159. Non-compliance is a release blocker. No escape hatch — no `--whole-screen-ok`, `--cast-only`, `--skip-vision-validation`, `--no-cleanup`, `--simulated-recording-ok`, `--unprefixed-recording` flag.

§11.4.160 — Vision-verified recording + HelixQA bridge mandate (User mandate, 2026-06-21). Compact summary: every video recording for feature/QA evidence MUST be processed through a vision/OCR pipeline that reads on-screen

content and confirms expected results BEFORE acceptance; the recording system MUST provide a bridge feeding captured frames to HelixQA's test infrastructure (or equivalent) for automated read-the-screen verification against SPECIFY-phase expected patterns (§11.4.159(J)); the bridge MUST capture frames at ≤5s intervals, run OCR/vision analysis with a self-validated golden-good/golden-bad analyzer (§11.4.107(10)), compare extracted text against specified patterns, produce a per-frame PASS/FAIL verdict with an evidence path to the frame, and surface failures immediately so the recording can be re-done per §11.4.159(L). The capture interval, OCR confidence floor, and pattern-matching thresholds are project-configured per §11.4.35, calibrated on the project's own fixtures (§11.4.6). Honest boundary: vision verification confirms on-screen content — does NOT replace §11.4.5 quality analysis nor §11.4.108 runtime-signature verification; FLAG_SECURE surfaces use the §11.4.117 proxy oracle per §11.4.112. Classification: universal (§11.4.17). Composes §11.4.5/.27/.69/.107(10)/.112/.117/.153/.158/.159(J)/§1.1. Propagation gate CM-COVENANT-114-160-PROPAGATION + recommended gate CM-VISION-VERIFIED-RECORDING-BRIDGE + paired §1.1 mutation.

§11.4.161 — Rootless container runtime mandate (User mandate, 2026-06-21).

Compact summary: every project MUST use Podman in rootless mode (or equivalent rootless container runtime) for ALL containerized workloads — Docker in rootful mode, sudo, or any escalation to root for container management is STRICTLY FORBIDDEN unless the target platform has no rootless option AND that constraint is documented per §11.4.112; the vasic-digital/containers Submodule (§11.4.76) MUST be used as the sole container orchestration layer — no ad-hoc docker/podman commands outside the Submodule's pkg/boot / pkg/compose / pkg/health primitives; if a missing capability forces a raw command, the containers Submodule MUST be extended upstream per §11.4.74 rather than worked around; all container-related integration tests MUST boot infra on-demand via the Submodule (the on-demand-infra invariant). Honest boundary: rootless execution eliminates the container-to-root privilege-escalation vector — it does NOT replace §11.4.10 credentials-handling (credentials in containers still require leak audits) nor §12.3 container hygiene (memory limits/OOM policies/restart backoff still apply). Classification: universal (§11.4.17). Composes §11.4.76/.74/.10/.112/§12.3/.6. Propagation gate CM-COVENANT-114-161-PROPAGATION + recommended gate CM-ROOTLESS-CONTAINER-RUNTIME + paired §1.1 mutation.

§11.4.162 — OpenDesign UI design system mandate (User mandate, 2026-06-21). Compact summary: every project producing user-facing interfaces (web, desktop, mobile, TUI) MUST use OpenDesign (<https://github.com/nexu-io/open-design>) as the mandatory UI design-and-refinement system — NOT ad-hoc CSS or one-off design tools; install as a project dependency and use its design tokens/themes system for: (a) all color palette definitions supporting BOTH light and dark themes from project brand assets (§11.4.35), (b) typography scale and spacing, (c) component-level design tokens; if a desired UI pattern is not supported, extend OpenDesign upstream per §11.4.74 (extend-don't-reimplement); every UI component MUST ship light + dark theme variants; elements MUST NOT overlap, fonts MUST NOT collide, labels MUST NOT overlay labels — any layout regression is a §11.4.162 violation; all UI changes MUST be covered by the project's standard test types including visual regression tests (before/after screenshots with per-pixel or perceptual-diff PASS/FAIL). Honest boundary: OpenDesign governs design tokens and themes — it does NOT replace functional testing (§11.4.27), WCAG accessibility assertions (§11.4.107), nor the §11.4.48/§11.4.49 UI-driven and dual-approach testing methodology. Classification: universal (§11.4.17). Composes §11.4.74/.25/.27/.4(b)/.107/.48/.49/.35/.69/§1.1. Propagation gate CM-COVENANT-114-162-PROPAGATION + recommended gate CM-OPENDSIGN-UI-SYSTEM + paired §1.1 mutation.

§11.4.163 — Universal Media Validation & Verification Mandate (User mandate, 2026-06-21). Compact summary: every recorded artifact (video/audio/screenshots/asciicast/text) MUST pass a MEDIA VALIDATION pipeline before acceptance — OCR for video/screenshots (§11.4.117/.107(12)), transcription for audio, text parsing; compare extracted content against SPECIFY-phase expected patterns (§11.4.159(J)); self-validated analyzer with golden-good/golden-bad fixture pair (§11.4.107(10)); produce structured verdict (PASS/FAIL + evidence path + matched/unmatched patterns + pinpoint data on FAIL — which frame/line/timestamp, expected vs actual); post-recording AND real-time trigger (§11.4.159(M)); paired §1.1 mutation ensures golden-bad fixture produces FAIL. Honest boundary: confirms artifact content matches patterns — does NOT replace §11.4.5 quality, §11.4.107 liveness, nor §11.4.108 runtime-signature. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-163-PROPAGATION + recommended gate CM-MEDIA-VALIDATION-PIPELINE + paired §1.1 mutation.

§11.4.164 — Universal Constitution Auto-Propagation & Hook System (User mandate, 2026-06-21). Compact summary: every fetch+pull of the constitution submodule MUST trigger `constitution/scripts/post_update_hook.sh` (inherited §11.4.28) that: detects changed files (governance MDs + `scripts/hooks/skills/mcp/plugins`); registers new/modified skills via consumer's `scripts/register_skills.sh` ; registers MCP servers via consumer's `scripts/register_mcp.sh` ; installs hooks into `.git/hooks/` ; validates scripts syntax per §11.4.67; emits summary log; failures logged per §11.4.6; tested by §1.1 mutation (add skill → hook detects it). Consumer MUST invoke after every pull. Honest boundary: installs components — does NOT guarantee runtime acceptance nor replace §11.4.32 sweep. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-164-PROPAGATION` + recommended gate `CM-CONSTITUTION-AUTO-PROPAGATION` + paired §1.1 mutation.

§11.4.165 — Universal Independent Verification Agent Mandate (User mandate, 2026-06-21). Compact summary: every code/media/docs/config output MUST pass INDEPENDENT verifier (§11.4.70/.20, structurally separate from author, §11.4.92 PRECEDES never satisfies), iterating to GO per §11.4.134. CODE: §11.4.142 review + build+test + paired §1.1 mutations + §11.4.108 runtime-signature. MEDIA: §11.4.163 pipeline + genuine-content check (§11.4.159(I)) + format check (§11.4.159(B)). DOCS: exports current §11.4.65 + revision header §11.4.44. CONFIG: syntax + schema + §11.4.10 leak check. Structured findings per class; verifier self-validated by §1.1 mutation. Honest boundary: source/artifact integrity — does NOT replace §11.4.108 runtime-signature nor §11.4.40 retest. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-165-PROPAGATION` + recommended gate `CM-INDEPENDENT-VERIFICATION-AGENT` + paired §1.1 mutation.

§11.4.166 — REPEALED (operator decision, 2026-06-22). The Universal Semgrep static-analysis mandate is repealed — Semgrep is NO LONGER mandatory. Static analysis remains encouraged, not mandated. Scaffolding, `submodules/semgrep` , MCP/pre-commit/PATH wiring, `docs_chain semgrep_status` context, and the `CM-COVENANT-114-166-PROPAGATION` / `CM-SEMGREP-WIRED` gates removed. Anchor 11.4.166 retired. See constitution `Constitution.md` §11.4.166.

§11.4.167 — Big-work-item feature work-stream lifecycle (User mandate, 2026-06-23). Compact summary: every BIG work item (new feature OR drastic/large fix) **MUST** be developed as its own isolated **feature work-stream** — its OWN full project copy in a sibling directory, its OWN `feature/<slug>` branch, its OWN per-feature flashable builds + feature release tags, kept SEPARATE from the trunk until the operator manually approves after testing, with the trunk regularly merged INTO every feature stream. (A) one big item = one feature work-stream in a sibling project copy (`<project>_<slug>` , §11.4.29 slug); small changes stay on the trunk. (B) disk-feasible duplication MANDATORY, naive deep copies FORBIDDEN — a CoW/reflink clone (`cp -a --reflink=auto` / FS snapshot) on btrfs/XFS/ZFS/APFS, else a shared-object-store mechanism (`git worktree` + externalized build output); per-stream `out/` excluded from the clone + one shared ccache; ~0-disk-per-fresh-stream verified by captured evidence (§11.4.69) before relied on (§11.4.6 no-guessing). (C) own branch + own tags greppable per §11.4.151 (`<base>-feat-<slug>[-<iter>]`), NEVER merged to trunk until §11.4.40 full retest GREEN on a clean target → §11.4.41 merge-first → §11.4.113 ff-only (no-force) merge. (D) trunk merged INTO every live stream FREQUENTLY (per trunk tag / daily, `git merge` never rebase a shared/tagged branch); a stale stream is flagged. (E) feature branch + mirrored feature tag cascades to EVERY touched submodule the moment a submodule is modified (§11.4.113 + tag-cascade); untouched submodules stay trunk-pinned. (F) single-builder build QUEUE (FIFO + advisory lock, §11.4.58 L1 / §12.7 / §12.8) + finite-device test queue (§11.4.119 per-device exclusive ownership); idle streams' regenerable build output GC'd (§11.4.77). (G) every change crosses the full quality gauntlet (§11.4.142/.125/.134 review + §11.4.145 impact research + §11.4 / .5 / .69 / .107 anti-bluff testing) — a feature stream is NOT a quality carve-out. (H) resumable, data-safe lifecycle — a stream registry (§11.4.116 / §11.4.147) tracks branch/base-tag/out-state/last-trunk-merge/merge-approval; atomic `.partial` → rename create + §11.4.84 quiescence resume; retire only after merged/approved + §9.2 backup; bounded by §12 / §12.6. Honest boundary (§11.4.6): guarantees ISOLATION + disk-feasibility + greppable identity + controlled trunk↔feature merge — NOT that any stream's work is correct (still crosses §11.4.108 + §11.4.40). Classification: universal (§11.4.17) — consumer supplies its CoW mechanism, sibling-dir + tag naming, build/device-queue counts, big-item threshold per §11.4.35. Propagation gate `CM-COVENANT-114-167-PROPAGATION` (literal `11.4.167`) + recommended gates `CM-FEATURE-WORKSTREAM-COW-CLONE` / `CM-FEATURE-WORKSTREAM-NO-MERGE-UNTIL-APPROVED` /

CM-FEATURE-WORKSTREAM-TRUNK-SYNC-CADENCE / CM-FEATURE-WORKSTREAM-SUBMODULE-CASCADE + paired §11.1 mutations. **Canonical authority:** constitution submodule Constitution.md §11.4.167. Non-compliance is a release blocker. No escape hatch — no `--full-clone-per-feature`, `--deep-copy-stream`, `--skip-cow-clone`, `--merge-feature-without-approval`, `--no-trunk-sync`, `--skip-submodule-cascade`, `--unbounded-parallel-builds`, `--shared-device-no-lock`, `--retire-unmerged-stream`, `--feature-stream-quality-carveout` flag.

§11.4.168 — Exported-document independent content + textual + full-visual validation mandate (User mandate, 2026-06-23). Compact summary: every generated/exported document (HTML / PDF / DOCX / any format in the §11.4.65 export scope) MUST pass INDEPENDENT validation by a review agent structurally separate from the generator (§11.4.70/§11.4.142, NEVER the author self-checking), on BOTH the source AND every exported artifact, ALWAYS, across THREE layers: (1) CONTENT — the export faithfully carries the source's intent + data, nothing dropped/truncated/garbled; (2) TEXTUAL — human-readable, NO raw markup / diagram-source (Mermaid `gantt` / `graph` / `flowchart` / `sequenceDiagram` etc.) / unrendered code-fence leaking as body text (the exact 2026-06-23 raw-gantt-in-PDF failure); (3) FULL VISUAL — embedded diagrams render as IMAGES not source, layout intact, no overlap/cut-off/garble/collapse, verified by RENDERING the export (`pdftotext` catches raw source as text + `pdfimages` confirms rendered images + `pdftoppm` → OCR confirms human-readable visual content) with captured evidence per §11.4.5/§11.4.69/§11.4.107. Anti-bluff: a rubber-stamp is not validation; the analyzer is self-validated golden-good/golden-bad per §11.4.107(10) (a golden-bad export seeded with raw `gantt` source MUST FAIL); iterate to a clean GO per §11.4.134. Forensic FACT (2026-06-23): a "green" Mermaid fix shipped raw `gantt` source as readable text into user-facing PDFs because validation grepped the HTML for `class="mermaid"` and NEVER checked the exported PDF — operator caught it (§11.4.138). A generated doc shipping without this content+textual+visual validation, or containing raw diagram source / unreadable garble, is a §11.4 PASS-bluff at the documentation layer. Honest boundary (§11.4.6): confirms the exported artifact a reader opens is faithful/readable/visually-rendered — does NOT prove the source content is correct (§11.4.142/§11.4.135) nor replace the §11.4.65 mtime-parity freshness gate (this is the READABILITY/FIDELITY layer the mtime gate cannot see); FLAG_SECURE/un-rasterisable surfaces document the gap per §11.4.112 + §11.4.117 proxy oracle,

never a faked visual PASS. Classification: universal (§11.4.17). Composes §11.4.65/.73/.107/.117/.135/.138/.142/.159/.163/.165/§1.1. Propagation gate `CM-COVENANT-114-168-PROPAGATION` (literal `11.4.168`) + recommended gate `CM-EXPORTED-DOC-VISUALLY-VALIDATED` + paired §1.1 mutation (a PDF/HTML/DOCX containing raw diagram source as body text → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.168. Non-compliance is a release blocker. No escape hatch — no `--skip-visual-validation`, `--raw-source-in-pdf-ok`, `--textual-check-suffices`, `--self-validate-export`, `--diagram-source-as-text-ok` flag.

§11.4.169 — Mandatory comprehensive test-type coverage with anti-bluff captured evidence (User mandate, 2026-06-25). Every project MUST have, with coverage aiming 100% and zero false results / rock-solid hard physical proof / zero bluff, this CLOSED enumerated test-type set: (1) **unit** (the ONLY layer mocks/stubs/fakes/placeholders are permitted §11.4.27(A)); (2) **integration** (real fully-wired components against real infra, no fakes beyond unit, infra booted on-demand via the **containers** submodule §11.4.76/§11.4.161); (3) **e2e** (full user journey across the real System §11.4.5/§11.4.107); (4) **full-automation** (fully autonomous, re-runnable, NO manual step after start §11.4.25/§11.4.52/§11.4.98, deterministic across N iterations §11.4.50); (5) **Challenges** (the **challenges** submodule `vasic-digital/challenges` — anti-bluff Challenge banks scoring PASS only on positive captured evidence §11.4.27(B)/§11.4.4(b) layer 4); (6) **HelixQA** (the **helix_qa** submodule with proper written **test banks (suites)** per application/service/platform AND comprehensive **fully-autonomous QA sessions** driving every registered bank with captured wire evidence per check §11.4.27); (7) **DDoS/load-flood** (sustained adversarial traffic — refuse cleanly OR degrade gracefully, never collapse; captured throughput/latency/error evidence); (8) **security** (authn/authz, injection/taint, secret-leak §11.4.10, transport+crypto, dependency CVEs, default-deny; composes the **security** submodule); (9) **stress+chaos** (sustained load + concurrent contention + boundary inputs + failure-injection [process-death/network-fault/input-corruption/resource-exhaustion/state-corruption] with categorised recovery + captured evidence §11.4.85); (10) **concurrency/atomicity** (correctness under concurrent callers, no lost updates, transactional atomicity proven, idempotency under replay); (11) **race-condition/deadlock** (race detector + lock-order/deadlock analysis under contention; no blocking op inside a shared-lock region per §1; captured detector output is the evidence); (12) **memory** (leak census over soak, peak-RSS ceilings e.g. the iOS-NEPacketTunnelProvider-class budget,

allocation/fragmentation profile; no unbounded growth across a 24h/N-iteration soak); (13) **benchmarking/performance** (p50/p95/p99 latency + throughput + resource cost vs a recorded baseline; a regression vs baseline is a finding). Each type is REQUIRED where the domain warrants it and each PASS cites rock-solid captured PHYSICAL evidence (§11.4.5/§11.4.69/§11.4.107); the ONLY permitted absence is an honest §11.4.3 SKIP-with-reason (topology/hardware/credential genuinely absent), NEVER a silent gap; a green result with no falsifiable captured evidence is a §11.4 PASS-bluff. Four-layer §11.4.4(b) enforcement of the suite (pre-build per-type-present+-executable + post-build + runtime + paired §1.1 mutation per type) + per-project coverage ledger (§11.4.25/§11.4.52) classifying every feature × test-type × evidence-state; a missing required type or an

`OPERATOR_ATTENDED_ONLY` row is a release blocker until promoted or honestly SKIP-justified. §11.4.169 is the strict explicitly-enumerated expansion of §11.4.27. Composes/strengthens

§11.4.4/.5/.6/.25/.27/.50/.52/.69/.76/.85/.98/.107/.123/.135/.146/§1.1. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-169-PROPAGATION` (literal `11.4.169`) + recommended gate `CM-MANDATORY-TEST-TYPES-COVERED` + paired §1.1 mutation (strip a required type or drop its evidence citation → gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.169. Non-compliance is a release blocker. No escape hatch — no `--skip-test-type`, `--partial-coverage-ok`, `--evidence-optional`, `--challenges-not-applicable`, `--skip-helixqa`, `--no-chaos`, `--no-ddos`, `--skip-memory-test`, `--skip-race-detector`, `--bench-optional` flag.

§11.4.170 — Device-independent host-side rendered-UI visual-proof mandate (User mandate, 2026-06-25). Compact summary: every change to ANY user-facing UI surface (screen/component/view/widget/layout/theme on any platform — Android-Compose, iOS-SwiftUI/UIKit, web, desktop, TUI) MUST be proven by DEVICE-INDEPENDENT host-side RENDERED PIXELS before it may be claimed correct — the real component rendered to a PNG ON THE HOST (NO device, NO emulator, NO running app) via a host-render harness (Compose → Roborazzi/Paparazzi on the JVM; web → Playwright/Storybook snapshot; SwiftUI → snapshot-testing; equivalent per stack), for EVERY relevant screen×state×{light,dark} theme, with the PNG validated by BOTH (i) golden image-diff (per-pixel/perceptual PASS/FAIL) AND (ii) an OCR/vision oracle reading rendered text+labels+control bounds to assert legibility + NO overlap / NO label-over-label / NO clipping / NO off-screen control / NO collapsed-or-giant-unbounded widget (the §11.4.162 layout invariants

PROVEN on real pixels). Forensic FACT (2026-06-25): UI work was "validated" by VALUE-EQUALITY unit tests (a hex colour string / an sp/dp size integer / a design-token field == a constant) that NEVER RENDERED A PIXEL — GREEN while the operator opened a broken unbounded giant-button screen. A value/token-equality / property-assertion UI test is FORBIDDEN as the PROOF a UI is correct (it may SUPPLEMENT, NEVER SUBSTITUTE the rendered-pixel proof); a "green" UI suite with no rendered-pixel artefact for the changed surface is a §11.4 PASS-bluff at the visual layer. Self-validated golden-good/golden-bad analyzer §11.4.107(10), thresholds calibrated on the project's own fixtures §11.4.6. "device offline" is NEVER a valid skip — host-render IS the device-independent path; honest §11.4.3 SKIP only where the platform genuinely has no host-render harness (tracked migration item, never a value-only fallback). Honest boundary §11.4.6: host-render proves the COMPONENT renders correctly device-independently per commit — it does NOT prove the running app on real hardware (that remains §11.4.153/.158/.159/.160's live-device recording + read-the-screen layer, which §11.4.170 COMPLEMENTS not replaces — both mandatory), does NOT replace §11.4.162 OpenDesign token/theme governance (§11.4.170 PROVES the rendered RESULT of those tokens), DISTINCT from §11.4.168 exported-document visual validation. Classification: universal (§11.4.17). Composes §11.4.3/.5/.27/.40/.69/.107/.108/.117/.153/.158/.159/.160/.162/.163/.168/.169/§1.1. Propagation gate CM-COVENANT-114-170-PROPAGATION (literal 11.4.170) + recommended gate CM-HOST-RENDERED-UI-VISUAL-PROOF + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.170. Non-compliance is a release blocker. No escape hatch — no --value-assertion-proves-ui , --skip-rendered-pixel-proof , --token-equality-suffices , --device-offline-skip-visual , --single-theme-only , --unvalidated-image-diff-OK , --no-ocr-layout-oracle flag.

§11.4.171 — Mandatory comprehensive human-readable workable-item descriptions (User mandate, 2026-06-29). Every workable item (ATM-NNN ticket) in the project's workable-items database, Issues.md, Fixed.md, Issues_Summary.md, Fixed_Summary.md, and ALL derived documents MUST carry a comprehensive human-readable description that explains the item in plain, non-technical language suitable for non-developers (project managers, stakeholders, team leads, business partners). The description MUST be 5-7 sentences minimum, written so that ANY person — regardless of technical background — can understand: (a) WHAT the item is (the feature, fix, or task in

simple terms); (b) WHY it matters (the user/business value); (c) HOW it works (the mechanism, without code references); (d) WHO benefits (the end user or stakeholder); (e) WHAT the expected outcome is (the measurable result). The description is MANDATORY — an item without a human-readable description is a §11.4 violation at the documentation layer. Descriptions MUST NOT use jargon, acronyms without expansion, code references, or technical implementation details as the primary explanation. Technical details MAY appear as supplementary context AFTER the plain-language explanation. The `description` column in the SQLite database (`docs/workable_items.db`) is the SINGLE SOURCE OF TRUTH for item descriptions; all derived documents (Issues.md, Fixed.md, summaries, exports) MUST carry the SAME description text. Pre-build gate `CM-HUMAN-READABLE-DESCRIPTION` (when implemented) walks every item in the DB and asserts the `description` field exists, is non-empty, and is ≥ 50 characters. Paired §1.1 meta-test mutation strips a description → gate FAILs. Classification: universal (§11.4.17). Composes §11.4.15 (item-status tracking), §11.4.16 (item-type tracking), §11.4.19 (column alignment), §11.4.91 (summary clarity), §11.4.93 (SQLite SSoT), §11.4.148 (item integrity), §11.4.65 (universal export). Propagation gate `CM-COVENANT-114-171-PROPAGATION` (literal `11.4.171`) + paired §1.1 mutation.

§11.4.172 — Mandatory production-readiness planning with realistic timeline projection (User mandate, 2026-06-29). Every project under this Constitution MUST maintain a living production-readiness planning document that: (a) provides REALISTIC timeline projections for all product milestones (MVP, beta, production-ready) based on actual development velocity data (items completed per week, reopening rate, blocking dependencies); (b) identifies ALL danger zones, weaknesses, and risk areas with mitigation strategies; (c) accounts for HW delivery timelines, licensing delays, and external dependency risks; (d) defines the critical path and identifies which items can slip without affecting MVP; (e) includes workstation/build-capacity analysis and optimization recommendations; (f) is updated at least monthly or whenever the item count changes by $\geq 10\%$; (g) is cross-referenced against the workable-items database for accuracy. The planning document MUST live at `docs/research/production_planning_<YYYYMMDD>/ANALYSIS.md` with HTML+PDF exports per §11.4.65. Honest boundary (§11.4.6): projections are estimates based on measured velocity — they guarantee the methodology is sound, NOT that specific dates will be met. Classification: universal (§11.4.17). Composes §11.4.6 (no-guessing — projections based on data, not

wishful thinking), §11.4.40 (full retest before tag), §11.4.42 (iteration discipline), §11.4.93 (SQLite SSoT), §11.4.108 (four-layer verification). Propagation gate `CM-COVENANT-114-172-PROPAGATION` (literal `11.4.172`) + paired §1.1 mutation.

§11.4.173 — Containerized + distributed build mandate (User mandate, 2026-06-29). EVERY build of EVERY component (source compile, artifact/package/installer/container-image production, codegen, asset render — for ANY language/platform: Go, Android/Gradle, desktop, web, native, firmware) MUST run INSIDE a specialized build container provisioned via the `digital.vasic.containers` submodule (§11.4.76) — NEVER on the bare host. The build containers MUST be DISTRIBUTED to the designated remote build host(s) (e.g. `thinker.local`) via the SAME containers-submodule distribution mechanism the infra uses (§11.4.76 Distributor / remote compose over SSH, §11.4.161 rootless), so the build EXECUTES on the remote build host (offloading the developer/main host); once the build completes the produced artifacts MUST be brought BACK to the originating main host (scp/rsync/volume copy) for use/flashing/distribution. Building outside a container, or on the bare host, is FORBIDDEN — a release blocker (the "works on my machine" / unreproducible-build class §11.4.76 exists to prevent). The build-host target + build-container definitions are config-injected (§11.4.28, never hardcoded in the submodule); a missing build-container capability is added by EXTENDING the containers submodule upstream (§11.4.74), never by an ad-hoc host build. Honest boundary (§11.4.6): the containerized+distributed build guarantees reproducibility + host-isolation + capacity offload — it does NOT replace the §11.4.40 full-suite retest, §11.4.108 four-layer artifact → runtime verification, or §11.4.38 installable-asset evidence (those still run against the brought-back artifact). Classification: universal (§11.4.17). Composes §11.4.76 (containers submodule — sole orchestration layer), §11.4.161 (rootless runtime), §11.4.74 (extend-don't-reimplement), §11.4.28 (config injection / decoupling), §11.4.24 (build-resource stats), §11.4.82 (iteration speedup — persistent caches in the container), §11.4.121 (no-commit-while-build-writes-artifacts), §11.4.38 (installable-asset evidence on the brought-back artifact), §11.4.108 (artifact → runtime verification), §12.6 (host memory ceiling — offloading the build preserves the main host). Propagation gate `CM-COVENANT-114-173-PROPAGATION` (literal `11.4.173`) + recommended gate `CM-CONTAINERIZED-DISTRIBUTED-BUILD` (every build runs via the containers submodule on the remote build host; a bare-host build is detected + FAILs) + paired §1.1 mutation. **Canonical authority:** constitution submodule

Constitution.md §11.4.173. Non-compliance is a release blocker. No escape hatch — no `--build-on-host` , `--skip-container-build` , `--local-build-ok` , `--no-distributed-build` , `--bare-host-build` flag.

§11.4.174 — Shared-host process-ownership verification mandate (User mandate, 2026-06-29). On any shared host where other projects/users/agents may run concurrently, every inspection of or action on a process, build, daemon, port, lock, or system-state signal MUST first positively VERIFY the target is OURS before diagnosing or acting. Attribute a process to us ONLY by a strong discriminator — cwd inside this project's tree, argv naming our scripts/artifacts/repo path, a PID we launched + recorded (§11.4.147), or a lock/port path under our tree — NEVER a loose `pgrep java/gradle/node` name match (it matches every project on the host). NEVER kill/signal/restart a process not positively ours (other projects' gradle/java/podman/zig are off-limits — operator-workload-safety, §12/§11.4.133); the §12.8 daemon-kill remediation MUST be scoped to OUR daemons. When our work waits on host contention from another project's process, surface it (§11.4.66/§11.4.101) — block-don't-break. Every `pgrep/ps` filter MUST exclude self-matches + other-project matches. Forensic anchor (FACT 2026-06-29): a 200%-CPU java+gradlew on the shared host was a separate `catalogizer` project's gradle test (cwd=Projects/catalogizer), nearly mis-attributed to our build. Composes §11.4.6/§11.4.66/§11.4.101/§11.4.147/§12/§12.8/§11.4.133. Classification: universal (§11.4.17). Propagation gate `CM-COVENANT-114-174-PROPAGATION` (literal `11.4.174`) + recommended gate `CM-PROCESS-OWNERSHIP-VERIFIED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.174. Non-compliance is a release blocker. No escape hatch — no `--assume-process-ours` , `--loose-pgrep-ok` , `--kill-any-matching-process` , `--skip-ownership-check` flag.

§11.4.176 — Conflict-free multi-track work-division + exactly-once claim registry + capability-aware deadlock-proof device-lock (User mandate, 2026-07-02). Two mandatory multi-track arbitration layers, conflict-free by construction: (A) exactly-once work-item/logical-group claim registry (extends §11.4.147/§11.4.116; disjoint file-scope §11.4.58 L3 + worktree/CoW §11.4.167; no codebase loss §9.2/§11.4.113/§11.4.84/§11.4.41); (B) capability-aware deadlock-proof device-lock (§11.4.119 single-owner; all-or-nothing + non-blocking + TTL-reap breaks all 4 Coffman conditions; append-only JSONL + atomic snapshot §11.4.116); (C) universal/decoupled (§11.4.17/§11.4.28/§11.4.35) + evidence-based resource-tuning (§11.4.24/§11.4.6, never guessed). Classification: universal

(§11.4.17). Gates CM-WORK-DIVISION-EXCLUSIVE-CLAIM + CM-DEVICE-LOCK-DEADLOCK-FREE + CM-COVENANT-114-176-PROPAGATION + paired §1.1 mutation. Composes §11.4.6 §11.4.17 §11.4.24 §11.4.28 §11.4.35 §11.4.41 §11.4.58 §11.4.84 §11.4.85 §11.4.103 §11.4.113 §11.4.116 §11.4.119 §11.4.147 §11.4.167 §12.6 §9.2 §11.4.176.

§12.11 — Maximal dynamic resource utilization for containerized builds (User mandate, 2026-07-03). The containerized build path (its OWN cgroup + own MemoryMax → OOM-kills ITSELF, never user.slice) MUST use the maximal SAFE fraction of host capacity, computed DYNAMICALLY per host: auto-detect nproc/MemTotal/RLIMIT_NPROC (never hardcode, §11.4.6), reserve explicit host headroom (cores+RAM), scale -j against BOTH the memory model AND the process rlimit (privileged nproc-raise for full -j; EAGAIN-safe -j otherwise), never break/bottleneck/wedge. NO fixed low -j for the containerized path. The §12.6 60% ceiling + bare-metal -j cap REMAIN, SCOPED to user.slice-resident work — §12.11 does NOT weaken §12.6, it scopes it. Classification: universal (§11.4.17). Gate CM-MAXRES-DYNAMIC-BUILD + CM-COVENANT-12-11-PROPAGATION + paired §1.1 mutation. Composes §12.1 §12.2 §12.3 §12.6 §12.10 §11.4.6 §11.4.24 §11.4.133 §9.2 §11.4.35 §12.11.

§11.4.177 — Developer-tooling project-decoupling + invocation-directory operation (User mandate, 2026-07-04). Compact summary: no project-specific script / hook / alias / binary may be wired (copied / symlinked / PATH-exported / aliased) into a global or shared developer-tooling PATH; shared tooling MUST be project-agnostic and operate on the INVOCATION directory (cwd / explicit path / config arg), NEVER a hardcoded project path; a project-specific helper lives in + runs from its own repo, a genuinely-reusable helper is promoted to the shared toolkit ONLY after being stripped of every project-specific assumption (paths, device serials, package names, region endpoints) + taking its target from the invocation context; a project script reachable on the global/shared PATH is a decoupling violation of equal severity to importing project-specific code into a shared submodule (§11.4.28); forensic FACT 2026-07-04 — a project cwd-hook symlinked into the global Claude-Toolkit PATH re-coupled every alias to one hardcoded checkout; honest boundary §11.4.6 — decoupling guarantees project-agnostic, not correct (still needs the tool's own tests). Classification: universal (§11.4.17). Composes §11.4.28/.29/.35/.11/.6. Propagation gate CM-COVENANT-114-177-PROPAGATION (literal 11.4.177) + recommended gate CM-TOOLING-PROJECT-DECOUPLED + paired §1.1 mutation. **Canonical authority:** constitution submodule

Constitution.md §11.4.177. Non-compliance is a release blocker. No escape hatch — no `--global-symlink-ok` , `--hardcode-project-path` , `--wire-into-shared-path` flag.

§11.4.178 — Track-qualified identity for parallel work streams (session-name collision ban) (User mandate, 2026-07-04). Compact summary: parallel work streams sharing hardware / a git backing store / a tmux-session namespace / a lock namespace / any single-owner resource MUST be addressed by a TRACK-QUALIFIED identity (`<project>__<track>__<role>`), NEVER a bare directory basename; multiple checkouts/clones/worktrees sharing basename `atmosphere` collapse into one session/lock/log key + cross-wire tmux targets, lock files, device claims, log dirs (observed collision 2026-07-04); every registry row / session name / lock path / log dir / device claim for a ≥ 2 -concurrent-claimant resource MUST carry the track-qualified key, a bare-basename identity for such a resource is a violation; honest boundary §11.4.6 — a unique identity prevents cross-wiring, it does not itself arbitrate ownership (that is §11.4.119 + **§11.4.176** (the multi-track work-division + exactly-once-claim + device-lock arbitration anchor)). Classification: universal (§11.4.17). Composes §11.4.111/.119/.29/.58 + **§11.4.176** (the multi-track work-division + exactly-once-claim + device-lock arbitration anchor). Propagation gate `CM-COVENANT-114-178-PROPAGATION` (literal `11.4.178`) + recommended gate `CM-TRACK-QUALIFIED-IDENTITY` + paired §1.1 mutation. **Multi-project-per-track layout clarification (2026-07-10):** a `/mnt/track<N>` track parent is NOT single-project — MULTIPLE projects coexist on the same track parents, each in its OWN project-named subdirectory `/mnt/track<N>/<project_name>/` (e.g. `/mnt/track1/atmosphere/` AND `/mnt/track1/helix_ota/` side by side); the per-track working copy of project P on track N is ALWAYS `/mnt/track<N>/<project_name>/` (canonical lowercase snake_case dir-name per §11.4.29), never the bare track root and never a shared basename; this is WHY the identity carries the `<project>` component (`<project>__<track>__<role>`) — so two coexisting projects' track-N sessions / locks / logs / device-claims / registry rows never collide; a project adopting the multi-track engine (§11.4.187) MUST create AND register its own `/mnt/track<N>/<project_name>/` per track and MUST NOT assume it owns the track root or that a track hosts only one project.

Canonical authority: constitution submodule Constitution.md §11.4.178. Non-compliance is a release blocker. No escape hatch — no `--bare-basename-ok` , `--skip-track-qualifier` , `--shared-session-name` flag.

§11.4.179 — Corruption-isolated parallel git streams (own-.git independent clones, not shared-common-dir worktrees) (User mandate, 2026-07-04).

Compact summary: when corruption-isolation is a requirement, parallel git work streams MUST each be an isolated repo with its OWN `.git` (own object store, own index, own lock namespace), NOT `git worktree` checkouts sharing one common `.git`; a shared common-dir is a single point of failure — one stale `index.lock` / `.commit_all.lock` / `.push_all.lock` freezes EVERY stream at once (observed 9-h all-track freeze 2026-07-04) + one corrupted object store loses every stream; each stream's `git rev-parse --git-common-dir` MUST resolve INSIDE its own tree, a common-dir resolving to a shared parent is a violation; where CoW/reflink disk-feasibility is the constraint use the §11.4.167 reflink-clone-with-own-`.git` path (btrfs/XFS/ZFS/APFS reflink shares extents on disk while keeping a per-stream `.git`, so isolation + disk-feasibility both hold); honest boundary §11.4.6 — own-`.git` contains lock/corruption blast radius, it does NOT remove per-repo stale-lock reaping (§11.4.180) nor ff-only no-force integration (§11.4.113). Classification: universal (§11.4.17). Composes §11.4.167/.119/.58/§9.2 + **§11.4.176** (the multi-track work-division + exactly-once-claim + device-lock arbitration anchor). Propagation gate `CM-COVENANT-114-179-PROPAGATION` (literal `11.4.179`) + recommended gate `CM-GIT-STREAMS-OWN-COMMON-DIR` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.179. Non-compliance is a release blocker. No escape hatch — no `--shared-common-git`, `--worktree-when-isolation-required`, `--skip-git-isolation` flag.

§11.4.180 — Commit/push (single-writer) wrappers MUST auto-reap provably-stale locks before acquiring (User mandate, 2026-07-04).

Compact summary: every commit / push / single-writer wrapper MUST, before acquiring its own lock, auto-reap any PROVABLY-stale git lock — one whose recorded holder PID is DEAD (`kill -0` fails), OR (no PID recorded) older than a defined threshold AND with no live wrapper/git process holding that repo; the reap covers the wrapper's own lock plus the git-internal existence-based locks (`index.lock`, `HEAD.lock`, `refs/**/*.*.lock`) git does NOT auto-release on holder death; it MUST NEVER remove a lock whose holder is ALIVE (removing a live-held lock corrupts a concurrent writer — §9.2); each reap decision (REAPED / KEPT + reason) is logged as captured evidence (§11.4.6 — liveness PROVEN via `kill -0`, never assumed); a wrapper that blocks indefinitely on a dead-holder lock (observed 9-h freeze 2026-07-04) is a violation; honest boundary §11.4.6 — reaping clears dead-

holder deadlock, it does NOT substitute for the own- .git isolation of §11.4.179. Classification: universal (§11.4.17). Composes §11.4.84/.88/.6/§9.2/§11.4.116/§11.4.179. Propagation gate `CM-COVENANT-114-180-PROPAGATION` (literal `11.4.180`) + recommended gate `CM-WRAPPER-STALE-LOCK-AUTO-REAP` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.180. Non-compliance is a release blocker. No escape hatch — no `--skip-stale-lock-reap`, `--force-remove-live-lock`, `--block-on-dead-holder` flag.

§11.4.181 — Consistent controlled feature-branch naming: one feature/logic-group ⇒ exactly ONE canonical branch name across the main repo + all owned submodules (User mandate, 2026-07-05). Compact summary: one feature / logical group of workable items MUST map to EXACTLY ONE canonical branch name used IDENTICALLY on the main repo AND every touched owned submodule — NEVER two differently-named branches for one feature/group, NEVER per-repo naming drift; (1) single canonical name per group minted ONCE per the §11.4.167 D scheme (`feature/<slug> branch + <prefix>-<base>-feat-<slug> tag, <slug> lowercase snake/kebab §11.4.29`), used identically across the main repo + every branched submodule (an untouched submodule stays on its base branch); (2) single source of truth — the canonical name is recorded ONCE in the §11.4.176 exactly-once claim registry / the §11.4.93 workable-items `logic_group → branch` mapping and LOOKED UP, never re-invented (creating a branch whose name ≠ the group's registered canonical name, or minting a 2nd name for a group that already has one, is a §11.4.6 no-guessing violation); (3) mechanical enforcement (§11.4.75) — recommended gate `CM-BRANCH-NAME-CONSISTENCY` FAILs on an unregistered feature-branch name / two divergent names for one group / a submodule branch name diverging from the group's main-repo canonical name, paired §1.1 mutation (register one group under two divergent names → gate FAILs); (4) risk-free reconciliation (§9.2/§11.4.113/§11.4.84) — a mis-named/duplicate branch is reconciled by MERGING its commits onto the canonical branch (union, no commit lost, no `-s ours /reset`), ff-only push to all upstreams (NEVER force-push), retiring the mis-named branch only after the merge is confirmed on all mirrors. Classification: universal (§11.4.17). Composes §11.4.167/.176/.178/.179/.28/.29/.93/.113/.84/.6/.75/§9.2. Propagation gate `CM-COVENANT-114-181-PROPAGATION` (literal `11.4.181`) + recommended gate `CM-BRANCH-NAME-CONSISTENCY` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.181. Non-compliance is a

release blocker. No escape hatch — no `--allow-divergent-branch-names` , `--skip-branch-registry` , `--rename-branch-freely` , `--per-repo-branch-name-OK` , `--two-names-one-feature-OK` flag.

§11.4.182 — Track+branch work-stream identity label (T<N>/<branch>) on every agent/subagent label + every operator-facing work-stream reference (User mandate, 2026-07-05). Compact summary: EVERY agent / subagent / work-stream label AND every operator-facing reference to a work stream MUST START with a (T<N>/<branch> - <alias>) prefix identifying the track + git branch + the claude-toolkit alias in use (e.g. (T1/main - claude1) ATM-312 MPV drm_prime investigation , (T2/feature/mistiq-vader - deepseek) SPK-XXX ...) so parallel-track work (/mnt/track1..4 , each on its own branch, each driven by a distinct toolkit alias) is never ambiguous; (1) universal label form on every agent-dispatch description, status report, resume/handoff doc, progress line; (2) DETERMINISTIC derivation never guessed (§11.4.6) — <N> from the checkout path /mnt/track<N>/... (honest ? off-track, never a guessed number), <branch> from git rev-parse --abbrev-ref HEAD , <alias> from CLAUDE_CONFIG_DIR basename .claude-<alias> → <alias> (honest ? when unset/non-matching, never a guessed name) (reference labeler scripts/multitrack/track_branch_label.sh); (3) mechanical enforcement (§11.4.75 / §11.4.109-class) — a PreToolUse guard hook (constitution/scripts/hooks/guard-track-branch-label.sh , inherited BY REFERENCE per §11.4.177, never copied) BLOCKS any Agent/Task/TaskCreate dispatch whose description (or subagent label) does not START with ^\([0-9]+\^[^)]+\) , non-agent tools untouched; pre-build gate CM-TRACK-BRANCH-LABEL (labeler+hook exist/exec/parseable, hook wired in settings, convention doc exists) + propagation gate CM-COVENANT-114-182-PROPAGATION (literal 11.4.182); (4) honest boundary (§11.4.6) — a ? in the track OR the alias field is a valid honest label, never a bluff, and the enforced numeric-track format surfaces (blocks) an off-track dispatch rather than mislabeling it. Classification: universal (§11.4.17). Composes §11.4.178 (track-qualified identity — the label-prefix instantiation applied to every agent/subagent label + operator-facing reference) / §11.4.29 / §11.4.75 / §11.4.109 / §11.4.177 / §11.4.6 / §11.4.58 / §11.4.103 / §11.4.119 / §11.4.116 / §11.4.147. Propagation gate CM-COVENANT-114-182-PROPAGATION (literal 11.4.182) + recommended gate CM-TRACK-BRANCH-LABEL + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.182. Non-compliance is a

release blocker. No escape hatch — no `--skip-track-label` , `--unlabeled-agent-OK` , `--bare-work-stream-reference` , `--no-track-branch-prefix` , `--label-from-memory` flag.

§11.4.183 — Maximal multi-agent utilization per work-stream + full-constitution-application + zero-bluff mandate (User mandate, 2026-07-07).

Compact summary: every track / work-stream MUST maximize multi-agent working approaches — subagent-driven development (§11.4.20/§11.4.70), independent review agents (§11.4.125/§11.4.142/§11.4.165/§11.4.134), parallel background streams with auto-backfill (§11.4.58/§11.4.103), and every other applicable agent form — so every track produces the MOST completed work at the HIGHEST quality in the LEAST time; every track MUST apply the ENTIRE constitution (every rule, mandatory constraint, guide, and derived technology — NOTHING ignored, skipped, avoided, or deemed less-relevant); there MUST NEVER be false / faulty / untested / unverified results or bluff of any kind anywhere (§11.4 anti-bluff covenant, captured evidence §11.4.5/§11.4.69/§11.4.107); honest boundary §11.4.6 — "maximize agents" is bounded by §12.6 60% memory + §11.4.58 parallel-agent cap + genuine parallelizability + §11.4.119 single-resource-owner, spawning contending/redundant agents is NOT maximization, the bar is maximum USEFUL parallel throughput not agent count. Classification: universal (§11.4.17). Composes §11.4.20/.58/.70/.92/.94/.97/.103/.119/.125/.126/.134/.142/.165/§12.6. Propagation gate `CM-COVENANT-114-183-PROPAGATION` (literal `11.4.183`) + recommended gate `CM-MAX-AGENTS-PER-TRACK` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.183. Non-compliance is a release blocker. No escape hatch — no `--single-agent-OK` , `--skip-review-agents` , `--partial-constitution-application` , `--serialise-parallelisable-work` flag.

§11.4.184 — Mandatory SonarQube static-analysis CLI + local tooling installed and PATH-discoverable (User mandate, 2026-07-06).

Compact summary: every project/host that runs static analysis MUST have the SonarQube scanner CLI (`sonar-scanner`) installed AND durably discoverable on the system PATH via the shell rc (`.bashrc` / `.zshrc`) — exactly like the Firebase/GitHub/GitLab CLIs — PLUS the shared `constitution/scripts/sonarqube/` tooling (`sonarqube_install_check.sh` / `sonarqube_lib.sh` / `sonarqube_run_scan.sh` / `sonarqube_container.sh` + `compose/`) consumed BY REFERENCE (§11.4.28 decoupled + §11.4.177 project-agnostic + §11.4.80-style inherited, NEVER copied per project); (1) installed + PATH-durable — the

scanner resolves in every fresh shell via a durable rc export, not an ephemeral session-only PATH; (2) shared tooling present + GREEN — `sonarqube_install_check.sh` exits 0 (scanner + rootless podman §11.4.161 + podman-compose + curl + ES-ready `vm.max_map_count`) before any scan-dependent work; (3) anti-bluff §11.4.6 — "installed/configured" PROVEN by `command -v sonar-scanner` resolving AND install-check exit 0, NEVER assumed, a claimed-installed-but-absent CLI is a §11.4 tooling-layer bluff; (4) rootless §11.4.161 — the local SonarQube server runs via rootless podman, never rootful docker/sudo; honest boundary §11.4.6 — names the SonarQube SCANNER CLI + shared local-server tooling, and per §11.4.166 (former universal Semgrep static-analysis mandate REPEALED) SonarQube is a DISTINCT operator-mandated tool not a Semgrep re-instatement, mandating TOOLING availability not a scan on every build. Classification: universal (§11.4.17). Composes §11.4.28/.36/.74/.75/.109/.161/.166/.177/.6. Propagation gate `CM-COVENANT-114-184-PROPAGATION` (literal `11.4.184`) + recommended gate `CM-SONARQUBE-CLI-INSTALLED` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.184. Non-compliance is a release blocker. No escape hatch — no `--skip-sonarqube-install` , `--sonar-cli-optional` , `--ad-hoc-sonar-path` , `--rootful-sonar-OK` flag.

§12.12 — Process/thread-limit (RLIMIT_NPROC) awareness for parallel subagent/multi-process work (User mandate, 2026-07-07). Compact summary: heavy parallel subagent/multi-process work is bounded by the OS per-user process/thread limit `ulimit -u` (`RLIMIT_NPROC` , Linux counts ALL threads+processes of the real UID, not per-process); the ambient tooling fleet (MCP servers, tokio/V8/libuv worker runtimes, `mongod` / `chrome` / `prisma` -class MCP servers, local LLM servers) consumes a large FIXED fraction before project work starts — FORENSIC FACT (2026-07-07): a host at `ulimit -u 4096` sat at ~3900-4005 threads from the ambient fleet alone (<150 headroom); a 4th Go-heavy subagent (`GOMAXPROCS=nproc` spawns dozens of threads) hit `EAGAIN` / `failed to create new OS thread (errno=11)` and crashed tooling; stopping 3 non-essential MCP servers freed ~1345 threads (3827 → 2482). Mandate: BEFORE scaling parallel subagents/processes, check thread headroom (`ulimit -u` `soft` + `ulimit -Hu` `hard` + `live ps -L --no-headers -u "$USER" | wc -l`) and treat exhaustion as a §12 host-safety event that YIELDS UNCONDITIONALLY — the ORTHOGONAL second exhaustion axis alongside §12.6's memory ceiling (abundant RAM ≠ abundant thread budget); signals `EAGAIN` / `fork: retry` /

failed to create new OS thread / cannot allocate memory on fork/clone (distinct from OOM, §11.4.6 no-guessing — never conflate); low headroom ⇒ SERIALIZE, avoid `GOMAXPROCS=nproc` -class spikes, never dispatch blind (a subagent hitting `EAGAIN` mid- git push is a §9.2 data-safety risk). Three raising techniques: (a) no-restart `prlimit --pid <PID> --nproc=<soft>:<hard>` (root, e.g. `su -c`) — new children of a running process inherit the raise; (b) persistent `/etc/security/limits.conf` (`<user> hard/soft nproc <N>`) or `systemd` `LimitNPROC=<N>`, next login; (c) self-raise `ulimit -u <N>` up to the existing hard ceiling within the current session — `su -c "ulimit -u N"` sets the limit ONLY inside that transient subshell, NOT the already-running session. Freeing threads (stopping non-essential MCP servers) REQUIRES operator authorization (§11.4.122, no autonomous tooling removal); CAUTION `pkill -f` / `pgrep -f` match the killer's OWN command line — always exclude the caller's `$$` / `$PPID` (and conductor PID) from any cmdline-pattern kill list to avoid self-kill.

Classification: universal (§11.4.17). Composes §12/.6/.7/§11.4.58/.101/.103/.122/ §9.2. Propagation gate `CM-COVENANT-12-12-PROPAGATION` (literal `12.12`) + recommended gate `CM-NPROC-HEADROOM-CHECK` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §12.12. Non-compliance is a release blocker. No escape hatch — no `--ignore-thread-limit`, `--dispatch-blind`, `--skip-headroom-check`, `--autonomous-tooling-kill`, `--assume-ulimit-scope` flag.

§11.4.185 — Manual QA-team testing as the mandatory FINAL confirmation of done (User mandate, 2026-07-07). Compact summary: no set of features / scope of work / release may be considered fully completed until it has received MANUAL TESTING confirmation by the project's QA team as the FINAL confirmation step — everywhere and always, never forgotten; every automated gate (§11.4 anti-bluff covenant, §11.4.5/§11.4.69 captured evidence, §11.4.52 autonomous validation, §11.4.40 full-suite retest, §11.4.108 four-layer verification, §11.4.132 risk-ordered validation) is NECESSARY but NOT SUFFICIENT — manual QA is the FINAL human sufficiency gate; ADDS to (never weakens) §11.4.52's mandatory autonomous-validation-path requirement — a feature needs BOTH the autonomous path AND the manual-QA final confirmation, never one substituting the other; pipeline: autonomous-GREEN → build → flash/deploy → autonomous on-device validation → hand off to QA team → ONLY after QA-team manual confirmation is the scope fully-completed / tag-eligible; the §11.4.126 autonomous-loop release-tag terminal condition now REQUIRES this confirmation; honest boundary §11.4.6 —

the agent MUST hand off + WAIT, never self-certify/simulate/infer the manual step, while continuing every other non-blocked item in parallel per §11.4.87/.94/.97/.101 (the wait parks one scope, not the loop). Classification: universal (§11.4.17). Composes §11.4/.5/.6/.40/.52/.69/.108/.126/.129/.130/.132. Propagation gate `CM-COVENANT-114-185-PROPAGATION` (literal `11.4.185`) + recommended gate `CM-MANUAL-QA-FINAL-CONFIRMATION` + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.185. Non-compliance is a release blocker. No escape hatch — no `--skip-manual-qa`, `--automation-suffices`, `--self-certify-manual-step`, `--assume-qa-confirmed`, `--autonomous-only-completion` flag.

§11.4.186 — Anti-divergence enforcement: cross-document consistency is a mandatory-before-export/commit gate, never an after-the-fact audit (research-derived, 2026-07-08). Compact summary: any project maintaining more than one representation of the same tracked data (a delivery plan + its MVP brief + a workable-items DB + their summaries + `.md` / `.html` / `.pdf` / `.docx` exports) MUST enforce cross-document CONSISTENCY as a deterministic gate that runs BEFORE any export render, any doc/DB sync `verify`, and any doc-set commit — NEVER as an after-the-fact human audit that runs only when an operator happens to ask (the forensic FACT: the same NanoKVM/phone-as-screen feature appeared in three plan clusters — one shared ticket SPK-596 across SIX rows split over TWO releases, and a carve-out due to ship 2026-08-31 whose own prerequisite does not START until 2026-10-05 — and NO gate caught it; only an operator's manual read of `Plan_v9.0.xlsx` surfaced it, after the divergent bytes had already exported). (1) **One no-divergence verdict, three seams** — a single deterministic PASS/FAIL/SKIP verdict (built by composing a cross-document integrity validator with the workable-items-DB internal `validate`) gates all three write-seams (export / sync-verify / commit); a FAIL at any seam refuses that seam, naming the exact offending records (§11.4.6 actionable). (2) **Five decidable check families** — DEDUP (no two records describe one feature with divergent timeline/status/release, keyed on ticket OR normalised `(subject, scope)`, NOT bare subject substring), TIMELINE (start ≤ deadline; no deadline-before-dependency; no dependency cycle; GATED exempt-but-marked), CROSS-DOC (the same item across every representation carries identical timeline/status/type against a NAMED authoritative source), INTEGRITY (no orphan refs; Status ↔ Type §11.4.33; location ↔ status), STRUCTURAL (required columns; ID uniqueness+monotonicity §11.4.54). (3) **Drift-proof trigger** — a §11.4.86 sha256-of-sorted-members fingerprint of the

authoritative inputs re-arms the gate on ANY input change, so a stale verify cannot pass on changed data. (4) **Self-validated analyzer (§11.4.107(10))** — the validator ships golden-good (MUST PASS) + one golden-bad per family (MUST FAIL, pinpointing the offender) + a negative-control (distinct same-subject tasks that MUST PASS — the false-positive guard); a validator that passes its golden-bad, or fails golden-good/the negative-control, is itself a §11.4 bluff and a release blocker. (5) **Reusable + decoupled (§11.4.28/31)** — packaged as a depth-1 reusable engine under the constitution submodule per the §11.4.28(C) carve-out with a `helix-deps.yaml` ; project-specific doc-sets + authoritative-source bindings live in a consumer-owned checkset (data, never engine code). (6) **Supersedes ad-hoc audits** — the per-cycle `*_INTEGRITY_FINDINGS.md` / `RECONCILIATION_*.md` after-the-fact audit docs are retired in favour of the gate; honest boundary §11.4.6 — the gate proves internal CONSISTENCY (no record diverges, no timeline is incoherent, every ref resolves), NOT plan CORRECTNESS (achievability/date-rightness stay operator/management decisions the gate SURFACES, never MAKES). Classification: universal (§11.4.17). Composes §11.4.6/.12/.33/.44/.50/.53/.54/.60/.65/.73/.75/.85/.86/.91/.93/.95/.106/.107/.108/.110/.148/.176. Propagation gate `CM-COVENANT-114-186-PROPAGATION` (literal `11.4.186`) + recommended gate `CM-DOC-INTEGRITY-VALIDATION` (5 invariants: validator present+executable; consumer checkset present; the no-divergence gate wired into each of the three seams; fingerprint sidecar fresh; `selfcheck` golden-good/golden-bad/negative-control wired into meta-test) + paired §1.1 mutation. **Canonical authority:** constitution submodule `Constitution.md` §11.4.186. Non-compliance is a release blocker. No escape hatch — no `--skip-divergence-gate` , `--audit-after-export-OK` , `--cross-doc-check-not-applicable` , `--unvalidated-analyzer-OK` , `--divergent-commit-OK` flag.

§11.4.187 — Automatic multi-track ruler orchestration (User mandate, 2026-07-04; landed 2026-07-09). Compact summary: every project incorporating this constitution MUST ship its multi-track parallel-development orchestration as a UNIVERSAL, AUTOMATIC, OUT-OF-THE-BOX, INHERITED capability — not a per-project reimplementation — via a single conductor session that programmatically SPAWNS, DRIVES, RESUMES, MONITORS, and CONTROLS per-track headless workers. (1) **Headless spawn primitive** — workers launch via native headless `claude -p --output-format stream-json --verbose` ; the worker's `session_id` is captured from the FIRST `init` event (never invented);

success/failure is read from the terminal event's `result.is_error` field, NEVER the process exit code (a CLI can exit non-zero on a successful `result`, or zero on a failed one — exit-code-as-success-signal is a §11.4.6 guess); continuation of an existing worker uses `--resume <session_id>` invoked from the SAME cwd + env the spawn used (the storage/resume-collision gotcha). (2) **Per-subscription auth + rate-limit fallback** — every spawn `unset` s any ambient `ANTHROPIC_API_KEY` (the precedence trap that silently routes a "subscription" worker onto pay-per-token API billing) and sets the selected alias's `CLAUDE_CODE_OAUTH_TOKEN` / `CLAUDE_CONFIG_DIR`; on a captured rate-limit signature (`isApiErrorMessage + apiErrorStatus:429`, quota-exhausted vs transient distinguished by evidence, never guessed) the ruler REBINDS the affected track onto the next healthy alias (cross-track reuse permitted), falls back to a provider alias where configured, and bounded-parks (§11.4.101) rather than hanging when every alias is cooled. (3) **Crash-resilient ruler self-supervisor** — the ruler's OWN state (per-track alias/session_id/worktree/last-verdict) is persisted durably (write-temp-then-rename, NEVER tmpfs) so an external watchdog detects a dead ruler PID and re-hydrates every track's binding from the durable state + the §11.4.116 event stream — no work is silently lost to a ruler crash (§11.4.147 at the ruler layer). (4) **Idempotent bootstrap + auto-install + conductor-stays-home** — one idempotent bootstrap command wires the whole system out of the box (cwd-hook install, config validation, reconciliation), auto-invoked on every constitution pull via the §11.4.164 `post_update_hook.sh` skill-register seam; the `conductor`: -designated alias/session is NEVER bound into a track worktree (the resolver returns nothing for it) so the conductor always runs from the main checkout. (5) **Project-agnostic engine, consumer-owned data** — the entire mechanism (config loader, alias → worktree resolver, cwd-hook, bind/fallback/cooldown orchestrator, session launcher, rate-limit monitor, supervisor) lives at `constitution/scripts/multitrack/`, inherited BY REFERENCE (§11.4.28(B)/§11.4.177), carrying NO project literal; every consuming project supplies its own per-host `config/multitrack/<hostname>.yaml` (tracks/mounts/serials/aliases/conductor) as DATA, never as an engine edit. Anti-bluff (§11.4/§11.4.108): every piece carries four-layer coverage (pre-build gate + on-host test + paired §1.1 mutation + captured evidence per §11.4.69), and each fix declares a runtime signature verified on a clean host — never a claim the mechanism "should work." Honest boundary (§11.4.6): this anchor mandates the MECHANISM; genuine per-subscription quota-isolation proof requires live tokens exercising real rate-limit

boundaries (an operator-scheduled live acceptance window) — a mock-driven GREEN proves the spawn/resume/fallback CONTRACT, never live-quota behaviour, and the two are never conflated or substituted for each other. Classification: universal (§11.4.17). Composes §11.4.20 (subagent-driven-by-default) / §11.4.28 (submodule decoupling + dependency layout) / §11.4.58 (parallel-development PWU pipeline) / §11.4.70 (subagent-driven execution default) / §11.4.101 (autonomous-decision-over-blocking — the bounded-park-on-all-cooled behaviour) / §11.4.103 (continuous parallel-stream routine) / §11.4.111 (resolve-by-stable-name — config by serial, never by index) / §11.4.116 (real-time conductor↔framework sync channel) / §11.4.119 (single-resource-owner partitioning) / §11.4.126 (default autonomous-loop mode) / §11.4.128 (always-on device-recording — the analogous host-safety discipline) / §11.4.131 (standing session-resumption file) / §11.4.147 (crashed-agent respawn-until-complete — the ruler-self-supervisor is its ruler-layer instantiation) / §11.4.164 (constitution auto-propagation hook seam) / §11.4.167 (feature work-stream lifecycle) / §11.4.176 (multi-track work-division + exactly-once claim + device-lock) / §12.6/§12.8 (host-safety memory + concurrency ceilings — the spawn/launch pool is budget-gated). Propagation gate `CM-COVENANT-114-187-PROPAGATION` (literal `11.4.187`) + recommended gates `CM-MULTITRACK-ENGINE-IN-CONSTITUTION` (the generic ruler scripts present under `constitution/scripts/multitrack/`, carrying no project literal, `bash -n + sh -n` clean per §11.4.67) / `CM-MULTITRACK-AUTOINSTALL-WIRED` / `CM-MULTITRACK-BOOTSTRAP-IDEMPOTENT` / `CM-MULTITRACK-BIND-ON-START` / `CM-MULTITRACK-FALLBACK-MONITOR` / `CM-MULTITRACK-HEADLESS-SPAWN` / `CM-MULTITRACK-RULER-SUPERVISOR` + paired §1.1 mutation (strip the literal → propagation gate FAILs; re-introduce a project literal into the engine, or downgrade a rate-limit rebind to exit-code-based success detection → the corresponding mechanism gate FAILs). **Canonical authority:** constitution submodule `Constitution.md` §11.4.187. Non-compliance is a release blocker. No escape hatch — no `--skip-multitrack`, `--manual-multitrack-only`, `--no-auto-install`, `--per-project-reimpl-OK`, `--exit-code-is-success-signal` flag.

§11.4.188 — Regular main → feature merge cadence: every track / feature-branch MUST FREQUENTLY merge canonical `main` into itself DURING the work, never only at the end (User mandate, 2026-07-09). Compact summary: every long-lived feature branch AND every parallel-development track (§11.4.58/ §11.4.103 parallel streams; §11.4.167 feature work-streams; §11.4.176/§11.4.178/

§11.4.179 multi-track) MUST regularly `git merge origin/main` (the canonical trunk/master) INTO its own branch THROUGHOUT the work — NOT only at the end — so no branch ever drifts far from main and the eventual back-merge stays a small, low-conflict, low-risk operation. GENERALISES §11.4.167(D) (trunk-merged-INTO-every-BIG-feature-STREAM, at minimum after every trunk tag / daily, `git merge` never rebase, stale-if-not-synced) from the big-work-item feature-stream lifecycle to EVERY feature branch on EVERY track — a small/medium feature branch, or any `git worktree` /CoW track that never earned a full §11.4.167 stream, is STILL bound to this cadence (the §11.4.167(D) discipline promoted to the general standalone rule; §11.4.167(D) remains its big-item specialisation). (1) **WHAT** — fetch the latest canonical `main` / `master` and merge it INTO the working branch, keeping the branch continuously close to trunk; a branch that has NOT integrated trunk within the cadence is flagged STALE (the exact long-lived-branch conflict-storm + integration-risk anti-pattern this forbids). (2) **CADENCE** — merge trunk in FREQUENTLY: after EVERY trunk tag, at minimum DAILY while the branch is live, AND before starting any significant new chunk of work on the branch — MANY SMALL merges, never one big deferred end-of-branch merge (small frequent merges minimise the divergence + conflict surface). (3) **HOW (safe-by-construction)** — MERGE, NEVER rebase a shared/tagged/pushed branch (no commit-loss, no history rewrite — §9/§11.4.113); FETCH-FIRST every remote before the merge (§11.4.37/§11.4.71 — remote state is unknowable without a fetch, §11.4.6); take a §9.2 hardlinked pre-op backup before any LARGE / conflict-heavy / risky merge; resolve every conflict CAREFULLY — ZERO conflict markers committed, ZERO file silently dropped, union of both sides preserved (§11.4.41 step 3 / §9 no-loss); integrate ONLY when the branch is QUIESCENT (§11.4.84 — no in-flight mutation gate, no unaccounted uncommitted work); prefer running the merge + its verification in the BACKGROUND parallel to the main stream so it never stalls other work (§11.4.88/§11.4.89/§11.4.103); NEVER force-push the merged result (§11.4.113 — the merged branch fast-forwards its own mirrors). (4) **CONSISTENCY + ANTI-BLUFF** — after EVERY merge the branch MUST be PROVEN consistent, never ASSUMED: a pre-build smoke gate runs post-merge (§11.4.42 step-3 smoke) and is GREEN, a `git grep` for conflict markers (`<<<<<<<` / `=====` / `>>>>>>>`) returns EMPTY, and no commit present pre-merge on either side is lost — captured evidence per §11.4.5/§11.4.69; a merge claimed "clean" without the post-merge smoke + marker-scan + no-lost-commit check is a §11.4/§11.4.1 bluff at the integration layer. (5) **HONEST BOUNDARY (§11.4.6)** — regular trunk-integration keeps the branch MERGEABLE + low-conflict;

it does NOT by itself prove the branch's own work is correct (each change still crosses §11.4.108 runtime-signature + §11.4.40 retest), and it is NOT the approval-gated back-merge TO trunk (that stays §11.4.167(I)/§11.4.40/§11.4.41 no-merge-until-approved). Classification: universal (§11.4.17) — the consuming project supplies its canonical-trunk ref name, cadence unit, post-merge smoke-gate command, and per-track worktree/CoW mechanism per §11.4.35. Composes §9/§9.2/§11.4.6/§11.4.17/§11.4.37/§11.4.41/§11.4.42/§11.4.71/§11.4.84/§11.4.88/§11.4.89/§11.4.103/§11.4.113/§11.4.167/§11.4.176/§11.4.178/§11.4.179/§11.4.181. Propagation gate `CM-COVENANT-114-188-PROPAGATION` (literal `11.4.188`) + recommended gate `CM-REGULAR-MAIN-MERGE-INTO-FEATURE` (every live feature branch / track has integrated the canonical trunk within the declared cadence; its last integration is a real `git merge` not a rebase; the post-merge smoke + zero-conflict-marker + no-lost-commit checks passed) + paired §1.1 mutation (let a live branch go stale past the cadence, OR record a rebase of a shared branch in place of a merge, OR commit a conflict marker → the gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.188. Non-compliance is a release blocker. No escape hatch — no `--skip-main-merge`, `--merge-only-at-end`, `--rebase-shared-branch`, `--defer-trunk-sync`, `--no-post-merge-smoke`, `--force-push-after-merge`, `--big-items-only-trunk-sync` flag.

§11.4.189 — Most-reopened cases get extra-depth live-testing scrutiny FIRST (User mandate, 2026-07-10). Verbatim operator mandate: "All live testing MUST pay extra attention to cases which have been reopened numerous times and those cases in depth retested, investigated, fully validated and verified with real physical results and no bluff of any kind anywhere!" All LIVE TESTING (on-device / on-target runtime validation, deploy-and-observe) MUST give EXTRA-DEPTH retest + in-depth investigation + FULL validation/verification with REAL PHYSICAL captured evidence (§11.4.5/§11.4.69/§11.4.107 — captured audio/video/sysfs/dumpsys/sink-side/runtime-signature) and NO bluff of any kind ANYWHERE to the cases that have been REOPENED THE MOST TIMES (highest §11.4.55 reopens-count) — the empirically-most-fragile set gets the DEEPEST live scrutiny, and it gets it FIRST. (1) **WHAT** — for the highest-reopens-count cases, live testing is NOT a single pass: it re-runs the case at EXTRA DEPTH (more iterations per §11.4.50 deterministic consistency, wider edge-case + regression coverage), RE-INVESTIGATES the underlying defect per §11.4.102 systematic-debugging + §11.4.146 reproduce-first, and CONFIRMS with rock-solid captured physical evidence per §11.4.123 — never

metadata-only / config-only / absence-of-error / grep-without-runtime (§11.4/§11.4.1), never a false result. (2) **ORDER** — the most-reopened set is scrutinised FIRST, ahead of the rest of the live-test suite (the §11.4.132 risk-descending discipline, with most-reopened the load-bearing key). (3) **KEY** — priority is keyed off the §11.4.55 reopens-count (the consuming project supplies its reopens-count source per §11.4.35, e.g. the §11.4.93 workable-items DB `reopens_count`); a high reopens-count is the strongest empirical fragility signal (§11.4.132(d)). (4) **PROOF** — each most-reopened case's live PASS cites its captured-evidence artefact path (§11.4.69/§11.4.107) on a CLEAN/fresh deployment (§11.4.108 runtime-signature), with the §11.4.115 RED → GREEN polarity flip captured where a permanent regression guard exists (§11.4.135). STRENGTHENS/REFINES §11.4.132 (risk-ordered validation factor (d) most-reopened — §11.4.189 promotes most-reopened into its OWN EXTRA-DEPTH mandate at the LIVE-TEST layer, not merely one ranking factor among four) + §11.4.55 (reopens-count as the priority key) + §11.4.129/§11.4.130 (validate-the-just-fixed / highest-risk-first after redeploy). Classification: universal (§11.4.17) — references no project-specific hardware; the consuming project supplies its reopens-count source per §11.4.35. Composes §11.4.5/§11.4.6/§11.4.55/§11.4.69/§11.4.107/§11.4.108/§11.4.115/§11.4.129/§11.4.130/§11.4.132/§11.4.135/§11.4.146. Propagation gate `CM-COVENANT-114-189-PROPAGATION` (literal `11.4.189`) + recommended gate `CM-MOST-REOPENED-EXTRA-DEPTH-LIVE-TEST` (the live-test suite orders + extra-depth-scrutinises the highest-reopens-count cases FIRST, each with captured physical evidence on a clean deployment) + paired §1.1 mutation (drop the most-reopened extra-depth pass, OR run it AFTER the rest of the suite, OR PASS a most-reopened case on metadata-only evidence → the gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.189. Non-compliance is a release blocker. No escape hatch — no `--skip-most-reopened-depth` , `--reopened-same-as-any` , `--live-test-any-order` , `--metadata-pass-for-reopened-OK` , `--defer-reopened-scrutiny` flag.

§11.4.190 — Website engineering-quality mandate: every project website MUST be fully responsive + completely SEO-optimized + uniquely OpenDesign-authored + bleeding-edge enterprise-quality, each PROVEN with captured evidence (User mandate, 2026-07-10). Verbatim operator mandate: "every website we create inside our projects MUST BE fully responsive - working on all browsers, all operating systems, all platforms and all devices and screen

sizes! Every Website MUST BE completely SEO optimized! Layouts and templates done MUST BE unique and everything fully designed from OpenDesign! Make sure this applies to our project Website as well! Everything must be bleeding-edge enterprise quality, stunning visual look and feel, innovative, creative and unique! There MUST BE no false or faulty result or bluff of any kind anywhere!" Every website / web-UI surface a project ships — INCLUDING this project's OWN website — MUST satisfy AND PROVE with captured physical evidence (§11.4.5/§11.4.69/ §11.4.107/§11.4.170 — never a bare claim, §11.4.6) ALL of: **(A) FULL RESPONSIVENESS** — correct, usable rendering across ALL major browser engines (Chromium/Firefox/WebKit), all operating systems, all device classes (desktop/laptop/tablet/phone/large-display) and the full range of screen sizes/ orientations/pixel-densities; NO broken layout / overflow / clipping / overlap / off-screen or unusable control at ANY breakpoint. **(B) COMPLETE SEO OPTIMIZATION** — semantic HTML + correct document outline; per-page unique `<title>` + meta description; Open Graph + Twitter cards; canonical URLs; structured data (schema.org / JSON-LD); `robots` + XML sitemap; crawlable / indexable markup; WCAG AA accessibility + Core Web Vitals performance (both feed ranking). **(C) UNIQUE OpenDesign-AUTHORED templates/layouts** — EVERY layout + template designed FROM the OpenDesign design-token system (§11.4.162), NOT generic / off-the-shelf / templated; visually unique, innovative, creative, brand-distinct. **(D) BLEEDING-EDGE ENTERPRISE VISUAL QUALITY** — stunning, polished, professional look-and-feel in BOTH light AND dark themes (§11.4.162 tokens). **(E) ANTI-BLUFF PROOF** — NONE of (A)–(D) may be CLAIMED without captured physical evidence: responsiveness PROVEN by device-independent host-rendered screenshots across the real breakpoint × engine matrix + an OCR/vision layout oracle (no overlap / label-over-label / clip / overflow / off-screen); SEO PROVEN by an automated audit (Lighthouse SEO + structured-data validation + meta / OG / sitemap / robots presence) meeting a defined score floor; visual quality PROVEN by the §11.4.170 device-independent host-rendered pixel proof per screen × state × {light,dark}; uniqueness PROVEN by OpenDesign-token provenance. A website reported responsive / SEO-optimized / unique / enterprise-quality WITHOUT these captured proofs is a §11.4 PASS-bluff at the web-UI layer. §11.4.190 is the strict WEB-quality expansion of §11.4.162 (OpenDesign — it applies §11.4.162's design-token mandate to the full responsive + SEO + enterprise-quality web surface) and binds §11.4.170's device-independent host-rendered pixel proof as the correctness oracle; sibling of §11.4.168 (exported-document independent visual validation) at the web-UI layer. Classification:

universal (§11.4.17) — references no project-specific hardware; the consuming project supplies its concrete breakpoint × engine matrix, SEO score-floor, and OpenDesign token set per §11.4.35. Composes §11.4.5/§11.4.6/§11.4.69/§11.4.107/§11.4.162/§11.4.168/§11.4.170. Propagation gate `CM-COVENANT-114-190-PROPAGATION` (literal `11.4.190`) + recommended gates `CM-WEBSITE-RESPONSIVE-PROVEN` (host-rendered screenshots across the breakpoint × engine matrix + layout oracle — no overlap/clip/overflow/off-screen) / `CM-WEBSITE-SEO-OPTIMIZED` (automated SEO audit meets the score floor + structured-data / meta / OG / sitemap / robots present) / `CM-WEBSITE-OPENDSIGN-UNIQUE` (every layout/template's OpenDesign-token provenance, no generic/off-the-shelf template) + paired §1.1 mutation (report a website responsive/SEO/unique/enterprise-quality with NO captured proof, OR ship a generic non-OpenDesign template, OR skip the light/dark host-rendered pixel proof → the recommended gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item). **Canonical authority:** constitution submodule `Constitution.md` §11.4.190. Non-compliance is a release blocker. No escape hatch — no `--skip-responsive-proof`, `--seo-optional`, `--generic-template-OK`, `--visual-quality-later`, `--claim-without-render` flag.

§11.4.191 — Work-to-track/branch binding enforcement: a logic-group's work can NEVER be committed OR dispatched onto the wrong track/branch (research-derived, 2026-07-10). Compact summary: every logical group of workable items binds to EXACTLY ONE canonical (`track`, `branch`) recorded ONCE in the single source of truth (the §11.4.93 workable-items DB `logic_groups.destination` = authoritative branch + a `canonical_track` column + a `group_paths` file-scope manifest), and NO change belonging to a group may LAND on, nor be DISPATCHED to, a track/branch other than that group's canonical one — the file-scope generalisation of §11.4.181/§11.4.182 from branch-NAME consistency to work-PLACEMENT consistency (forensic FACT 2026-07-10: `srccpy/Remote-app` work — group `mistiq-vader`, canonical `feature/mistiq-vader` /T2 — was committed onto `main` /T1; §11.4.181's `guard-branch-consistency.sh` never fired because committing onto an existing branch is not a feature-branch CREATE, and no gate checked which FILES land on which BRANCH). (1) **Single source of truth** — the `logic_group` → (`branch`, `track`, `path-globs`) binding is recorded ONCE in the DB and LOOKED UP, never re-invented (a commit/dispatch placing a group's file-scope on a non-canonical branch/track is a §11.4.6 no-guessing violation). (2) **Preventive hook**

(§11.4.109/§11.4.177) — a PreToolUse guard `guard-work-track-binding.sh` (inherited BY REFERENCE, never copied) BLOCKS (exit 2) a git-commit-class Bash call whose to-be-committed file set contains a path owned by a group whose canonical branch/track $\neq$ the current checkout, AND BLOCKS an Agent/Task dispatch whose §11.4.182 label track/branch $\neq$ the canonical (track, branch) of the ticket's group; fail-closed on unreadable registry, `# guardrails:allow <reason>` audited escape. (3) **Detective gate (§11.4.110/§11.4.108)** — `CM-WORK-TRACK-BINDING-ENFORCED` asserts, on the current branch's own added commits (`merge-base(HEAD,main)..HEAD`), that every `group_paths`-owned file's group destination == current branch (and canonical_track == current track when pinned) — the actual enforcement boundary regardless of how the commit was made, the runtime signature being FAIL-on-wrong-placement / PASS-on-canonical-placement on a clean checkout. (4) **Risk-free reconciliation (§9.2/§11.4.113/§11.4.84)** — a mis-placed change is reconciled by MERGING its commits onto the group's canonical branch (union, no commit lost, no `-s ours /reset`), ff-only push to all upstreams (NEVER force-push), then removing it from the wrong branch only after the merge is confirmed on all mirrors. Classification: universal (§11.4.17) — references no project hardware; the consuming project supplies its `(track, branch)` map, `group_paths` seed, and commit-wrapper wiring per §11.4.35. Composes

§11.4.6/.28/.29/.35/.58/.75/.84/.93/.95/.109/.110/.113/.119/.167/.176/.177/.178/.181/.182/

§9.2. Propagation gate `CM-COVENANT-114-191-PROPAGATION` (literal `11.4.191`) + recommended gate `CM-WORK-TRACK-BINDING-ENFORCED` + paired §1.1 mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.191. Non-compliance is a release blocker. No escape hatch — no `--allow-cross-track-commit`, `--skip-work-binding`, `--commit-any-branch-OK`, `--dispatch-any-track-OK`, `--no-file-scope-check` flag.

§11.4.192 — Continuous multi-track auto-backfill: a FREE track MUST be IMMEDIATELY re-assigned its next-highest-priority domain work-set, never left idle while its domain has actionable items (User mandate, 2026-07-11).

Verbatim operator mandate: "once a Track is free, we shall assign next set of workable items to it ... full continuous multi-track development ... no stopping ... this is MANDATORY". Compact summary: the moment any track's worker COMPLETES its current work-set (its item(s) closed with anti-bluff evidence, or merged), that FREE track MUST be IMMEDIATELY assigned the NEXT-highest-priority actionable work-set from its OWN domain (its §11.4.191 logic_group /

canonical branch) — a track is NEVER left idle while its domain still holds actionable workable items, and the conductor/ruler CONTINUOUSLY re-assigns with NO stopping (full continuous multi-track development); PROMOTES the §11.4.103(B) ≥3-parallel-background-streams-with-auto-backfill invariant into the multi-track (§11.4.176/§11.4.178/§11.4.179/§11.4.187) layer (§11.4.103(B) backfills a STREAM the moment it is done; §11.4.192 backfills a TRACK the moment its worker is done, keyed to that track's canonical (track, branch) domain per §11.4.191). (1) **WHAT** — on any track-worker completion the conductor/ruler (§11.4.187) MUST, per that track's domain, look up the NEXT actionable work-set, CLAIM it exactly-once (§11.4.176 exactly-once claim registry, no two tracks claiming one item), and DISPATCH it as the track's new §11.4.182-labelled work stream — without waiting for a scheduled wake or an operator prompt. (2) **PRIORITY** — the next work-set is the track domain's highest-priority actionable item (§11.4.42/§11.4.72 priority order, audio-first where in scope), never an arbitrary pick. (3) **SOURCE-SIDE ADVANCES EVEN WHEN VALIDATION IS GATED** — when the next item's on-device VALIDATION is blocked on a pending build / device availability (§11.4.108 clean-target / §11.4.185 manual-QA / operator-gated flash), the track MUST still advance the item's SOURCE-SIDE work (investigation, RED-test §11.4.115, source patch, pre-build gate + paired §1.1 mutation, docs) and honestly §11.4.3/§11.4.52 SKIP-with-reason ONLY the genuinely device-gated validation portion — a build/device gate on the validation step is NOT a reason to idle the whole track. (4) **HONEST BOUNDARY (§11.4.6)** — a track idles ONLY when its domain GENUINELY has zero remaining actionable workable items OR every remaining item is externally blocked with no source-side advancement possible (§11.4.94(A)/§11.4.97/§11.4.101 idle-only-when-blocked closed set); "the merges are done" / "the current item is done" / "nothing visible right now" is NEVER a valid idle reason while the domain still holds progressable items (a §11.4.94+§11.4.97 violation); an unavoidable per-item block PARKS one item, it does NOT stop the track (§11.4.101 — the track backfills its next non-blocked domain item). Classification: universal (§11.4.17) — references no project hardware; the consuming project supplies its track → domain map + priority source per §11.4.35. Composes §11.4.42/.58/.72/.94/.97/.101/.103/.167/.176/.178/.179/.182/.187. Propagation gate CM-COVENANT-114-192-PROPAGATION (literal 11.4.192) + recommended gate CM-CONTINUOUS-MULTITRACK-BACKFILL (on a track-worker completion the ruler re-assigns that track's next-highest-priority domain work-set; a track with a completed worker AND remaining actionable domain items but no new dispatch →

the gate FAILs) + paired §1.1 mutation (let a free track sit idle while its domain holds an actionable item, OR treat "merges done" as a valid idle reason → the gate FAILs; strip the literal → propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.192. Non-compliance is a release blocker. No escape hatch — no `--allow-idle-track` , `--skip-backfill` , `--stop-multitrack` , `--merges-done-is-idle` , `--defer-track-reassign` flag.

§11.4.193 — Anti-blind-typing mandate: every UI interaction MUST be "seen" and "understood" via OCR/vision/screenshot proof, NEVER blind-typed (User mandate, 2026-07-13). Blind typing is STRICTLY FORBIDDEN everywhere — every agent, every tool, every automation. See-before-you-type, verify-after-you-type, OCR/vision oracle mandatory, WebView fallback required, credential redaction per §11.4.10. No escape hatch. Canonical authority: constitution submodule `Constitution.md` §11.4.193.